

Mapping Class Groups

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1 Introduction

The *mapping class group* Mod_S of an orientable surface S is defined as the group of isotopy classes of orientation-preserving diffeomorphisms $S \rightarrow S$. In addition to being a central object of the topology of surfaces (cf. 2.1), these groups also play an important role in the theory of Teichmüller spaces and in algebraic geometry, where they are known under the name *Teichmüller modular*

groups or simply *modular groups*. Our notations are derived from the latter terminology. There are several closely related groups, which also deserve the name of the mapping class groups (or Teichmüller modular groups). First of all, one may consider the *extended mapping class group* Mod_S° of S defined as the group of the isotopy classes of *all* diffeomorphisms $S \rightarrow S$. The *pure mapping class group* PMod_S of S is defined as the group of isotopy classes of all orientation-preserving diffeomorphisms $S \rightarrow S$ preserving setwise all boundary components of S . Finally, one may consider the group $\mathcal{M}_S$ of all (orientation-preserving) diffeomorphisms $S \rightarrow S$ fixed on the boundary ∂S , considered up to isotopies fixed on the boundary. If $\partial S \neq \emptyset$, then diffeomorphisms fixed on ∂S are automatically orientation-preserving. If $\partial S = \emptyset$, then, of course, $\mathcal{M}_S = \text{Mod}_S$. All these groups could be also defined as the 0-th homotopy groups of suitable diffeomorphisms groups of S . For example, $\text{Mod}_S = \pi_0(\text{Diff}(S))$, where $\text{Diff}(S)$ is the group of all orientation-preserving diffeomorphisms of S considered with, for example, C^∞ -topology (or any other reasonable topology).

The study of the mapping class groups was initiated in the 1920-ies by M. Dehn [D], [D1], [D2] and J. Nielsen [Ni1]–[Ni3]. Although their work had some common themes (cf., for example, the Dehn-Nielsen-Baer theorems in 2.9), in general their approaches were fairly different. M. Dehn was interested in the properties of the mapping class group as a whole, addressing, for example, such questions as the existence of a finite set of generators. He developed and exploited an important tool for this purpose: the action of the mapping class group Mod_S on the collection of the isotopy classes of all circles on S . He called this collection the *arithmetic field* of S . On the contrary, J. Nielsen was mainly interested in understanding the fine structure of the individual elements of Mod_S . His methods draw heavily on hyperbolic geometry (a tool also favored by M. Dehn). For quite a while, the work of both M. Dehn and J. Nielsen was apparently forgotten. The ideas of M. Dehn related to the arithmetic field of a surface found a natural continuation in the ideas of W. Harvey [Harv1], [Harv2] about the *complex of curves* of a surface, which is nothing else but the arithmetic field made in a simplicial complex in a natural way (see 3.1). A closely related object was considered in an influential paper of A. Hatcher and W. Thurston [HatT]. The ideas of J. Nielsen were partially rediscovered, extended and brought to an essentially complete form by W. Thurston in his theory of surface diffeomorphisms [Th1], [FLP] (see Chapter 7). Later on, his theory was applied also to the structure of the mapping class group (and not only to its individual elements).

In the present notes we have adopted a point of view going back to that of M. Dehn. Compared to the other thread, going back to J. Nielsen, it is more elementary and allows us to reach more quickly some really deep results about mapping class groups. Everyone interested in the mapping class groups eventually should learn the other point of view also (especially Thurston's theory of surface diffeomorphisms). We hardly do justice to it in a chapter devoted to Thurston's theory and some of its applications. If one takes M. Dehn's

terminology seriously, our exposition is slanted toward the *arithmetic* of our subject, which seems to be only natural for an introduction. The analogy with the *arithmetic groups* (see 9.1 for a definition), briefly touched on in Chapter 9, fits nicely in this approach (for example, the main results of Chapter 6 are, in fact, analogues of some fundamental theorems about arithmetic groups), giving a deeper meaning to the word *arithmetic* in this context.

Our exposition is centered around three topics: the Dehn-Lickorish theorem providing a finite set of generators of Mod_S for closed S ; J. Harer's theorem computing the so-called virtual cohomology dimension of Mod_S ; and the author's theorem describing all automorphisms of W. Harvey's complex of curves. The first two topics are presented with complete proofs and a detailed review of the prerequisites; for the third topic only an outline is given. The proofs we give for the theorems of Dehn-Lickorish and Harer contain some new ideas and may be interesting even for experts. The table of contents gives a good idea of the topics included in these notes. The limitations of time and space prevented us from giving a more detailed treatment of many included topics and from discussing other topics, equally or, may be, even more important. As we already mentioned, our treatment of Thurston's theory is hardly adequate (although we hope that it is still useful); the same applies to the aspects of the theory of the Teichmüller spaces related to the mapping class groups. Nothing is said about relations with algebraic geometry and mathematical physics; we refer to the recent survey of R. Hain and E. Looijenga [HL] for this. Two omitted topics are very closely related to the ones included. In order not to pass them in complete silence, we very briefly discuss them here.

The first one is the study of the so-called *Torelli group*, the subgroup of Mod_S consisting of the isotopy classes of diffeomorphisms of S acting trivially on $H_1(S, \mathbf{Z})$. The pioneering work in this field is due to D. Johnson. In [J3] he proved that the Torelli group of S is finitely generated if the genus of S is at least 3 and S has no more than 1 boundary component, and in [J4], [J5] he computed the first homology group of the Torelli group (under the same restrictions). See [J2] for a survey of this work. It was continued, in particular, in the algebro-geometric work of R. Hain; see [H2], and especially the remarkable paper [H3], where R. Hain proved the existence of a finite presentation for a Lie algebraic analogue of the Torelli group. Note that the question about the existence of a finite presentation of the Torelli group itself remains open.

The second topic is the theory and applications of the characteristic classes of surface bundles (which are nothing other than the elements of the cohomology group $H^*(\text{Mod}_S)$), constructed independently by D. Mumford [Mumf3], Sh. Morita [Mor1], [Mor2] and E. Miller [Mi]. Sh. Morita [Mor1], [Mor2] and E. Miller [Mi] proved the nontriviality of these classes, and Sh. Morita [Mor1], [Mor2] found a spectacular application of this nontriviality: he proved that the canonical homomorphism $\text{Diff}(S) \rightarrow \text{Mod}_S$ admits no section. In his further papers (see, for example, [Mor3]–[Mor6], [Mor10]–[Mor12]) Sh. Morita studied these characteristic classes and applied them to the theory of the Torelli group

(extending in some respects the above mentioned work of D. Johnson) and to the Casson invariant of homology 3-spheres and related problems of the topology of 3-manifolds. We refer to his own surveys [Mor7]–[Mor9] for a discussion of these results.

I am very grateful to R. Sher for the careful reading of this paper.

2 Topology of surfaces

In this section we attempt to outline the material which should be at the foundation of the theory of the mapping class groups. All this material is fairly classical, although some results (for example, the ones about spaces of diffeomorphisms and embeddings) are not so widely known as they deserve. Our discussion is more broad than is needed for later applications, in order to get closer to a coherent picture; still, our treatment is far from being complete. The proofs in this section are not to be taken too seriously; mostly, they are only outlines or just indications of the ideas involved in the real proofs.

2.1 The dimension 2

In order to put the topology of surfaces in a perspective, let us start with a particular view of the topology of manifolds in general.

The central problem of the topology of manifolds is the problem of classification of manifolds. It is usually considered to be more or less solved for at least simply connected manifolds of dimension ≥ 5 . Another major problem is the problem of isotopy classification of diffeomorphisms. Consideration of diffeomorphisms up to isotopy distinguishes topology from other subjects, especially from the theory of smooth dynamical systems, which concentrates on the classification of diffeomorphisms up to conjugacy. A satisfactory solution of the isotopy classification problem for diffeomorphisms is known only for some special classes of manifolds. The problem which naturally comes after the classification of diffeomorphisms is the problem of understanding of the homotopy type of the diffeomorphism group of a manifold. It is extremely difficult even for such manifolds as spheres. These problems actually form a natural series. Namely, the classification of manifolds, the isotopy classification of diffeomorphisms, and the computation of homotopy groups $\pi_i(\text{Diff}(M))$ for $i > 0$, where $\text{Diff}(M)$ is the group of all diffeomorphisms $M \rightarrow M$ (with, say, C^∞ -topology), form a series of problems, parallel and related to the study of functors K_1 , K_2 , and K_i for $i > 2$ of the algebraic K -theory. We can add also to this row the problem of the existence of a smooth manifold within a given homotopy type and the functor K_0 .

In the dimension 2, the solution of the classification problem is well known and is, in fact, a part of any introductory course in topology. The homotopy groups $\pi_i(\text{Diff}(M))$ with $i > 0$ are known for all compact two dimensional

manifolds, and, moreover, most of them are equal to 0. We will discuss these results in 2.7. This leaves us in the dimension 2 with only one of the above problems—the problem of the classification of diffeomorphisms. Interpreted broadly, this problem naturally includes a search for an understanding of the mapping class groups $\pi_0(\text{Diff}(M))$, which is the main focus of our survey. But, before we concentrate on $\pi_0(\text{Diff}(M))$, we will review some other parts of the topology of surfaces, partly because they are needed to understand $\pi_0(\text{Diff}(M))$ and partly for their own sake.

2.2 Classification of surfaces

Throughout the paper, all surfaces are assumed to be orientable and, unless the contrary is obvious from the context, compact.

There is no doubt that the reader is familiar with the classification of surfaces. Let us recall, for the sake of completeness, that an orientable connected surface S is determined up to a diffeomorphism by its genus g and the number of the boundary components b . Following Riemann, the genus can be defined as the maximal number of disjoint circles on S such that these circles do not separate S , i.e. the complement in S of the union of these circles is connected. The genus g and the number of boundary components b are related to the Euler characteristic $\chi(S)$ of S by the well known formula $\chi(S) = 2 - 2g - b$. A surface of genus g with b boundary components is also known as a *sphere with g handles and b holes*; the terminology is justified by a well known picture.

The standard proof of this classification theorem, based on triangulations and cutting and pasting arguments, provides a canonical model of a surface of genus g with b boundary components. Namely, such a surface can be obtained from a $4g$ -gon by a well known identification of sides and removing b disjoint discs from the interior. The identification of sides is described by the word $a_1 b_1 a_1^{-1} b_1^{-1} \dots a_g b_g a_g^{-1} b_g^{-1}$ and this leads to a presentation of $\pi_1(S)$:

$$\pi_1(S) = \langle a_1, b_1, \dots, a_g, b_g, d_1, \dots, d_b \mid a_1 b_1 a_1^{-1} b_1^{-1} \dots a_g b_g a_g^{-1} b_g^{-1} d_1 \dots d_b = 1 \rangle.$$

If $b > 0$, then one can eliminate one d_i from this presentation and conclude that $\pi_1(S)$ is a free group. One can obtain the same result in a more topological manner, by observing (using the above $4g$ -gon model, for example), that S has a 1-dimensional *CW*-complex as a deformation retract. But considering $\pi_1(S)$ for surfaces with boundary as just a free group is to a great extent misleading, because in this way we lose all information about the boundary. It is much better to keep the track of the boundary by introducing the so-called *peripheral structure* — the set of $2b$ conjugacy classes of elements of $\pi_1(S)$ corresponding to loops going once round a boundary component. It is also useful to consider the *oriented peripheral structure* — the set of b conjugacy classes of elements of $\pi_1(S)$ corresponding to loops going once round a boundary component in the

direction prescribed by a fixed orientation of S (which induces an orientation of the boundary ∂S).

Note that all generators in the above presentation can be obviously represented by *embedded* loops, i.e. by loops $f : [0, 1] \rightarrow S$ such that $f(x) = f(y)$ only if $x = y$ or $\{x, y\} = \{0, 1\}$ (of course $f(0) = f(1)$ is the base point).

2.3 Circles and arcs on surfaces

Circles and arcs on surfaces and the action of diffeomorphisms on them are some of the main tools (and objects) of the topology of surfaces. By *circles* we understand the *embedded* circles, i.e. submanifolds diffeomorphic to the standard circle. Similarly, by *arcs* we understand submanifolds diffeomorphic to the interval $[0, 1]$. Usually, one restricts attention to properly embedded arcs; an arc I on a surface S is called *properly embedded* (or simply *proper*) if $\partial I = I \cap \partial S$ and I is transverse to ∂S .

It is convenient to single out circles and arcs considered to be trivial. We call a circle C on a surface S *trivial* if C is either contractible in S , or is homotopic to some boundary component of S . Similarly, we call an arc I *trivial* if it can be deformed into the boundary of S in such a way that its boundary ∂I stays in ∂S . Also, it is important to distinguish between separating and nonseparating circles. We call a circle C on S *nonseparating* if $S \setminus C$ is connected, and *separating* otherwise.

As we noticed in 2.2 all generators in the standard presentation of $\pi_1(S)$ can be represented by embedded loops. The image of an embedded loop is a circle, and we call an embedded loop *separating* or *nonseparating* if this image is separating, or, respectively, nonseparating. The obvious loops representing $a_1, b_1, \dots, a_g, b_g$ are nonseparating, while the obvious loops representing $d_1, \dots, d_b$ are separating (in fact, their images can be deformed into the corresponding boundary components). But one can easily represent elements $a_1 d_1, \dots, a_1 d_b$ by embedded nonseparating loops. It follows that $\pi_1(S)$ can be generated by elements represented by nonseparating embedded loops. This remark will be used in our proof of the Dehn-Lickorish theorem (see the proof of Theorem 4.2.C).

If C is a circle on S or, more generally, union of several disjoint circles, we can cut S along C and get a new surface, which will be denoted by S_C . If C is a nonseparating circle, then S_C is connected. If C is a separating circle, then S_C consists of two components. Note that we can always reconstruct S from S_C by gluing along the boundary components of S_C resulting from C . More generally one can cut S also along proper arcs and, more generally, disjoint unions of circles and proper arcs. If (proper) arcs are involved, the result of cutting is a smooth surface with corners.

Suppose now that C is a nonseparating circle. Then S_C is a connected surface having two more boundary components than S . The additivity of the Euler characteristic implies that S_C has the same Euler characteristic as S .

Hence, by the classification of surfaces, S_C is determined up to a diffeomorphism by S (and the fact that C is nonseparating).

Lemma 2.3.A *If C, C' are two nonseparating circles on S , then there is a diffeomorphism $f : S \rightarrow S$ such that $f(C) = C'$.*

Proof. By the remarks preceding the Lemma, S_C and $S_{C'}$ are diffeomorphic. Let us fix an orientation of S ; it induces an orientation of both S_C and $S_{C'}$. Since every orientable surface obviously admits an orientation-reversing diffeomorphism, we can always choose an orientation-preserving diffeomorphism $F : S_C \rightarrow S_{C'}$. Let C_- and C_+ be two boundary components of S_C resulting from C , and let $C'_- = F(C_-)$, $C'_+ = F(C_+)$. Let $I : C_- \rightarrow C_+$ and $I' : C'_- \rightarrow C'_+$ be two gluing diffeomorphisms giving S back from S_C and $S_{C'}$, respectively. If F agrees with these gluings, i.e. if $F \circ I = I' \circ F$ on C_- , then F gives rise (by gluing) to a diffeomorphism $f : S \rightarrow S$ such that $f(C) = C'$. Note that both I and I' are orientation-reversing (assuming that C_-, C_+, C'_-, C'_+ are oriented as boundary components of $S_C, S_{C'}$) and F is orientation-preserving on both C_- and C_+ . It follows that diffeomorphisms $F \circ I, I' \circ F : C_- \rightarrow C'_+$ are both orientation-reversing. Since every two orientation-reversing (as also every two orientation-preserving) diffeomorphisms of a circle are isotopic, we can find a diffeomorphism $F' : C_+ \rightarrow C'_+$ isotopic to $F|_{C_+}$ and such that $F' \circ I = I' \circ F|_{C_-}$. By extending the isotopy between $F|_{C_+}$ and F' to an isotopy of F (and without changing $F|_{C_-}$), we get a diffeomorphism $F : S_C \rightarrow S_{C'}$ which agrees with gluings. As we noted above, this is sufficient to complete the proof. $\square$

If C is nonseparating, then there is a circle C' transversely intersecting C at exactly one point. (To construct such a C' , take a small arc I intersecting C transversely at one point in the interior of I and connect the two ends of I by an arc in $S \setminus C$.) In particular, the homological intersection number $C \cdot C'$ is ± 1 . This immediately implies that C is nontrivial.

Lemma 2.3.B *A circle C contractible in S bounds a disc in S . A circle C homotopic to a boundary component B of S , bounds, together with B , an annulus in S . In particular, a circle is non-trivial if and only if it does not bound a disc in S and does not bound an annulus in S together with a boundary component.*

Proof. First, notice that a trivial circle C is always separating by the remarks preceding the Lemma. For such a C , we can represent S as the union of two surfaces S_1, S_2 intersecting only along their common boundary component C . If C is contractible in S , then van Kampen's theorem (together with the basic properties of the amalgamated products) implies that C is contractible in one of the surfaces S_1, S_2 , say in S_1 . In view of 2.2, any embedded loop with image C is freely homotopic to a loop representing d_i or d_i^{-1} for some generator d_i of

$\pi_1(S_1)$. Because C is contractible in S_1 , this implies that $d_i = 1$. Obviously, this is possible only if the genus of S_1 is 0 and S_1 has only one boundary component (namely, C), i.e. only if S_1 is a disc. In this case C bounds a disc in S , namely S_1 . A similar argument applies if C is homotopic to a boundary component. We leave it to the reader. $\square$

2.4 Pants decompositions

With few exceptions, a surface has negative Euler characteristic (by the classification of surfaces). The simplest such surface is a disc with two holes, often called a *pair of pants*, or simply *pants*; see Fig. 2.1 (a). As the next theorem shows, pairs of pants are the elementary building blocks of all surfaces of negative Euler characteristic.

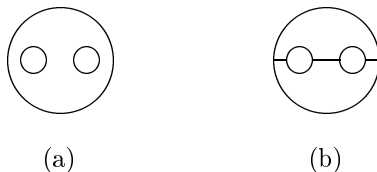


Fig. 2.1

Theorem 2.4.A *If S is a surface of negative Euler characteristic (i.e., if S is not a sphere, a torus, a disc, or an annulus), then there is a collection of disjoint circles on S such that the result S_C of cutting S along their union C is a disjoint union of pairs of pants. In other words, C decomposes S into pairs of pants.*

Proof. Using the classification of surfaces, one can easily exhibit the required collection of circles in each case. Let us outline a proof independent of the classification. Choose a Morse function $f : S \rightarrow \mathbf{R}$ on S such that f is equal to 0 on ∂S , f is nonnegative and the values of f at different critical points are different. Let $0 = a_0 < a_1, \dots < a_n$ be a sequence of noncritical values of f such that every interval $[a_{i-1}, a_i]$, $1 \leq i \leq n$ contains exactly one critical value of f and $\text{Im } f \subset [0, a_n]$. Consider a component of the preimage $f^{-1}([a_{i-1}, a_i])$. If it does not contain a critical point of f , it is an annulus. If it contains a critical point of f of index 0 or 2, then it is a disc. Finally, if it contains a critical point of f of index 1, then it is easily seen to be a disc with two holes. So, if we cut S along the union of the components of all preimages $f^{-1}(a_i)$, $i \geq 1$, the resulting surface will consist of discs, annuli, and pairs of pants. We can simplify this decomposition of S into discs, annuli, and pairs of pants in the following manner. Note that gluing a disc to a boundary component of a disc, annulus

or a pair of pants results in a sphere, a disc, or an annulus respectively. Also, gluing an annulus to a boundary component of any surface does not change it up to a diffeomorphism. Hence, by repeatedly gluing discs and annuli to other pieces of the decomposition, we will eventually reach a decomposition of S either into several pairs of pants, or into two discs glued together (in this case S is a sphere), or into an annulus glued to itself (in this case S is a torus), or into just one disc or annulus. To complete the proof, note that a gluing operation corresponds to removing a circle from our collection of cutting circles. $\square$

This theorem can be used to give an alternative proof of the classification of surfaces. Let us cut S along the union of several circles into a finite collection of pairs of pants, as in the theorem. Then let us reassemble S from this collection in the following manner. Start with some pair of pants and glue to it the other pairs of pants, one at a time, and only along one boundary component at a time, as long as this is possible. At every stage we will have, as one can easily see by induction, a disc with several holes. When no further such gluing is possible, the remaining gluings identify some pairs of boundary components of our disc with holes. If there are g such pairs and b boundary components not in any such pair, then the result of the gluing of these pairs is, obviously, a sphere with g handles and b holes, i.e. a surface of genus g with b boundary components.

Sometimes it is useful to decompose pairs of pants further. Namely, three nontrivial arcs, shown in Fig. 2.1(b), decompose a pair of pants into two “topological hexagons” (which are smooth manifolds with corners and are homeomorphic, but not diffeomorphic to a disc). Combining this decomposition with Theorem 2.4.A, we can decompose S into such topological hexagons.

From the perspective of 3-dimensional topology, nontrivial circles and arcs are analogues of incompressible surfaces, and the above decomposition of S into hexagons is an analogue of the Haken decomposition of a Haken manifold into topological balls. In fact, since pairs of pants are simple enough, one rarely needs to use nontrivial arcs to decompose a surface into simpler pieces. Note that from the point of view of this analogy all surfaces, with the exception of discs and spheres, are Haken.

2.5 Geometric structures on surfaces

By a *geometric structure* on a surface S we understand simply a riemannian metric on S having constant curvature and geodesic boundary. By scaling the metric, if necessary, we can assume that its curvature is equal to 1, 0 or -1 . It turns out that every surface admits a geometric structure. For surfaces of nonnegative Euler characteristic this is very simple. For a sphere, it is sufficient to take the metric of the standard unit sphere in $\mathbf{R}^3$. For a disc, it is sufficient to take the metric of a hemisphere of a standard sphere. For an annulus, one may take a product metric on $S^1 \times [0, 1]$. For a torus, one may take the metric on $\mathbf{R}^2/\mathbf{Z}^2$ induced from the standard Euclidian metric on $\mathbf{R}^2$. In particular,

spheres and discs admit metrics of (constant) positive curvature, and annuli and tori admit metrics of curvature 0. The really interesting case is the case of surfaces of negative Euler characteristic. They all admit *hyperbolic structures*, i.e. geometric structures with metric of curvature -1 ; see Theorem 2.5.B below.

Geometric structures on surfaces can be used on two levels. On the first level, the very existence of a geometric structure is useful. For example, fixing a geometric structure on a surface S allows us to represent isotopy classes of circles by geodesics; for hyperbolic structures such representatives are unique. Moreover, the following lemma holds.

Lemma 2.5.A *Suppose that S is endowed with a geometric structure. Then every nontrivial circle on S is isotopic to a geodesic circle on S . If S is endowed with a hyperbolic structure, then such a geodesic circle is unique. If two nontrivial circles are disjoint, then any two geodesic circles isotopic to them are either disjoint or equal.*

We refer to [FLP], Exposé 3, §§I and III for a proof. On a second level, one considers all possible geometric structures on S simultaneously. It is convenient to identify geometric structures on S which differ one from another in a trivial way; namely, to identify two geometric structures if one is the pull-back of the other by a diffeomorphism $S \rightarrow S$ isotopic to the identity. The resulting set of equivalence classes is the *Teichmüller space of S* . We explore this point of view in Section 5.

Theorem 2.5.B *If a surface has negative Euler characteristic, then it admits a hyperbolic structure.*

Proof. There are (at least) two ways to prove this result. The first one works best for closed surfaces S . Represent S as a result of the standard identification of sides of a $4g$ -gon; cf 2.2. One can realize this $4g$ -gon as a regular polygon on the hyperbolic plane with all angles equal to $2\pi/4g = \pi/2g$. Note that the hyperbolic plane has a canonical riemannian metric of constant negative curvature (in the upper half plane model, it is the metric $(dx^2 + dy^2)/y^2$ on $\{(x, y) \in \mathbf{R}^2 : y > 0\}$); this metric induces a riemannian metric on S because the lengths of glued edges are equal and the sum of angles is equal to 2π (notice that all vertices of our $4g$ -gon are identified in the gluing process). Clearly, this metric on S has constant negative curvature. Note that it is not necessary to consider a regular $4g$ -gon; a weaker condition would suffice. For the details of this construction we refer to [St], Sections 5.5 and 5.6.

Now, one can use Lemma 2.5.A to deal with the surfaces with boundary. Given a surface with boundary S , we can endow the double dS of S with a hyperbolic structure. Recall that dS is the result of gluing S with a copy of itself along the boundary; in particular, $dS \supset S$. Lemma 2.5.A implies that ∂S is isotopic in dS to a union of disjoint geodesic circles. If we cut dS along this

union, we get two surfaces diffeomorphic to S , each of them endowed with a hyperbolic structure.

Another construction works equally well both for closed surfaces and for surfaces with boundary and is based on Theorem 2.4.A. Let us prove first that a pair of pants P admits a hyperbolic structure such that all three boundary components have the same length. To this end, we represent P as the union of two topological hexagons as in Fig. 2.1(b). Let us realize these hexagons as two (isometric) regular hexagons in the hyperbolic plane with all angles equal to $\pi/2$. Then gluing P from these hexagons endows P with a hyperbolic structure (one should take isometries as the gluing maps; this is possible because lengths of edges are all equal); the boundary is geodesic because angles of hexagons are equal to $\pi/2$. Clearly, the lengths of all boundary components of P are equal to twice the length of an edge of our hexagons; in particular, the lengths of all boundary components of P are equal. Now, given a surface S of negative Euler characteristic, we can decompose S into pairs of pants as in Theorem 2.4.A. Let us endow all pairs of pants of this decomposition with the hyperbolic structure we just constructed. Since the lengths of all boundary components of these pairs of pants are equal, one can take isometries as gluing maps and then these hyperbolic structures on pieces of S induce a hyperbolic structure on S itself. For the details, and a more general version of this approach, we refer to [Bu], Section 1.7 and Chapter 3. It can be easily generalized to give a description of all hyperbolic structures on S ; see [Bu], Chapters 3 and 6, and [FLP], Exposé 7 and Section 5. $\square$

2.6 Spaces of diffeomorphisms and embeddings: generalities

For an orientable manifold M we denote by $\text{Diff}^\diamond(M)$ the group of all diffeomorphisms $M \rightarrow M$, and by $\text{Diff}(M)$ the subgroup of all orientation-preserving diffeomorphisms. (Since we are interested mainly in the groups $\text{Diff}(M)$, we chose for them a simpler notation.) For $X \subset M$ we denote by $\text{Diff}^\diamond(M \text{ fix } X)$ the subgroup of diffeomorphisms $f : M \rightarrow M$ fixed on X , i.e. such that $f(x) = x$ for all $x \in X$. We denote by $\text{Diff}(M \text{ fix } X)$ the intersection $\text{Diff}^\diamond(M \text{ fix } X) \cap \text{Diff}(M)$. The most important case is that of $X = \partial M$; we denote the groups $\text{Diff}^\diamond(M \text{ fix } \partial M)$ and $\text{Diff}(M \text{ fix } \partial M)$ by $\text{Diff}^\diamond(M \text{ fix } \partial)$ and $\text{Diff}(M \text{ fix } \partial)$ respectively.

If N is a submanifold of M , we denote by $\text{Emb}^\diamond(N, M)$ the set of all embeddings $N \rightarrow M$ which can be extended to a diffeomorphism $M \rightarrow M$. If N has codimension 0 in M (the main case is that of a disc in the interior of M), then it makes sense to speak about orientation-preserving embeddings $N \rightarrow M$, and we denote by $\text{Emb}(N, M)$ the subset of all orientation-preserving embeddings. By $\text{Sub}(N, M)$ we denote the set of submanifolds of M diffeomorphic to N by a diffeomorphism which extends to M (i.e., the set of all images of embeddings from $\text{Emb}^\diamond(N, M)$).

As usual, all diffeomorphisms and embeddings are assumed to be of the class C^∞ , and the above groups of diffeomorphisms and sets of embeddings are considered together with their C^∞ -topology. $\text{Sub}(M, N)$ is considered with the topology of a quotient space of $\text{Emb}^\circ(N, M)$. As is well known, groups of diffeomorphisms with the C^∞ -topology are topological groups. Also, for a submanifold N of M the natural map $\text{Diff}^\circ(M) \rightarrow \text{Emb}^\circ(N, M)$ given by the restriction of diffeomorphisms to N is continuous.

It is easy to see that if M is a surface, then $\text{Mod}_S = \pi_0(\text{Diff}(S))$, $\text{Mod}_S^\circ = \pi_0(\text{Diff}^\circ(S))$ and $\mathcal{M}_S = \pi_0(\text{Diff}(S \text{ fix } \partial))$.

Theorem 2.6.A *Let N be submanifold of M such that ∂N intersects ∂M along several components of ∂N and N is transverse to ∂M . Then the natural map $\text{Diff}^\circ(M) \rightarrow \text{Emb}^\circ(N, M)$ is a Serre fibration. If N is of codimension 0 in M , then the natural map $\text{Diff}(M) \rightarrow \text{Emb}(N, M)$ is a Serre fibration. If N is of codimension 0 in M and is contained in the interior of M , then the natural map $\text{Diff}(M \text{ fix } \partial) \rightarrow \text{Emb}(N, M)$ is also a Serre fibration.*

Proof. Recall that a continuous map is a Serre fibration if the homotopy lifting property holds for cubes. For $\text{Diff}^\circ(M) \rightarrow \text{Emb}^\circ(N, M)$ the homotopy lifting property for 0-dimensional cubes amounts to the extension of isotopies of N in M to isotopies of the whole M . This extension property is well known and easy to prove; see for example, [Kos], Theorem II.5.2. The homotopy lifting property for higher dimensional cubes is essentially a multiparameter version of this isotopy extension property and can be proved by a direct generalization of the standard proof the usual isotopy extension property. Also, the second statement of the theorem obviously follows from the first. The last statement of the theorem can be proved in exactly the same manner as the first one. $\square$

Theorem 2.6.B *The natural map $\text{Emb}^\circ(N, M) \rightarrow \text{Sub}(N, M)$, $\text{Emb}(N, M) \rightarrow \text{Sub}(N, M)$, given by the formula $f \mapsto f(N)$, are Serre fibrations.*

This theorem can be proved using a variant of the ideas of the proof of Theorem 2.6.A.

The fibers of the fibrations $\text{Diff}^\circ(M) \rightarrow \text{Emb}^\circ(N, M)$ and $\text{Diff}(M) \rightarrow \text{Emb}(N, M)$ are easy to identify. They are $\text{Diff}^\circ(M \text{ fix } N)$ and $\text{Diff}(M \text{ fix } N)$ respectively. In addition, if N is a submanifold of codimension 0 contained in the interior of M , then we can identify $\text{Diff}^\circ(M \text{ fix } N)$ and $\text{Diff}(M \text{ fix } N)$ with $\text{Diff}^\circ(M \setminus \text{int } N \text{ fix } \partial N)$ and $\text{Diff}(M \setminus \text{int } N \text{ fix } \partial N)$ respectively. The fibers of the maps $\text{Emb}^\circ(N, M) \rightarrow \text{Sub}(N, M)$, $\text{Emb}(N, M) \rightarrow \text{Sub}(N, M)$ are, obviously, $\text{Diff}^\circ(N)$ and $\text{Diff}(N)$ respectively.

Now, let us fix a point $m \in M$ and consider the space FM of all orientation-preserving isomorphisms $T_m M \rightarrow T_x M$, $x \in M$ (where $T_x M$ is, as usual, the tangent space to M at x). There is an obvious projection $FM \rightarrow M$; moreover, FM is a principal $GL^+(n, \mathbf{R})$ -bundle over M , where $n = \dim M$. Let

$d : \text{Diff}(M) \rightarrow FM$ be the map assigning to a diffeomorphism $M \rightarrow M$ its differential at m . If N is a submanifold of codimension 0 and $m \in N$, then there is a similar map $d : \text{Emb}(N, M) \rightarrow FM$.

Theorem 2.6.C *The maps $d : \text{Diff}(M) \rightarrow FM$ and $d : \text{Emb}(N, M) \rightarrow FM$ are Serre fibrations. If N is a disc of codimension 0 in the interior of M , then the second map is a weak homotopy equivalence.*

Proof. We may consider points of FM as framed points in M . Then the homotopy lifting property for d becomes a version of the (multiparameter) isotopy extension property for isotopies of points, which is not much more difficult to prove the usual isotopy extension property. The second statement of the theorem is essentially a multiparameter version of a well known theorem to the effect that there is exactly one isotopy class of orientation-preserving embeddings of a codimension 0 disc in an orientable manifold. See, for example, [Kos], Corollary III.3.6. Again, it is easy to add parameters to the usual proof. $\square$

If $\dim M = 2$, it is convenient to fix a riemannian metric on M and consider the unit tangent bundle $UT(M)$ instead of FM . Let us fix a vector $v \in T_m M$. Then the evaluation at v followed by the normalization defines a map $FM \rightarrow UT(M)$, which is a homotopy equivalence (if $\dim M = 2$). Also, define $u : \text{Diff}(M) \rightarrow UT(M)$ as the map assigning to a diffeomorphism $M \rightarrow M$ the normalized image of v under its differential at m . Similarly define $u : \text{Emb}(N, M) \rightarrow UT(M)$.

Theorem 2.6.D *The maps $u : \text{Diff}(M) \rightarrow UT(M)$ and $u : \text{Emb}(N, M) \rightarrow UT(M)$ are Serre fibrations. If N is a disc of codimension 0 in the interior of M (and $\dim M = 2$), then the second map is a weak homotopy equivalence.*

Proof. The first statement is similar to the first statement of Theorem 2.6.C and, in fact, holds in all dimensions. The second statement follows from the second statement of Theorem 2.6.C and the fact that the above map $FM \rightarrow UT(M)$ is a homotopy equivalence. $\square$

The results of this section go back to J. Cerf [Ce1] and, for closed manifolds, R. Palais [P]. In fact, they proved that our Serre fibrations are actually fiber bundles. But we need only the Serre fibration property, because we are interested only in the homotopy groups of spaces of diffeomorphisms and embeddings (in fact, only in π_0 and π_1), and this property is much easier to prove.

2.7 Spaces of diffeomorphisms and embeddings: surfaces

In this section, we describe the weak homotopy type of some spaces of diffeomorphisms of surfaces and embeddings of circles. While all these results are known to be true for the usual homotopy type instead of the weak one, the

results about the weak homotopy type are easier to prove (in particular, there is no need to know that Serre fibrations from 2.6 are fiber bundles), and only they are needed for the applications we have in mind.

We start with two preliminary results about 1-dimensional manifolds.

Lemma 2.7.A *$\text{Diff}(D^1 \text{ fix } \partial)$ is contractible, where D^1 is the 1-dimensional disc.*

Proof. We may assume that $D^1 = [0, 1]$. Now, if $f : [0, 1] \rightarrow [0, 1]$ is a diffeomorphism fixed on $\{0, 1\}$, then $x \mapsto (1 - t)f(x) + tx$ is a diffeomorphism for every $t \in [0, 1]$, because its derivative is always > 0 . $\square$

Corollary 2.7.B *$\text{Diff}(S^1)$ is homotopy equivalent to S^1 and, moreover, contains S^1 as a deformation retract.*

Proof. Note that S^1 is naturally contained in $\text{Diff}(S^1)$ as the group of rotations. Fix $x \in S^1$ and consider the Serre fibration $\text{Diff}(S^1) \rightarrow \text{Emb}(\{x\}, S^1)$. Its base can be obviously identified with S^1 , and $S^1 \subset \text{Diff}(S^1)$ defines a section. Using this section, it is easy to see that this Serre fibration is actually a fiber bundle. If we notice that its fiber is homotopy equivalent to $\text{Diff}(D^1 \text{ fix } \partial)$, the corollary follows. $\square$

The first basic result about the groups of diffeomorphisms of surfaces is the following theorem of S. Smale [Sm].

Theorem 2.7.C *$\text{Diff}(D^2 \text{ fix } \partial)$ is contractible, where D^2 is the 2-dimensional disc.*

In that follows, we understand by a *weak deformation retract* of a topological space X a subspace $Y \subset X$ such that the inclusion $Y \rightarrow X$ is a weak homotopy equivalence. The following corollary is also due to S. Smale [Sm] (who actually proved it with all adjectives “weak” omitted).

Corollary 2.7.D *$\text{Diff}(D^2)$ is weakly homotopy equivalent to a circle. $\text{Diff}(S^2)$ is weakly homotopy equivalent to $SO(3)$ and, moreover, contains $SO(3)$ as a weak deformation retract.*

Proof. Consider the (Serre) fibration $\text{Diff}(D^2) \rightarrow \text{Diff}(\partial D^2)$. Its base is homotopy equivalent to S^1 by Corollary 2.7.B and its fiber is homotopy equivalent to a point by Theorem 2.7.C. The first statement follows. Now, fix a disc $D^2 \subset S^2$ and consider the fibration $\text{Diff}(S^2) \rightarrow \text{Emb}(D^2, S^2)$. By Theorem 2.6.D its base $\text{Emb}(D^2, S^2)$ is weakly homotopy equivalent to the unit tangent bundle of S^2 , which can be identified with $SO(3)$. The fiber is $\text{Diff}(S^2 \text{ fix } D^2)$, which can be identified with $\text{Diff}(S^2 \setminus \text{int } D^2 \text{ fix } \partial D^2)$. But

$S^2 \setminus \text{int } D^2$ is a disc, hence by Theorem 2.7.C the fiber is contractible. It follows that $\text{Diff}(S^2)$ is weakly homotopy equivalent to $SO(3)$. Moreover, the fibration $u : \text{Diff}(S^2) \rightarrow UT(S^2) = SO(3)$ from 2.6 is a weak homotopy equivalence. The inclusion of $SO(3)$ in $\text{Diff}(S^2)$ as the subgroup of rotations defines a section of this fibration. Clearly, the image of this section, i.e. $SO(3)$, is a weak deformation retract of $\text{Diff}(S^2)$. $\square$

Corollary 2.7.E *Let A be an annulus (i.e. a manifold diffeomorphic to $S^1 \times [0, 1]$). Then $\text{Diff}(A \text{ fix } \partial)$ has weakly contractible components, and the group of components $\pi_0(\text{Diff}(A \text{ fix } \partial))$ is isomorphic to $\mathbf{Z}$.*

Proof. Consider the fibration $\text{Diff}(D^2 \text{ fix } \partial) \rightarrow \text{Emb}(B^2, D^2)$, where D^2 is a disc and B^2 is a disc in the interior of D^2 . Its total space is contractible by Theorem 2.7.C. By Theorem 2.6.D its base is weakly homotopy equivalent to the unit tangent bundle of D^2 and, hence, to S^1 . Its fiber is $\text{Diff } D^2 \setminus \text{int } B^2 \text{ fix } \partial$, which can be identified with $\text{Diff}(A \text{ fix } \partial)$. The corollary easily follows from these facts and the homotopy sequence of our fibration. $\square$

Theorem 2.7.F *Let $T = S^1 \times S^1$ be a torus. The components of $\text{Diff}(T)$ are weakly homotopy equivalent to T and, in fact, contain T as a weak deformation retract.*

Note that since T is a Lie group, T is naturally contained in $\text{Diff}(T)$. In fact, this subgroup $T \subset \text{Diff}(T)$ is a (weak) deformation retract of the component of the identity of $\text{Diff}(T)$.

Theorem 2.7.G *If S is a surface of negative Euler characteristic, then the components of $\text{Diff}(S \text{ fix } \partial)$ and of $\text{Diff}(S)$ are weakly contractible.*

Theorems 2.7.F and 2.7.G (in fact, with adjectives “weak” omitted) are due to C. Earle and J. Eells [EE]. The proofs in [EE] are analytical and are based on the theory of Beltrami equations. The corresponding results about spaces of homeomorphisms are due to M.-E. Hamstrom [Ham1], [Ham2]; her proofs are purely topological. The corresponding results in the piecewise linear category were obtained by G. P. Scott [Sco]. Nowadays the easiest way to obtain Theorems 2.7.F and 2.7.G is, probably, to consider them as 2-dimensional versions of 3-dimensional results of A. Hatcher [Hat1] and the author [II] about Haken manifolds (cf. end of 2.5). Unfortunately, such an approach is not written down yet.

Theorem 2.7.H *Let S be a surface of negative Euler characteristic and let C be a nontrivial circle on S . Then the components of $\text{Emb}(C, S)$ are weakly homotopy equivalent to a circle and the components of $\text{Sub}(C, S)$ are weakly homotopy equivalent to a point.*

This theorem can be deduced from the above results about spaces of diffeomorphisms using fibrations from 2.6. Alternatively, if one proves Theorems 2.7.F and 2.7.G along the lines of theorems of [Hat1] and [I1] about Haken manifolds, then this theorem is, essentially, the main step of the proof.

Finally, we give an application of these theorems which will be needed later (see 6.3).

Theorem 2.7.I *Let R be a connected subsurface of a connected surface S of negative Euler characteristic. Suppose that the boundary of R is connected and nontrivial in S . Then the natural map $\mathcal{M}_R \rightarrow \mathcal{M}_S$, given by the extension of diffeomorphisms of R fixed on ∂R by the identity to diffeomorphisms of S , is injective.*

Proof. Let T be the closure of $S \setminus R$. Clearly, T is connected. The natural map $\text{Diff}(S \text{ fix } \partial) \rightarrow \text{Emb}(T, S)$ is a Serre fibration with fiber $\text{Diff}(R \text{ fix } \partial)$. The map $\pi_0(\text{Diff}(R \text{ fix } \partial)) \rightarrow \pi_0(\text{Diff}(S \text{ fix } \partial))$ induced by the inclusion of the fiber in the total space is nothing other as our map $\mathcal{M}_R \rightarrow \mathcal{M}_S$. In view of the homotopy sequence of this fibration, this map will be injective if $\pi_1(\text{Emb}(T, S)) = 0$. Note that since ∂R is nontrivial, T has negative Euler characteristic and hence $\pi_1(\text{Diff}(T)) = 0$. Let us consider now the natural map $\text{Emb}(T, S) \rightarrow \text{Sub}(T, S)$ from Theorem 2.6.B, which is a Serre fibration with fiber $\text{Diff}(T)$. In view of the homotopy sequence of this fibration, $\pi_1(\text{Emb}(T, S)) = 0$ if $\pi_1(\text{Sub}(T, S)) = 0$. But a submanifold of S diffeomorphic to T is determined by its boundary (may be up to two possibilities, if R is diffeomorphic to T). It follows that $\pi_1(\text{Sub}(T, S)) = \pi_1(\text{Sub}(\partial T, S))$. The latter group is 0 in view of Theorem 2.7.H. This completes the proof. $\square$

2.8 Gluing discs to a surface

If a surface has nonempty boundary, then we can construct a closed surface or a surface with a smaller number of boundary components by gluing discs to boundary components. Very often this gluing procedure allows us to reduce theorems about surfaces with possibly nonempty boundary to the case of closed surfaces, or use the induction on the number of boundary components. In this section we discuss some basic constructions and results used in such arguments.

Let R be a surface with nonempty boundary and let Q be the result of gluing discs $D_1, \dots, D_n$ to n different boundary components of R . Given a diffeomorphism $f : R \rightarrow R$ preserving setwise all components of ∂R , there exists an extension $f' : Q \rightarrow Q$ of f . Such an extension is obviously not unique, but it is unique up to isotopy, because the spaces $\text{Diff}(D_i \text{ fix } \partial)$ are connected. Hence, extension of diffeomorphisms from R to Q defines a map $\rho : \text{PMod}_R \rightarrow \text{PMod}_Q$. Clearly, ρ is a homomorphism. We are interested in particular in its kernel, and

to understand it we need the notion of the pure braid groups of surfaces and the properties of these groups discussed below.

For any surface S , consider $S^{[m]} = \{(y_1, \dots, y_m) \in S^m : y_i \neq y_j \text{ for } i \neq j\}$. The *pure braid group* $PB_m(S)$ is defined as $\pi_1(S^{[m]})$, where we tacitly assume that some base point $(x_1, \dots, x_m) \in S^{[m]}$ have been chosen.

Let $S_{(i)} = S \setminus \{x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_m\}$ (note that $S_{(i)}$ is a noncompact surface). Let x_i be the base point of $S_{(i)}$. The map $S_{(i)} \rightarrow S^{[m]}$ given by the formula $x \mapsto (x_1, \dots, x_{i-1}, x, x_{i+1}, \dots, x_m)$ induces a map $\pi_1(S_{(i)}, x_i) \rightarrow \pi_1(S^{[m]}) = PB_m(S)$.

Theorem 2.8.A *The group $PB_m(S)$ is generated by the images of the above maps $\pi_1(S_{(i)}, x_i) \rightarrow \pi_1(S^{[m]}) = PB_m(S)$.*

Proof. Use induction on m and the natural fibrations $S^{[l]} \rightarrow S^{[l-1]}$ given by the formula $(y_1, \dots, y_l) \rightarrow (y_1, \dots, y_{l-1})$. $\square$

We are mainly interested in the group $PB_n(Q)$. We will assume that the base point $(x_1, \dots, x_n) \in Q^{[n]}$ has been chosen in such a way that $x_i \in \text{int } D_i$, $1 \leq i \leq n$. Let Q_i be the result of gluing only the disc D_i to R . Clearly, Q_i is contained in $Q_{(i)}$ and hence natural maps $\pi_1(Q_i, x_i) \rightarrow PB_n(Q)$ are defined. In fact, Q_i is a deformation retract of $Q_{(i)}$.

Corollary 2.8.B *The group $PB_n(Q)$ is generated by the images of the natural maps $\pi_1(Q_i, x_i) \rightarrow PB_n(Q)$.*

Now we construct a homomorphism $j : PB_n(Q) \rightarrow \text{PMod}_R$. Let $\beta \in PB_n(Q)$; represent β by a loop in $Q^{[n]}$. Such a loop may be considered as an isotopy of the 0-dimensional submanifold $\{x_1, \dots, x_n\}$ of Q . Let us extend this isotopy to an isotopy $\{f_t : Q \rightarrow Q\}_{0 \leq t \leq 1}$ of the whole manifold Q ; in particular, $f_0 = \text{id}_Q$, $f_1(x_i) = x_i$ for all i . We can easily arrange that, in addition, $f_1(D_i) = D_i$ for all i . (It follows from Theorem 2.6.C that the space of discs in a surface, containing a given point in the interior, is connected.) Then f_1 preserves R and preserves all boundary components of R . We take as $j(\beta)$ the isotopy class of the restriction of f_1 on R . One can easily check that j is correctly defined (using Theorem 2.6.C again). We leave it as an exercise to check that j is a homomorphism. If $n = 1$, then $PB_n(Q) = PB_1(Q)$ is nothing more than the fundamental group $\pi_1(Q)$ and j is a homomorphism $\pi_1(Q) \rightarrow \text{PMod}_R$.

The first main property of j is contained in the following theorem.

Theorem 2.8.C *The homomorphism ρ is surjective and $\text{Ker } \rho = \text{Im } j$; in other words, the sequence*

$$PB_n(Q) \longrightarrow \text{PMod}_R \longrightarrow \text{PMod}_Q \longrightarrow 1$$

is exact. If Q has negative Euler characteristic, then j is injective; in other words, the sequence

$$1 \longrightarrow PB_n(Q) \longrightarrow \text{PMod}_R \longrightarrow \text{PMod}_Q \longrightarrow 1$$

is exact.

Proof. Let us prove the surjectivity of $\rho : \text{PMod}_R \rightarrow \text{PMod}_Q$ first. Represent an element α of PMod_Q by a diffeomorphism $g : Q \rightarrow Q$. Connectedness of the spaces of (orientation-preserving) embeddings of discs into a surface easily implies that g is isotopic to a diffeomorphism g' such that $g'(D_i) = D_i$, $1 \leq i \leq n$. Let γ be the isotopy class of the restriction of g' to R . Clearly, $\rho(\gamma) = \alpha$. This proves the surjectivity of ρ .

Now, let $\gamma \in \text{Ker } \rho$. Let us represent γ by a diffeomorphism $f : R \rightarrow R$ and extend f to a diffeomorphism $f' : Q \rightarrow Q$. We may assume that $f'(x_i) = x_i$, $1 \leq i \leq n$. Since $\gamma \in \text{Ker } \rho$, the diffeomorphism f' is isotopic to the identity id_Q . Let $\{f'_t : Q \rightarrow Q\}_{0 \leq t \leq 1}$ be some isotopy between $f' = f'_0$ and $f'_1 = \text{id}_Q$. Then $t \mapsto (f'_t(x_1), \dots, f'_t(x_n))$ defines a loop in $Q^{[n]}$ with the base point $(x_1, \dots, x_n)$ and, hence, an element $\beta \in \pi_1(Q_{[n]}) = PB_n(Q)$. Obviously, $j(\beta) = \gamma$. This proves that $\text{Ker } \rho \subset \text{Im } j$. The opposite inclusion follows directly from the definitions.

Suppose now that the Euler characteristic of Q is negative. Let $\beta \in \text{Ker } j$. Let us represent β by a loop in $Q^{[n]}$ and construct an isotopy $\{f_t : Q \rightarrow Q\}_{0 \leq t \leq 1}$ as in the definition of j . In particular, $f_1(D_i) = (D_i)$ for all i . Since $\beta \in \text{Ker } j$, the restriction of f_1 on R is isotopic to id_R . It follows that after changing the isotopy, if necessary, we may assume that the restriction of f_1 on R is equal to id_R . Next, in view of Theorem 2.7.C, we may assume that $f_1 = \text{id}_Q$. Now, $\{f_t : Q \rightarrow Q\}_{0 \leq t \leq 1}$ defines a loop in $\text{Diff}(Q)$. By Theorem 2.7.G this loop is contractible. Clearly, this implies that the original loop representing β is contractible, and hence $\beta = 1$. $\square$

This Theorem is due to J. Birman [Bir1]. See also [Bir3], Theorem 4.2. In general, $\text{Ker } j$ is isomorphic to the quotient of the group $PB_n(Q)$ by its center, which can be easily identified; see [Bir3], Chapter 4.

Now, let $\alpha \in \text{Mod}_R$. Suppose that some (and then every) diffeomorphism $f : R \rightarrow R$ representing α preserves (setwise) the union of circles ∂D_i , $1 \leq i \leq n$. Then f can be extended to a diffeomorphism $f' : Q \rightarrow Q$. Such extension is unique up to isotopy (cf. the construction of ρ). We may assume that $f'(x_i) = x_{\sigma(i)}$ for some permutation $\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$. Extension f' is unique up to isotopy even in the class of such diffeomorphisms (this follows from the fact that discs are connected and the results of 2.6). Such an extension f' induces a map $Q^{[n]} \rightarrow Q^{[n]}$ mapping the base point $* = \{x_1, \dots, x_n\}$ to $\sigma(*) = \{x_{\sigma(1)}, \dots, x_{\sigma(n)}\}$. In particular, f' induces a homomorphism of the fundamental groups $\pi_1(Q_{[n]}, *) \rightarrow \pi_1(Q_{[n]}, \sigma(*))$. It is easy to see that this

homomorphism depends only on α . By a slight abuse of notations, we will denote it, even if $\sigma \neq \text{id}$, by $\alpha_* : PB_n(Q) \rightarrow PB_n(Q)$ (keeping in mind that the second pure braid group is defined using a different base point).

Note that, obviously, α_* maps the image of $\pi_1(Q_i, x_i)$ in $PB_n(Q)$ to the image of $\pi_1(Q_{\sigma(i)}, x_{\sigma(i)})$.

The homomorphisms α_* and j are related by the following fundamental formula:

$$\alpha j(\beta) \alpha^{-1} = j(\alpha_*(\beta)),$$

which follows immediately from the constructions of α_* and j .

As the last application of the extension of diffeomorphisms from R to Q , consider the situation when only one disc, D_1 , is added to R . Then the extension of diffeomorphisms defines a map $\text{Mod}'_R \rightarrow \pi_0(\text{Diff}(Q \text{ fix } x_1))$, where Mod'_R is the subgroup of Mod_R consisting of isotopy classes of diffeomorphisms preserving ∂D_1 . It turns out that this map is an isomorphism. (Again, this follows from 2.6.)

2.9 The Dehn-Nielsen-Baer Theorems

For a group π , let $\text{Aut}(\pi)$ be the group of automorphisms of π , and let $\text{Inn}(\pi)$ be the subgroup of inner automorphisms of π . A trivial check shows that $\text{Inn}(\pi)$ is a normal subgroup of $\text{Aut}(\pi)$. The quotient group $\text{Aut}(\pi)/\text{Inn}(\pi)$ is denoted by $\text{Out}(\pi)$ and called the *outer automorphisms group* of π . These groups naturally show up when one deals with homotopy self-equivalences of a (connected) space X with $\pi_1(X) = \pi$. Namely, any self-map $f : X \rightarrow X$ defines a homomorphism $f_* : \pi_1(X) \rightarrow \pi_1(X)$ and if f is a homotopy equivalence, then f_* is an isomorphism. But, if we do not assume a base point to be fixed, f_* is well defined only up to an inner automorphism of $\pi_1(X) = \pi$. Hence, a homotopy equivalence $f : X \rightarrow X$ defines an element $f_* \in \text{Out}(\pi)$. Obviously, f_* depends only on the homotopy class of f .

Now, let $X = S$ be a compact (connected) orientable surface. In view of the above remarks, there is a natural homomorphism $\text{Mod}_S^\circ \rightarrow \text{Out}(\pi_1(S))$. If S has nonempty boundary, then the elements of the image of this homomorphism obviously preserve the peripheral structure of $\pi_1(S)$ from 2.2.

Theorem 2.9.A *If S is closed and is not a sphere, then the natural homomorphism $\text{Mod}_S^\circ \rightarrow \text{Out}(\pi_1(S))$ is an isomorphism. If S has nonempty boundary and negative Euler characteristic (i.e., S is not a disc or an annulus), then the natural homomorphism $\text{Mod}_S^\circ \rightarrow \text{Out}(\pi_1(S))$ is an isomorphism onto the subgroup of $\text{Out}(\pi_1(S))$ consisting of elements preserving the peripheral structure.*

One can complement this theorem by a description of the subgroup of $\text{Out}(\pi_1(S))$ corresponding to Mod_S . Suppose first that S is closed (and is not a sphere). (For a review of the cohomology of groups, used below, see 6.1.) Then S is an Eilenberg-MacLane space, and hence $H^2(\pi_1(S), \mathbf{Z}) = H^2(S, \mathbf{Z}) = \mathbf{Z}$.

The group $\text{Out}(\pi_1(S))$ naturally acts on $H^*(\pi_1(S), \mathbf{Z})$, because inner automorphisms act trivially on cohomology groups. Let $\text{Out}^+(\pi_1(S))$ be the subgroup of $\text{Out}(\pi_1(S))$ consisting of elements acting trivially on $H^2(\pi_1(S), \mathbf{Z})$. Clearly, the image of Mod_S under the homomorphism $\text{Mod}_S^\diamond \rightarrow \text{Out}(\pi_1(S))$ is contained in $\text{Out}^+(\pi_1(S))$. If S has nonempty boundary (and negative Euler characteristic), then the image of Mod_S in $\text{Out}(\pi_1(S))$ is obviously contained in the subgroup of $\text{Out}(\pi_1(S))$ consisting of elements preserving the *oriented* peripheral structure from 2.2.

Theorem 2.9.B *If S is closed and is not a sphere, then the natural homomorphism $\text{Mod}_S \rightarrow \text{Out}^+(\pi_1(S))$ is an isomorphism. If S has nonempty boundary and negative Euler characteristic, then the natural homomorphism $\text{Mod}_S \rightarrow \text{Out}(\pi_1(S))$ is an isomorphism onto the subgroup of $\text{Out}(\pi_1(S))$ consisting of elements preserving the oriented peripheral structure.*

For closed S , the surjectivity part of these theorems is due to M. Dehn, who did not publish his proof, and to J. Nielsen [Ni1], who published a proof partially based on Dehn's ideas. Another proof was suggested by H. Seifert [Sei]. The injectivity part is due to R. Baer [Ba]. The surjectivity part for surfaces with boundary was proved by W. Magnus [M], and the injectivity part — much later — by H. Zieschang [Zi]. Proofs and a detailed discussion of these results can be found in [ZiVC]; see [ZiVC], Theorem 3.3.11 (surjectivity for closed surfaces following Seifert), Theorem 5.7.1 (surjectivity for surfaces with boundary) and Theorem 5.13.1 (injectivity).

In principle, these results give a purely algebraic description of mapping class groups and allow one to reduce every question about them to a purely algebraic question. Surprisingly, this reduction turned out to be not very successful. Probably, the only algebraic property of the mapping class groups which for a long time could be proved only in terms of the outer automorphisms groups is the residually finiteness.

Theorem 2.9.C *The groups $\text{Mod}_S^\diamond$ are residually finite, i.e. for every nontrivial element $f \in \text{Mod}_S^\diamond$, $f \neq 1$, there is a homomorphism $h : \text{Mod}_S^\diamond \rightarrow G$ onto a finite group G such that $h(f) \neq 1$.*

This theorem is due to E. Grossman [Gros], whose proof was based on the description of mapping class groups in terms of outer automorphisms groups and fairly complicated combinatorial group theory arguments. A more conceptual proof, still based on the description in terms of outer automorphisms groups was given by H. Bass and A. Lubotzky [BasL]. Now a simple direct proof, not based on the results of Dehn-Nielsen-Baer, is available; see [I14] or [I15], Exercise 1.

In one case, namely, if S is a closed torus, the Dehn-Nielsen-Baer theorems provide a complete and efficient description of Mod_S and $\text{Mod}_S^\diamond$. In this case $\pi_1(S)$ is abelian and, moreover, is isomorphic to $\mathbf{Z}^2$, and hence $\text{Out}(\pi_1(S)) = \text{Aut}(\pi_1(S)) = \text{Aut}(\mathbf{Z}^2) = GL_2(\mathbf{Z})$. This chain of isomorphisms clearly maps

the subgroup of $\text{Out}^+(\pi_1(S))$ to $SL_2(\mathbf{Z})$. It follows that $\text{Mod}_S^\diamond = GL_2(\mathbf{Z})$ and $\text{Mod}_S = SL_2(\mathbf{Z})$ for a closed torus S .

3 Complexes of curves

Complexes of curves are, probably, the most fundamental geometric objects on which the mapping class groups act (but the Teichmüller spaces are, certainly, the most important). They were discovered by W. Harvey [Harv1], [Harv2], and have played an ever increasing role since then. In this section we give the definition and discuss the basic properties of them. We prove that complexes of curves are connected, and do this in a way which can be generalized to prove much stronger homotopy properties of them. After this, we discuss the homotopy type and the hyperbolic properties of them. Later sections will amply illustrate the usefulness of complexes of curves.

3.1 Definitions

Complexes of curves are simplicial complexes in the sense of [Sp], Chapter 3 or [Bro3], for example. Thus, a *simplicial complex* consists of a set of *vertices* and a set of *simplices*. Simplices are non empty finite sets of vertices, subject only to the following two conditions: a non empty subset of a simplex is a simplex; every vertex belongs to some simplex. The *dimension* of a simplex is the number of vertices in it minus 1. One dimensional simplices play a special role and are called *edges*.

The vertices of the *complex of curves* $C(S)$ of a compact orientable surface S are the isotopy classes of simple closed curves (as usual, we call them *circles*) on S , which are *non-trivial* in the sense of 2.3, i.e. not contractible in S into a point or into the boundary ∂S . We denote the isotopy class of a circle C by $\langle C \rangle$. A set of vertices $\{\gamma_0, \dots, \gamma_n\}$ is declared to be a simplex if and only if $\gamma_0 = \langle C_0 \rangle, \dots, \gamma_n = \langle C_n \rangle$ for some pairwise disjoint circles $C_0, \dots, C_n$. The extended mapping class group $\text{Mod}_S^\diamond$ acts on $C(S)$ in an obvious way: the isotopy class $f \in \text{Mod}_S$ of a diffeomorphism $F : S \rightarrow S$ maps $\langle C \rangle$ to $\langle F(C) \rangle$.

$C(S)$ can be defined in another way. According to 2.5, we can equip S with a riemannian metric of constant curvature with geodesic boundary. Then we can take as the set of vertices of $C(S)$ the set of geodesic circles in $S \setminus \partial S$ and as simplices the sets of pairwise disjoint geodesic circles. In view of Lemma 2.5.A, this definition is equivalent to the previous one. This definition makes clear that a set of vertices is a simplex if and only if all its 2-element subsets are simplices (i.e., any two its vertices are connected by an edge). In the language of the theory of buildings (cf. [Bro3], for example) this means that $C(S)$ is a *flag complex*. Thus, $C(S)$ is, in fact, completely determined by its 1-skeleton. Higher dimensional simplices do not contribute any new combinatorial information, but they make the homotopy theory of $C(S)$ much more relevant and interesting.

Speaking about the homotopy properties of $C(S)$ we, of course, have in mind the homotopy properties of its *geometric realization* (cf. [Sp]), which is a topological space and even a CW-complex.

Obviously, $C(S)$ is an infinite complex. It is even *locally infinite* when it has positive dimension: every vertex is connected by an edge with an infinite number of vertices. But it is finitely dimensional: its dimension is equal to the maximal number of pairwise non-isotopic disjoint non-trivial circles on S minus 1. It follows that if S is a connected orientable surface of genus g with b boundary components, then the dimension of $C(S)$ is equal to $3g - 4 + b$, except in the case $g = 1$ and $b = 0$, when the dimension is zero (if this formula leads to a negative number, the complex of curves is empty).

3.2 Connectivity of $C(S)$ and the Morse-Cerf theory

Before discussing other homotopy properties of $C(S)$, we will present a proof of the connectedness of $C(S)$. This proof can be generalized to give the best possible connectivity result for closed surfaces S and surfaces S with one boundary component; cf. Theorem 3.2.C (the case of surfaces with more boundary components can be reduced to the case of closed surfaces in at least two different ways; cf. 3.3). It is based on some elementary facts from the theory of singularities, more precisely, on a generalization of the Morse theory to families of functions due to J. Cerf [Ce2], [Ce3]. Namely, we need the following result.

Lemma 3.2.A *Any family of functions $\{f_t : S \rightarrow \mathbf{R}\}_{0 \leq t \leq 1}$ can be approximated (say, in the C^∞ -topology) by a family of functions $\{g_t\}_{0 \leq t \leq 1}$ such that all functions g_t belong to one of the following three classes:*

- (i) *Morse functions with all critical values (i.e., values at critical points) different;*
- (ii) *Morse functions with two equal critical values and all other critical values different from them and from each other;*
- (iii) *functions having all critical values different and exactly one non-Morse critical point such that in appropriate local coordinates (x, y) around this critical point the function has the form $x^3 \pm y^2 + c$ for some constant $c \in \mathbf{R}$.*

This lemma was originally proved in [Ce2], Chapter II, and has become well known since then. The proof shows that one can also require that all but finitely many functions g_t belong to the first class, but we do not need this fact. For a general introduction to the relevant ideas of the theory of singularities one can recommend a nice short book by T. Poston and I. Stewart [PoS]; cf. [PoS], Chapters 1, 2.

Theorem 3.2.B *Suppose that S is neither a sphere with ≤ 4 holes, nor a torus with ≤ 1 holes. Then $C(S)$ is connected.*

Proof. If S is a torus with two holes, it is easy to see directly that $C(S)$ is connected. We leave this as an exercise to the reader, with the following hints: there are two types of non-trivial circles, namely, non separating circles and circles separating S into a torus with one hole and a disc with two holes; the circles of the first type are never disjoint, but any two of them can be made (by an isotopy) disjoint from a circle of the second type.

In the remaining part of the proof we assume that S is neither a sphere with ≤ 4 holes, nor a torus with ≤ 2 holes.

Let $\gamma = \langle D \rangle$, $\gamma' = \langle D' \rangle$ be two vertices of $C(S)$. We need to prove that there is a sequence of vertices $\gamma = \gamma_0, \dots, \gamma_n = \gamma'$ such that γ_i is connected by an edge with γ_{i+1} (or is equal to γ_{i+1}) for all $i = 0, \dots, n-1$.

To begin with, let us choose two smooth functions $f, f' : S \rightarrow \mathbf{R}$ such that D (respectively D') is a component of a level set $f^{-1}(a)$ of f (respectively, of f') and, moreover, f and f' have no critical points on D and D' respectively. In addition, we may assume that both f and f' are Morse functions having different critical values. Clearly, we can connect f and f' by a path $\{f_t : S \rightarrow \mathbf{R}\}$, $0 \leq t \leq 1$ in the space of smooth functions, $f = f_0$, $f' = f_1$.

Let $\{g_t\}_{0 \leq t \leq 1}$ be some approximation of $\{f_t\}_{0 \leq t \leq 1}$ provided by Lemma 3.2.A. Since g_0 is C^∞ -close to f_0 , some component C of the level set $g^{-1}(a)$ is a circle close and isotopic to D . In particular, this circle belongs to the isotopy class γ . Similarly, some component C' of some level set of g_1 belongs to the isotopy class γ' . The following claim provides similar components of level sets for all functions g_t .

Claim. If a function $g : S \rightarrow \mathbf{R}$ belongs to one of the three classes of Lemma 3.2.A, then some component of some level set $g^{-1}(r)$ of g contains no critical points and is a *non-trivial* circle.

In order to prove this claim, let us choose a critical point x of g which is not a local maximum or minimum (such a critical value exists, because S is not a sphere or a disc). Let $c = g(x)$ and let L be the component of the level set $g^{-1}(c)$ containing x . Choose an $\varepsilon > 0$ such that c is the only critical value in $[c - \varepsilon, c + \varepsilon]$. Then exactly one component of the set $g^{-1}([c - \varepsilon, c + \varepsilon])$ contains L . Let us denote this component by L_ε . If one of the boundary components of L_ε is non-trivial, then we are done. Suppose that all of them are trivial, i.e. bound in S either a disc, or an annulus together with a boundary component of S .

If one of these discs or annuli contains L_ε , then we replace g by a function g' equal to g outside this disc (or annulus) X and having only one Morse critical point on it if it is a disc (or no critical points if it is an annulus). We may assume that if g' has a critical point in X (i.e., if X is a disc), then the corresponding critical value is different from all other critical values of g' . Note that if X is a disc, then g has at least two critical points on X (the point x and a minimum

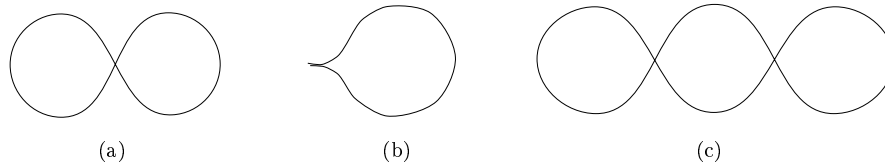


Fig. 3.1

or a maximum), and g' has only one. If X is an annulus, then g has at least one critical point on X (namely, the point x), and g' has none. In both cases g' has fewer critical points than g . If some component of some level set of g' is non-trivial, then the same is true for g itself, because all components of level sets of g' contained in X are trivial. So, in this case our problem is reduced to a similar problem for the function g' , which has fewer critical points than g . Since g' obviously belongs to one of three classes of Lemma 3.2.A, we can use induction.

If none of these discs or annuli contain L_ε , then S is equal to the union of L_ε with these discs or annuli. It follows that the Euler characteristic $\chi(S)$ is equal to $\chi(L_\varepsilon) + d$, where $d \geq 0$ is the number of the discs. Clearly, L is a deformation retract of L_ε and, hence, $\chi(S) = \chi(L) + d$. But, $\chi(L) = -1$ if L contains one Morse critical point, $\chi(L) = -2$ if L contains two Morse critical points and $\chi(L) = 0$ if L contains a non-Morse critical point (in the last case L does not contain any other critical points and is a topological circle with a cusp at the critical point). The various possibilities for L are presented in Fig. 3.1. It follows that $\chi(S) \geq -2$ in this case. But any such S is either excluded by the assumptions of the theorem, or is a torus with two holes, which is also excluded now. This completes the proof of the claim.

Now we return to the proof of the theorem. The above claim implies that for any t , $0 \leq t \leq 1$ the function g_t has a component C_t of a level set $g_t^{-1}(a_t)$ such that C_t does not contain any critical point of g_t and is a non-trivial circle. We may assume that $C_0 = C$ and $C_1 = C'$. If u is sufficiently close to t , say u belongs to a neighborhood U_t of t , then the level set $g_u^{-1}(a_t)$ of the function g_u has a component C_{tu} close and isotopic to C_t , because g_t has no critical points on C_t . The family of neighborhoods $\{U_t\}_{0 \leq t \leq 1}$ forms an open cover of the interval $[0, 1]$. By using a well known Lebesgue lemma we can divide the interval $[0, 1]$ by several points $0 = x_0 \leq x_1 \leq \dots \leq x_n = 1$ into intervals $[x_i, x_{i+1}]$ such that any interval $[x_i, x_{i+1}]$, $0 \leq i \leq n-1$ is contained in a neighborhood U_{t_i} . We may assume that $t_0 = 0$ and $t_{n-1} = 1$.

Let $\gamma_i = \langle C_{t_i} \rangle$. We claim that $\gamma_0, \dots, \gamma_n$ is the required sequence of vertices. By our choice, $\gamma_0 = \gamma$ and $\gamma_n = \gamma'$. Let us prove that γ_i is connected by an edge with γ_{i+1} for $i = 0, \dots, n-1$. Let $0 \leq i \leq n-1$, and let $x = x_i$,

$y = x_{i+1}$, $z = x_{i+2}$ and $v = t_i$, $w = t_{i+1}$. Note that C_v is isotopic to C_{vy} because $y \in [x, y] \subset U_v$ and C_w is isotopic to C_{wy} because $y \in [y, z] \subset U_w$. Hence, $\gamma_i = \langle C_{t_i} \rangle = \langle C_v \rangle = \langle C_{vy} \rangle$ and $\gamma_{i+1} = \langle C_{t_{i+1}} \rangle = \langle C_w \rangle = \langle C_{wy} \rangle$. But, C_{vy} and C_{wy} are components of the level sets of the *same* function g_y . Hence, they are either equal or disjoint and their isotopy classes γ_i and γ_{i+1} are either equal or connected by an edge. This completes the proof. $\square$

This theorem for closed surfaces S was originally deduced by the author [I3] from the results of A. Hatcher and W. Thurston [HatT], which were also based on the ideas of the Morse-Cerf theory. Later on, the method was streamlined and extended to cover the higher connectivity of $C(S)$; cf. the discussion below. In the meantime, J. Harer [Har3], [Har4] proved that $C(S)$ is homotopy equivalent to a bouquet of spheres, all having the same dimension, and computed this dimension, giving, in particular, the best possible estimate of the higher-connectedness. Cf. 3.3.

In [Ce2] J. Cerf also proved a version of Lemma 3.2.A for 2-parameter families of functions. Here, some new types of functions appear, such as Morse functions with three equal critical values or functions having critical points of the form $\pm x^4 \pm y^2 + c$. Using this version of Lemma 3.2.A, one can modify the proof of Theorem 3.2.B and prove the simply connectivity of $C(S)$ in most cases. Instead of a subdivision of the interval $[0, 1]$ into subintervals, one should use a sufficiently fine triangulation of a disc.

The Morse-Cerf theory is based on a complete classification of possible singularities of functions in *generic* families. Such a classification is unknown for families with sufficiently many parameters. But, in fact, one can carry out the main arguments of the proof of Theorem 3.2.B without complete classification. One needs to prove only some restrictions on the possible complexity of singularities of functions in generic families (cf. [I5], Lemmas 2.3 and 2.4). This program is carried out in [I5], Sections 1 and 2 and leads to the following result.

Theorem 3.2.C *If S is a closed surface (respectively, a surface 1 boundary component, a surface with ≥ 2 boundary components), then the geometric realization of $C(S)$ is $-\chi(S) - 1 = 2g - 3$ -connected (respectively, $-\chi(S) - 2$ -connected, $-\chi(S) - 3$ -connected), where $\chi(S)$ is the Euler characteristic of S .*

This result is not the best possible for surfaces with ≥ 2 boundary components (the best is the $-\chi(S) - 2$ -connectedness), but the closed case is sufficient for the main applications (cf., for example, 6.4), and, in fact, the general case can be reduced to it.

3.3 The homotopy type of the complexes of curves

The following theorem gives an almost complete description of the homotopy type of $C(S)$.

Theorem 3.3.A *The geometric realization of $C(S)$ is homotopy equivalent to a bouquet of spheres of dimension $-\chi(S)$ if S is closed and of dimension $-\chi(S)-1$ if S has non empty boundary, where $\chi(S)$ is the Euler characteristic of S .*

This theorem is due to J. Harer [Har4]. His proof is based on a connection of the complexes of curves with ideal triangulations of Teichmüller spaces (cf. 5.5) and a combinatorial argument used to give an upper estimate of the *homotopy dimension* of $C(S)$, i.e., of the minimal dimension of a CW-complex homotopy equivalent to it. This upper estimate, in fact, matches the connectivity results, and together they imply that $C(S)$ is homotopy equivalent to a bouquet of spheres. Another proof, based on Theorem 3.2.C, a duality between the properties of $C(S)$ and Mod_S (cf. 6.1 and 5.4) and Harer's combinatorial argument giving an upper estimate of homotopy dimension was suggested by the author [I5]. We refer to [I5] for a detailed presentation of this proof and a discussion of its relations with Harer's approach; cf. [I5], Section 6.

Both approaches use an induction on the number of boundary components reducing, in fact, the general case to the closed one. Harer's inductive argument [Har4] is made on the geometric level of complexes of curves. He shows that making a hole in a surface with non empty boundary increases the connectivity of $C(S)$ by 1 (making the first hole is more complicated). In [I5], this inductive argument is made on the algebraic level of the mapping class groups, where the exact sequences of 2.8 together with theory of groups with duality [BieE] provide a convenient tool. An improved version of this argument is used in the proof of Theorem 6.4.B.

The following table, summarizing the dependence of the homotopy dimension $\text{hdim } C(S)$ and the usual dimension $\dim C(S)$ of $C(S)$ on the genus g and the number b of boundary components of S , is often quite useful.

g	b	$\text{hdim } C(S)$	$\dim C(S)$
$g = 0$	$b \leq 3$	-1	-1
$g = 0$	$b \geq 3$	$b - 4$	$b - 4$
$g = 1$	$b = 0$	0	0
$g = 1$	$b \geq 1$	$b - 1$	$b - 1$
$g \geq 2$	$b = 0$	$2g - 2$	$3g - 4$
$g \geq 2$	$b \geq 1$	$2g - 3 + b$	$3g - 4 + b$

Corollary 3.3.B *Suppose that S is connected and has non empty boundary. Then the structure of the simplicial complex $C(S)$ (considered up to isomorphism) determines if the genus of S is ≤ 1 , and the whole topological type of S if it turns out that the genus of S is ≥ 2 .*

Proof. The table shows that $\text{hdim } C(S) = \dim C(S)$ if and only if the genus is ≤ 1 or the genus is 2 and the boundary is empty. The last line of the table allows us to find g and b if we know already that $g \geq 2$ and $b \geq 1$. $\square$

3.4 Hyperbolic properties

Now we turn from the homotopy properties to the geometry of complexes of curves. Let $C_1(S)$ be the 1-skeleton of $C(S)$, i.e. the graph with the same vertices and edges as $C(S)$. As we noted in 3.1, $C(S)$ is a flag complex and, hence, $C_1(S)$ completely determines the whole structure of $C(S)$.

Any graph or, rather, its geometric realization, has a canonical structure of a metric space, which is constructed as follows: any edge is made isometric to the interval $[0, 1]$ and then the distance between two points is defined to be the infimum of the lengths of paths connecting them. Clearly, the distance between two vertices γ, γ' is equal to the length n of the shortest chain $\gamma = \gamma_0, \dots, \gamma_n = \gamma'$ such that γ_i is connected by an edge with γ_{i+1} (compare the proof of Theorem 3.2.B). This structure of a metric space turns any graph into a *geodesic space*. This means that any two points x, y can be connected by an isometric image of an interval in the real line $\mathbf{R}$. Any such image is called a *geodesic* and is denoted by $[xy]$, despite the fact that it is usually not unique.

A geodesic metric space X is called δ -*hyperbolic*, where $\delta \geq 0$ is some real number, if for any three points $x, y, z \in X$ any geodesic $[xz]$ is contained in a δ -neighborhood of the union $[xy] \cup [yz]$. A geodesic metric space is called *hyperbolic* if it is δ -hyperbolic for some $\delta \geq 0$. These notions are due to M. Gromov [Gro] and serve as the starting point of his theory of hyperbolic spaces and groups. The main examples are provided by the classical hyperbolic spaces, (infinite) trees (i.e., graphs without cycles), and Cayley graphs of fundamental groups of closed negatively curved manifolds. (For trees one can take $\delta = 0$.) Note also that any geodesic metric space of finite diameter D is trivially δ -hyperbolic with $\delta = D$.

Theorem 3.4.A *If $C_1(S)$ is connected, then its geometric realization is a hyperbolic space of infinite diameter.*

This remarkable and completely unexpected theorem was recently proved by H. Masur and Y. Minsky [MasM1]. An overview of the proof is provided by Y. Minsky [Min1]. The replacement of $C(S)$ by $C_1(S)$ is not significant: we can turn the geometric realization of $C(S)$ into a metric space by making every simplex be isometric to a regular simplex with edges of length 1. Then the resulting metric space will also be hyperbolic. This easily follows from Theorem 3.4.A and basic properties of hyperbolic spaces. It simply seems that it is more convenient to work with $C_1(S)$ in this context.

Even the fact that $C_1(S)$ is of infinite diameter is non-trivial and interesting. It means that for any natural number N there are two vertices of $C(S)$ which cannot be connected by a chain of edges shorter than N . This seems to be nearly obvious, but, apparently, was not proved before [MasM1].

The fact that $C(S)$ is not locally finite presents a serious obstacle if one attempts to use Theorem 3.4.A along the lines of M. Gromov's theory of hyperbolic groups and spaces [Gro]. In [MasM2], H. Masur and Y. Minsky developed

some tools to overcome this obstacle. As a first application to the mapping class groups, they proved the following theorem about pseudo-Anosov elements (for the definition, see 7.1) of the mapping class groups.

Theorem 3.4.B *Let us fix a finite set of generators of Mod_S and let us denote by $|\cdot|_W$ the minimal word length with respect to these generators. If f_1, f_2 are two conjugate pseudo-Anosov elements of Mod_S , then $f_1 = gf_2g^{-1}$ for an element $g \in \text{Mod}_S$ such that*

$$|g|_W \leq C(|f_1|_W + |f_2|_W),$$

where the constant C depends only on S and the generating set.

To put this theorem into proper perspective, one should note that G. Hemion [He] proved that the conjugacy problem for the mapping class group is (algorithmically) solvable. Pseudo-Anosov elements are, in a definite sense, typical elements of Mod_S , and L. Mosher [Mos1] gave an explicit algorithm for determining the conjugacy for pseudo-Anosov elements. In both [He] and [Mos1] no explicit bounds on the time required by the algorithm was given. Theorem 3.4.B obviously implies an explicit bound for the conjugacy problem, because it provides a bound on the word length of a conjugating element.

4 Dehn twists, generators, and relations

The main examples of nontrivial elements of mapping class groups, and also the main building blocks in various constructions of other elements, are the so-called Dehn twists. They were introduced for the first time by M. Dehn [D1], and then played a central role in his fundamental paper [D2]. This section starts with a definition and some basic properties of Dehn twists, and then proceeds to one of their main applications, namely, to their role as generators of mapping class groups. The material of this section is crucial for developing an intuitive feeling of the mapping class groups.

In this section S is always assumed to be a compact *oriented* surface.

4.1 Dehn twists

The goal of this section is to define Dehn twists and to prove their basic properties. Dehn twists are, by definition, the isotopy classes of some special diffeomorphisms called twist diffeomorphisms. We begin with a discussion of the latter.

We start with a description of a standard twist diffeomorphism of an annulus. Let A be the annulus in $\mathbf{R}^2$ given by the inequality $1 \leq r \leq 2$ in the standard polar coordinates (r, θ) in $\mathbf{R}^2$. Its boundary ∂A consists of two components $\partial_1 A$, $\partial_2 A$ defined by the equations $r = 1$, $r = 2$ respectively. Let us fix a

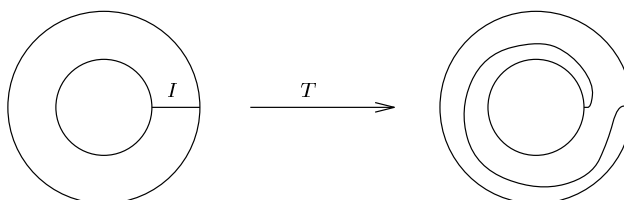


Fig. 4.1

smooth function $\varphi : \mathbf{R} \rightarrow \mathbf{R}$ such that $\varphi(x) = 0$ for $x \leq 1$, $\varphi(x) = 2\pi$ for $x \geq 2$, $0 \leq \varphi \leq 2\pi$, and the derivative φ' is ≥ 0 . Let us define a diffeomorphism $T : A \rightarrow A$ by the formula $T(r, \theta) = (r, \theta + \varphi(r))$ in our polar coordinates (r, θ) . We call T the *standard twist diffeomorphism* of the annulus A . It is easy to see that, up to an isotopy fixed on the boundary ∂A , the diffeomorphism T does not depend on the choice of φ . Fig. 4.1 illustrates the action of T . In fact, up to an isotopy fixed on the boundary, T is the unique diffeomorphism fixed on the boundary and taking the radial arc I in Fig. 4.1(a) into an arc isotopic to the arc in Fig. 4.1(b). (Using the fact that $\text{Diff}(I \text{ fix } \partial I)$ is connected, any two such diffeomorphisms can be made equal on I by an isotopy. By an additional small isotopy, these two diffeomorphisms can be made equal on a neighborhood N of $I \cup \partial A$. Consider a disc D contained in A with the boundary contained in N . Since $\text{Diff}(D \text{ fix } \partial)$ is connected by Theorem 2.7.C, any two such diffeomorphisms can be made equal by an isotopy.) So, Fig. 4.1 defines T up to an isotopy fixed on ∂A .

Now, let $e : A \rightarrow S$ be an *orientation-preserving* embedding of A into our surface S . We may use e to transplant T from A to S as follows: take the diffeomorphism $e \circ T \circ e^{-1} : e(A) \rightarrow e(A)$ and extend it by the identity to a diffeomorphism $T_e : S \rightarrow S$. We call any such diffeomorphism T_e a *twist diffeomorphism* of S . Clearly, up to isotopy fixed on ∂S , the diffeomorphism T_e depends only on the isotopy class of the embedding e . The isotopy class of an embedding $e : A \rightarrow S$ is determined by the isotopy class of the (oriented) image $e(a)$ of the *oriented* (say, counterclockwise) axis $a = \{(r, \theta) : r = 3/2\}$ of the annulus A and by which boundary component of A is mapped to the right side of $e(a)$ (the notion of the *right side* is determined by the orientations of $e(a)$ and S). So, given the isotopy class of the *unoriented* image $e(a)$, there are four possible isotopy classes of e , but only two of them are orientation-preserving. In fact, if e is orientation-preserving, then e has to map $\partial_2 A$ to the right of $e(a)$. It is easy to see (draw a picture!) that both orientation-preserving embeddings e with the same isotopy class of $e(a)$ lead to isotopic diffeomorphisms T_e .

In particular, up to isotopy fixed on ∂S , the diffeomorphism T_e depends only on the image $C = e(a)$ of the axis a of A , and even only on its isotopy class $\gamma = \langle C \rangle$. By a slight abuse of the language, we call this diffeomorphism

a twist diffeomorphism *about* C . Clearly, any twist diffeomorphism preserves orientation and is fixed on the boundary of S . We call the isotopy class of a twist diffeomorphism about a circle C the *Dehn twist about* C . Since this isotopy class depends only on the isotopy class $\gamma = \langle C \rangle$ of C , we usually denote it by t_γ and call it also the *Dehn twist about* γ . There is an ambiguity in these notations and terminology; namely, one may consider only isotopies fixed on the boundary and then $t_\gamma \in \mathcal{M}_S$, or one may consider all isotopies and then $t_\gamma \in \text{PMod}_S$. The context usually resolves this ambiguity.

The Dehn twists we defined are often called the *left* Dehn twists, and the elements of the form t_γ^{-1} are called the *right* Dehn twists. Similarly, one may speak about the *left* and the *right* twist diffeomorphisms. This terminology is justified, first, by the fact that the right twist diffeomorphisms and Dehn twists can be constructed in a manner completely similar to the construction of the left ones, but using orientation-reversing embeddings $A \rightarrow S$ instead of orientation-preserving ones (we leave this as an exercise). Second, one can distinguish between the right and the left twist diffeomorphisms in the following way. Let $e : A \rightarrow S$ be an embedding and T_e be the corresponding (left or right) twist diffeomorphism. If J is an arc in S transversely intersecting $C_1 = e(\partial_1 A)$, then as we approach the annulus $e(A)$ along $T_e(J)$ and pass the circle C_1 , we will be turning left if T_e is a left twist diffeomorphism and we will be turning right if T_e is a right twist diffeomorphism. Of course, the meaning of turning left or right is derived from the orientation of S . A useful exercise is to convince yourself that we will turn to the same direction if we approach $e(A)$ from the other side crossing $C_2 = e(\partial_2 A)$. Curiously enough, M. Dehn himself [D1], [D2] did not distinguished between left and right twists in the notations (or otherwise), but always described explicitly which twist he deals with.

Let us consider now twist diffeomorphisms about trivial circles. If $e(a)$ bounds a disc D in S , we may assume that D contains $\partial_1 A$ (and not $\partial_2 A$), replacing, if necessary, e by $e \circ i$, where $i : A \rightarrow A$ is defined by the formula $i(r, \theta) = (3 - r, 2\pi - \theta)$ (clearly, $e(a) = e \circ i(a)$, but e and $e \circ i$ belong to different isotopy classes). Then we can extend e to an embedding $e' : D_2 \rightarrow S$ of the disc $D_2 = \{(r, \theta) : r \leq 2\}$ such that $e'(D_{3/2}) = D$, where $D_{3/2} = \{(r, \theta) : r \leq 3/2\}$. We can use e' to transplant the isotopy $\{T'_u : D_2 \rightarrow D_2\}_{0 \leq u \leq 1}$ of D_2 defined by the formula $T'_u(r, \theta) = (r, \theta + \varphi(r) + u(2\pi - \varphi(r)))$ from D_2 to S in the same manner as we transplanted T from A to S . Clearly, the isotopy $\{T'_u\}_{0 \leq u \leq 1}$ is fixed on the boundary $\partial D_2 = \partial_2 A$ and connects the extension T'_0 of T to D_2 by the identity with $T'_1 = \text{id}_{D_2}$ (note that $(r, \theta) = (r, \theta + 2\pi)$). Hence, our transplanted isotopy connects T_e with the identity diffeomorphism of S and is fixed on ∂S . Therefore, any twist diffeomorphism about a circle bounding a disc in S is isotopic to the identity by an isotopy fixed on ∂S . A similar argument shows that any twist diffeomorphism about a circle C bounding an annulus together with a boundary component of S (or equal to a such component) is isotopic to the identity; this time by an isotopy moving this boundary component.

It follows that, in PMod_S , any Dehn twist about a trivial circle (i.e. a circle

bounding a disc or homotopic to a boundary component) is trivial (i.e. equal to 1). But in $\mathcal{M}_S$ only the Dehn twists about circles bounding a disc are trivial and the Dehn twists about circles homotopic to a boundary component are nontrivial (of course, the last statement requires a proof).

Now, we prove some basic properties of Dehn twists. We start with a refinement of the second statement of Corollary 2.7.E

Lemma 4.1.A *The group $\pi_0(\text{Diff}(A \text{ fix } \partial A))$ is an infinite cyclic group generated by the isotopy class of the standard twist diffeomorphism T .*

Proof. Consider the fibration $\text{Diff}(D_2 \text{ fix } \partial) \rightarrow \text{Emb}(D_1, D_2)$ from Theorem 2.6.A (recall that $D_i = \{(r, \theta) : r \leq i\}$; $i = 1, 2$). Since $\text{Diff}(D_2 \text{ fix } \partial)$ is contractible (by Theorem 2.7.C) and since $\text{Emb}(D_1, D_2)$ is weakly homotopy equivalent to the unit tangent bundle of D^2 (by Theorem 2.6.C) and hence to a circle, the homotopy sequence of our fibration implies that $\pi_0(\text{Diff}(A \text{ fix } \partial A))$ is isomorphic to $\mathbf{Z}$. Obviously, $\pi_1(\text{Emb}(D_1, D_2))$ is generated by the homotopy class of the loop uniformly rotating D_1 once around its center. To compute the image of this generator in $\pi_0(\text{Diff}(A \text{ fix } \partial A))$ we need to extend this rotation to an isotopy of D_2 fixed on ∂D_2 . This was done, in fact, in our discussion of twist diffeomorphisms about trivial circles above. The extended isotopy ends in a diffeomorphism equal to the identity on D_1 and to the standard twist diffeomorphism on A . The lemma follows. $\square$

Corollary 4.1.B *Let C be circle on S , and let $H, H' : S \rightarrow S$ be two orientation-preserving diffeomorphisms leaving C invariant and preserving the orientation of C . If the results of cutting H and H' along C are isotopic (with free boundary), then, up to isotopy, H differs from H' by a power of a twist diffeomorphism about C .*

Proof. Changing H and H' , if necessary, by an isotopy, we may assume that both H and H' are equal to the identity on C (because $\text{Diff}(C)$ is connected). Obviously, both H and H' preserve the sides of C , and hence H and H' may be assumed to be equal to the identity on some annulus X having C as one of the boundary components. If we remove the interior of X from S , we get a surface S_X diffeomorphic to the result S_C of cutting S along C . Obviously, the diffeomorphisms $H_X, H'_X : S_X \rightarrow S_X$ induced by H, H' are isotopic (after a natural identification of S_X with S_C) to the results of cutting H, H' along C . Hence, H_X, H'_X are isotopic. By extending this isotopy to S , we get an isotopy between H and some diffeomorphism H'' which is equal to H' outside X and hence is equal to $H' \circ G$, where G is some diffeomorphism supported* in X . Since X is an annulus, an application of the lemma completes the proof. $\square$

*The support of a diffeomorphism f is the closure of the set $\{x : f(x) \neq x\}$.

Lemma 4.1.C *If $f \in \text{Mod}_S$ and γ is the isotopy class of a circle on S , then*

$$ft_\gamma f^{-1} = t_{f(\gamma)}.$$

Proof. If $e : A \rightarrow S$ is an embedding and $F : S \rightarrow S$ is a diffeomorphism, then, clearly, $F \circ T_e \circ F^{-1} = T_{F \circ e}$. The lemma follows. $\square$

Corollary 4.1.D *Any two Dehn twists about nonseparating circles on S are conjugate.*

Proof. It is sufficient to apply Lemma 2.3.A. $\square$

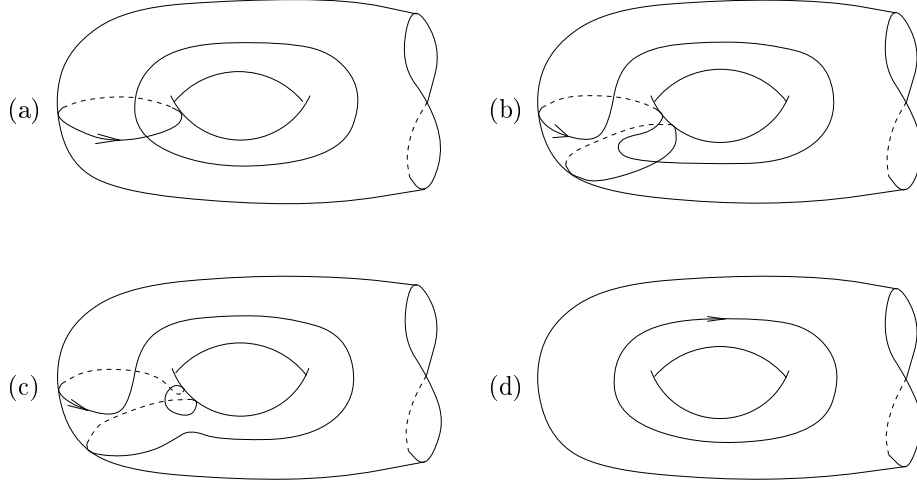


Fig. 4.2

Lemma 4.1.E *Let C and D be two circles on S transversely intersecting at one point and let γ, δ , respectively, be their isotopy classes.*

- (i) $t_\gamma t_\delta(\gamma) = \delta$.
- (ii) $t_\delta t_\gamma^2 t_\delta(\gamma) = \gamma$, and $t_\delta t_\gamma^2 t_\delta$ can be represented by a diffeomorphism taking C to C and reversing the orientation of C (or, what is the same, interchanging the sides of C).

Proof. First, notice that for any two pairs of circles transversely intersecting at one point there is a diffeomorphism of S taking one pair to the other. So, it is sufficient to prove the lemma for one such pair; for example, for the pair C, D in Fig. 4.2(a) (the orientations of circles play a role only in the proof of (ii)).

- (i) The image E of C under a twist diffeomorphism about D is illustrated in Fig. 4.2(b), together with a circle isotopic to C . The image of E under a

twist diffeomorphism about this last circle is illustrated in Fig. 4.2(c). Clearly, the circle in Fig. 4.2(c) is isotopic to the circle in Fig. 4.2(d), which is nothing other than D . This proves (i).

(ii) In order to prove that $t_\delta t_\gamma^2 t_\delta(\gamma) = \gamma$, it is sufficient to apply (i) twice. In order to prove the remaining part of (ii), one needs to trace the orientations of our circles. The image of C oriented as in Fig. 4.2 (a) under the composition of two twist diffeomorphisms from the proof of (i) is illustrated in Fig. 4.2(d). This image is isotopic to D with the orientation given by turning counterclockwise the orientation of C at the (unique) intersection point of C and D . If we apply these two diffeomorphisms once more, but in the other order (so we get an oriented representative of $t_\delta t_\gamma^2 t_\delta(\gamma)$), we get C with the orientation given by turning the last orientation of D again counterclockwise. Clearly, this orientation of C is opposite to the original one. This proves (ii). $\square$

I learned Lemma 4.1.E from J. Birman's survey [Bir4], but the assertion (i) is definitely due to M. Dehn [D2] (cf. [D2], §4 a)), as are also Lemma 4.1.C and Corollary 4.1.D. Now we prove several fundamental relations between Dehn twists.

Lemma 4.1.F *Let C and D be two circles on S and let γ, δ , respectively, be their isotopy classes.*

(i) *If $C \cap D = \emptyset$, then $t_\gamma t_\delta = t_\delta t_\gamma$.*

(ii) *If C and D intersect transversely at exactly one point, then the following Artin relation holds:*

$$t_\gamma t_\delta t_\gamma = t_\delta t_\gamma t_\delta.$$

Proof. (i) If $C \cap D = \emptyset$, then, obviously, t_γ and t_δ can be represented by twist diffeomorphisms with disjoint supports. Such diffeomorphisms obviously commute.

(ii) $t_\gamma t_\delta t_\gamma = t_\delta t_\gamma t_\delta$ is obviously equivalent to $t_\delta t_\gamma t_\delta^{-1} = t_\gamma^{-1} t_\delta t_\gamma$. By Lemma 4.1.C, the latter equation is equivalent to

$$t_{t_\delta(\gamma)} = t_{t_\gamma^{-1}(\delta)}.$$

Hence, it is sufficient to prove that $t_\delta(\gamma) = t_\gamma^{-1}(\delta)$, or, what is the same, $t_\gamma t_\delta(\gamma) = \delta$. But this is the content of Lemma 4.1.E(ii). $\square$

Lemma 4.1.G *Let C and D be two circles on S transversely intersecting at one point and let γ, δ , respectively, be their isotopy classes. Let N be a regular neighborhood of the union $C \cup D$ (this means that N cut along $C \cup D$ is an annulus with corners) and let $B = \partial N$, $\beta = \langle B \rangle$. Then*

$$(t_\gamma t_\delta)^6 = t_\beta.$$

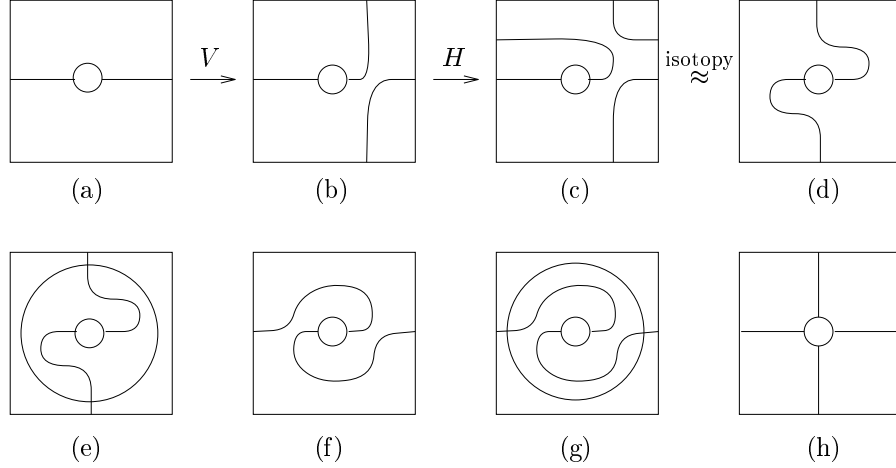


Fig. 4.3

Proof. Instead of N we may consider the standard torus with one hole resulting from the identification of the opposite sides of a square having a small hole around its center. As C and D we may take circles represented by a vertical and a horizontal segment in this square, respectively. Let V and H be the corresponding twist diffeomorphisms.

Consider the horizontal arc h in Fig. 4.3(a). Fig. 4.3(a)-(d) compute $H \circ V(h)$ (up to isotopy). If we apply V to the arc in Fig. 4.3(d), we get the same arc. So, the arc in Fig. 4.3(d) is isotopic to $V \circ H \circ V(h)$. Outside the big circle in Fig. 4.3(e) the arc in Fig. 4.3(d) looks exactly as the arc in Fig. 4.3(a) turned 90° . Such a turn preserves orientation and hence interchanges V and H (more precisely, conjugates one to the other). So, by turning Fig. 4.3(a)-(d), we can find the image of the arc in Fig. 4.3(d) under $H \circ V \circ H$, i.e. we can find $(H \circ V)^3(h) = (H \circ V \circ H) \circ (V \circ H \circ V)(h)$. The result is the arc in Fig. 4.3(f). This arc coincides with h outside the big circle in Fig. 4.3(g). This allows us to find the image of the arc in Fig. 4.3(f) under $(H \circ V)^3$ or, what is the same, $(H \circ V)^6(h)$. Clearly, the result is isotopic to the image of h under a twist diffeomorphism about the boundary (the small circle in our pictures).

Next, we would like to find $(H \circ V)^6(v)$, where v is a vertical arc similar to h ; cf. Fig. 4.3(h). Using a 90° turn as above, we see that it is sufficient to find $(V \circ H)^6(h)$. Note that the first H in $(V \circ H)^6$ acts on h trivially, so $(V \circ H)^6(h) = (V \circ H)^5 \circ V(h) = V \circ (H \circ V)^5(h)$. But $V \circ (H \circ V)^5(h) = (H \circ V)^6(h)$, because in the computation of $(H \circ V)^6(h)$ the last H acts trivially (like the last V in the computation of $V \circ H \circ V(h)$). Hence, $(V \circ H)^6(h) = (H \circ V)^6(h)$, and hence $(V \circ H)^6(h)$ is equal to the image of h under a twist diffeomorphism about the

boundary. It follows that the image $(H \circ V)^6(v)$ is equal to the image of v under a twist diffeomorphism about the boundary.

It remains to notice that if we cut our torus with a hole along h and v , we get a disc (with corners), and hence a diffeomorphism of our torus (fixed on the boundary) is determined, up to an isotopy fixed on the boundary, by its action on h and v (recall that $\text{Diff}(D \text{ fix } \partial D)$ is connected for any disc D). $\square$

This lemma is also due (in a slightly different form) to M. Dehn [D2] (cf. [D2], §6 c)).

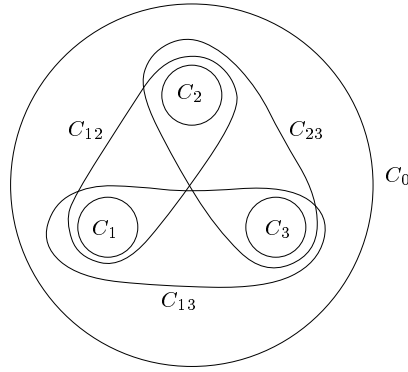


Fig. 4.4

Lemma 4.1.H *Let S_0^4 be a sphere with four holes. Let the boundary components of S_0^4 be $C_0, \dots, C_3$ and for $1 \leq i < j \leq 3$, let C_{ij} denote a circle encircling C_i and C_j as in Fig. 4.4. Suppose that S_0^4 is embedded in S . Let $t_i \in \mathcal{M}_S$ be the Dehn twist about C_i , $0 \leq i \leq 3$, and let $t_{ij} \in \mathcal{M}_S$ be the Dehn twist about C_{ij} , $1 \leq i < j \leq 3$. Then the following lantern relation holds in $\mathcal{M}_S$ (and hence in Mod_S):*

$$t_0 t_1 t_2 t_3 = t_{12} t_{13} t_{23}.$$

Proof. Connect C_0 with C_1, C_2, C_3 by three disjoint arcs I_1, I_2, I_3 . Clearly, if we cut S_0^4 along these three arcs, we get a disc (with corners). Hence (as in the proof of Lemma 4.1.G), a diffeomorphism of S_0^4 fixed on ∂S_0^4 is determined up to an isotopy fixed on ∂S_0^4 by its action on I_1, I_2, I_3 . Therefore, in order to prove our relation it is sufficient to compute the action of some representatives of both sides on the arcs I_1, I_2, I_3 . Of course, we will take compositions of twist diffeomorphisms as our representatives, and then their action on the arcs can be computed in a straightforward manner, as in the proof of Lemma 4.1.E. One just needs to draw several pictures, and we leave this task to the reader. $\square$

The relation of Lemma 4.1.H was discovered by M. Dehn [D2] (cf. [D2] §7 g) 1)) and much later rediscovered and popularized under the name *lantern relation* by D. Johnson [J1].

Finally, we will prove a lemma borrowed from [Bir4] (cf. [Bir4], the discussion after Corollary 3.3). It is crucial for the induction step in some arguments using the induction on the number of boundary components. Cf. Section 4.2.

Lemma 4.1.I *Suppose that $\partial S \neq \emptyset$. Let R be the result of gluing a disc D to one of the boundary components of S . Let $\beta \in \pi_1(R, x)$ be an element represented by an embedded loop b . Without any loss of generality we may assume that the base point x is contained in the interior of D and the loop b intersects D in a single arc containing x . Then we can choose an annulus B containing D and b in its interior and having boundary components isotopic to b . Let C_1 be the component of ∂B lying at the left side of b , and let C_2 be the component of ∂B lying at the right side of b , where the meaning of left and right is determined by the direction of the loop b and the orientation of S . Then*

$$j(\beta) = t_{\gamma_1} t_{\gamma_2}^{-1},$$

where $j : \pi_1(R, x) \rightarrow \text{PMod}_S$ is the homomorphism from 2.8, and $\gamma_1 = \langle C_1 \rangle$, $\gamma_2 = \langle C_2 \rangle$.

Proof. $j(\beta)$ can be obtained in the following manner (cf. 2.8). Consider the loop b as an isotopy of the base point x and extend this isotopy to an isotopy $\{F_t : R \rightarrow R\}_{0 \leq t \leq 1}$, $F_0 = \text{id}_R$. We may assume that the ending diffeomorphism F_1 of this isotopy is fixed on D . Then it induces a diffeomorphism $F : S \rightarrow S$, and $j(\beta)$ is equal to the isotopy class of F . Obviously, the isotopy $\{F_t\}_{0 \leq t \leq 1}$ may be chosen to be fixed outside of the annulus B . Moreover, it can be chosen in such a way that F_1 is equal to the composition of a left twist diffeomorphism supported in an annulus (contained in B) with the boundary C_1 and a right twist diffeomorphism supported in an annulus (disjoint from the first one and also contained in B) with the boundary C_2 . One can see this by drawing a couple of simple pictures or, alternatively, by defining $\{F_t\}_{0 \leq t \leq 1}$ by explicit formulas (compare our discussion of twist diffeomorphisms about trivial circles). We leave this as an exercise to the reader (the pictures can be found in [I8], Section (6.1)). Clearly, this description of F_1 implies the lemma. $\square$

4.2 The Dehn-Lickorish theorem

The main result of this section is Theorem 4.2.D, known as the Dehn-Lickorish theorem. It provides an explicit finite set of Dehn twists generating Mod_S (for closed S). We also discuss some corollaries of this result. Let us start with a couple of simple lemmas.

Lemma 4.2.A *The Dehn twists about the longitude and the meridian of a closed torus S generate Mod_S .*

Proof. By the discussion after Theorem 2.9.A, Mod_S is isomorphic to $SL_2(\mathbf{Z})$. This isomorphism, of course, depends on the choice of an isomorphism $\pi_1(S) \rightarrow \mathbf{Z}^2$. If the latter isomorphism takes (the elements represented by) the longitude and the meridian into the standard basis of $\mathbf{Z}^2$, then, as it is easy to see, the Dehn twists about the longitude and the meridian correspond to matrices

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \text{ and } \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

On the other hand, it is well known that these matrices generate $SL_2(\mathbf{Z})$. The lemma follows. $\square$

Lemma 4.2.B *Suppose that S is a closed surface of genus ≥ 2 . Let γ, γ' be two vertices of $C(S)$ represented by nonseparating circles on S . Then there is a sequence $\gamma = \gamma_1, \gamma_2, \dots, \gamma_n = \gamma'$ of vertices of $C(S)$ such that any two consecutive vertices γ_i, γ_{i+1} are connected by an edge in $C(S)$ and, moreover, if they are represented by two disjoint circles C_i, C_{i+1} , then the union $C_i \cup C_{i+1}$ does not separate S .*

Proof. Since $C(S)$ is connected, there is a sequence $\gamma = \delta_1, \delta_2, \dots, \delta_m = \gamma'$ of vertices of $C(S)$ such that any two consecutive vertices δ_i, δ_{i+1} are connected by an edge in $C(S)$. It may happen that some δ_i is the isotopy class of a circle D_i separating S into two parts S_1, S_2 . In this case $\delta_{i-1}, \delta_{i+1}$ can be represented by two circles D_{i-1}, D_{i+1} , respectively, disjoint from D_i . If D_{i-1} and D_{i+1} are contained in different parts of S , then $D_{i-1} \cap D_{i+1} = \emptyset$ and we can delete δ_i from our sequence. Suppose now that both D_{i-1} and D_{i+1} are contained in the same part of S , say in S_1 . Since S is closed and D_i is nontrivial, the genus of S_2 is ≥ 1 and hence S_2 contains some nonseparating circle D'_i . In this case we replace δ_i by $\langle D'_i \rangle$. After repeating this procedure several times, we will get a new sequence $\gamma = \delta_1, \delta_2, \dots, \delta_m = \gamma'$ such that all δ_i are the isotopy classes of nonseparating circles (and any two consecutive vertices δ_i, δ_{i+1} are connected by an edge).

Now, let $1 \leq i \leq m-1$ and let D_i, D_{i+1} be two disjoint circles representing δ_i, δ_{i+1} respectively. Suppose that $D_i \cup D_{i+1}$ separates S into two parts S_1, S_2 . Since S is closed, both these parts have genus ≥ 1 (we assume that $\delta_i \neq \delta_{i+1}$). It follows that S_1 (as also S_2) contains some nonseparating circle D'_i . Clearly, both $D_i \cup D'_i$ and $D'_i \cup D_{i+1}$ do not separate S . Let us replace the part δ_i, δ_{i+1} of our sequence by $\delta_i = \langle D_i \rangle, \delta'_i = \langle D'_i \rangle, \delta_{i+1} = \langle D_{i+1} \rangle$. After repeating this procedure several times, we will get the required sequence. $\square$

Theorem 4.2.C *If S is a compact surface of genus ≥ 1 , then PMod_S is generated by Dehn twists about nonseparating circles.*

Proof. We use the double induction on the genus and the number of boundary components. The induction starts with the closed torus. In this case the theorem follows from Lemma 4.2.A.

Suppose that $\partial S \neq \emptyset$. Let R be the result of gluing a disc to one of the boundary components of S . By the inductive assumption, PMod_R is generated by twists about nonseparating circles. Obviously, every circle C on R is isotopic to a circle C' contained in S , and $t_C = \rho(t_{C'})$, where $\rho : \text{PMod}_S \rightarrow \text{PMod}_R$ is the homomorphism from 2.8. Moreover, if C is nonseparating, then C' is also nonseparating. It follows that in order to prove the theorem for S it is sufficient to prove that $\ker \rho$ is generated by Dehn twists about nonseparating circles. The latter follows from Lemma 4.1.I and the fact that $\pi_1(R)$ is generated by the homotopy classes of (embedded) nonseparating loops. This completes the induction step of the induction on the number of boundary components.

Next, suppose that S is closed. Let $f \in \text{Mod}_S$. Let us choose a nonseparating circle C on S . Consider two vertices $\gamma = \langle C \rangle$, $\gamma' = f(\gamma)$ of $C(S)$ and connect them by a sequence $\gamma = \gamma_1, \gamma_2, \dots, \gamma_m = \gamma'$ as in Lemma 4.2.B. For every $i = 1, 2, \dots, m-1$ we can represent γ_i, γ_{i+1} by disjoint circles C_i, C_{i+1} . Since $C_i \cup C_{i+1}$ does not separate S , we can choose a circle D_i transversely intersecting each of C_i and C_{i+1} at exactly one point. Let $\delta_i = \langle D_i \rangle$. By Lemma 4.1.E(i), $t_{\gamma_{i+1}} t_{\delta_i}(\gamma_{i+1}) = \delta_i$ and $t_{\delta_i} t_{\gamma_i}(\delta_i) = \gamma_i$. Hence, $g(\gamma') = \gamma$ and $g \circ f(\gamma) = \gamma$, where g is the product of elements $t_{\delta_i} t_{\gamma_i} t_{\gamma_{i+1}} t_{\delta_i}$, $i = 1, 2, \dots, m-1$. Since $g \circ f(\gamma) = \gamma$, we can represent $g \circ f$ by a diffeomorphism $H : S \rightarrow S$ preserving C . If H interchanges the sides of C , we replace g by $t_{\delta_1} t_{\gamma_1}^2 t_{\delta_1} g$. Lemma 4.1.E(ii) implies that for the new g the element $g \circ f$ can be represented by a diffeomorphism $H : S \rightarrow S$ preserving C and preserving the sides of C . Now, let us cut S and H along C . We will get a diffeomorphism $H_C : S_C \rightarrow S_C$ preserving both boundary components of S_C . Hence, the isotopy class h_C of H_C belongs to PMod_{S_C} . Since the genus of S_C is less than the genus of S , h_C can be presented, by the inductive assumption, as a product of Dehn twists about nonseparating circles on S_C . Every circle on S_C is at the same time a circle on S , and hence these Dehn twists in PMod_{S_C} naturally define Dehn twists in PMod_S . The product of these Dehn twists in PMod_S can be represented by a diffeomorphism $H' : S \rightarrow S$ preserving C and isotopic to H_C after cutting along C . It follows that H differs from H' (up to isotopy) by some power of a Dehn twist about C (cf. Corollary 4.1.B). Hence, $g \circ f$ is equal to a product of Dehn twists about nonseparating circles. Since g is also such a product, by its construction, we conclude that f is a product of Dehn twists about nonseparating circles. This completes the induction step of the induction on genus, and hence the proof. $\square$

This theorem is due to M. Dehn [D2]. His work was forgotten for a while, and the theorem was rediscovered by W. B. R. Lickorish [Li1]. To a certain extent, our proof follows the proof of J. Birman [Bir4] (cf. [Bir4], Theorem 4.1), who simplified the proof of Lickorish. The use of the connectedness of $C(S)$

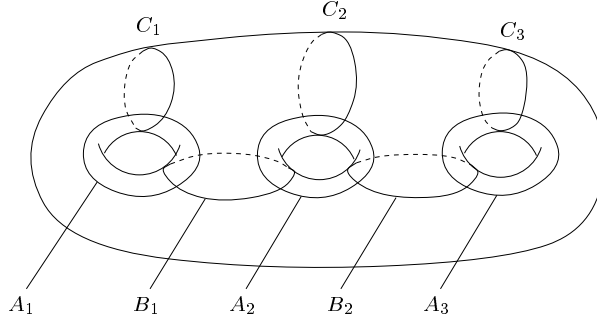


Fig. 4.5

leads to further radical simplifications, and is, in fact, closer in spirit to Dehn's original approach than to the one of Lickorish. The main tool of M. Dehn [D2] was the action of Mod_S on the set of (the isotopy classes of) systems of circles. What was lacking in [D2] (and is crucial to our approach) is an appropriate structure on this set, namely, the structure of a simplicial complex introduced much later by W. Harvey [Harv2].

Now we are ready to prove the Dehn-Lickorish theorem.

Theorem 4.2.D *If S is a closed surface of genus g , then Mod_S is generated by Dehn twists about the $3g - 1$ circles $A_1, \dots, A_g, B_1, \dots, B_{g-1}, C_1, \dots, C_g$ presented in Fig. 4.5.*

Proof. By Lemma 4.2.A the theorem holds for $g = 1$. Using induction on g , we may assume that the theorem holds for all surfaces of genus $\leq g - 1$.

Let us denote by $\alpha_1, \dots, \alpha_g, \beta_1, \dots, \beta_{g-1}, \gamma_1, \dots, \gamma_g$ the isotopy classes of the circles $A_1, \dots, A_g, B_1, \dots, B_{g-1}, C_1, \dots, C_g$ respectively. Let us denote by G_S the subgroup of Mod_S generated by Dehn twists about these $3g - 1$ circles. So, we have to prove that $G_S = \text{Mod}_S$. In view of Theorem 4.2.C it is sufficient to prove that G_S contains all Dehn twists about nonseparating circles. By its definition, the subgroup G_S contains some such twists. Hence, by Lemma 4.1.C it is sufficient to prove that G_S acts transitively on the set of isotopy classes of nonseparating circles. Moreover, since C_1 is nonseparating, it is sufficient to prove that for every nonseparating circle C' there is an element $f \in G_S$ such that $f(\gamma_1) = \gamma'$, where $\gamma' = \langle C' \rangle$.

Now we claim that it is sufficient to prove the last statement only for circles C' such that $C_1 \cap C' = \emptyset$ and $C_1 \cup C'$ does not separate S . In fact, suppose that our statement is proved for such special circles C' and consider an arbitrary nonseparating circle C' . Let us connect γ_1 with $\gamma' = \langle C' \rangle$ by a sequence $\gamma_1 = \gamma_{11}, \gamma_{12}, \dots, \gamma_{1n} = \gamma'$ as in Lemma 4.2.B. By our assumption, there is $g \in G_S$ such that $g(\gamma_1) = g(\gamma_{11}) = \gamma_{12}$. Note that $\gamma_1 = \gamma_{11}$ is connected with $g^{-1}(\gamma')$

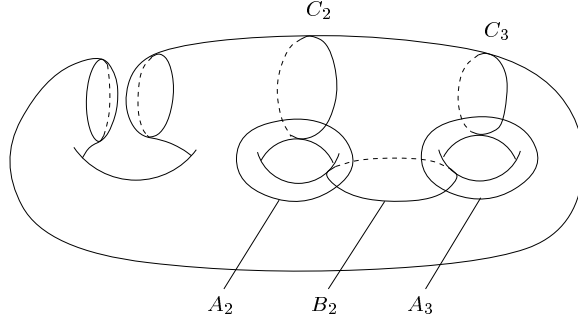


Fig. 4.6

by a shorter sequence $\gamma_1 = \gamma_{11} = g^{-1}(\gamma_{12}), g^{-1}(\gamma_{13}), \dots, g^{-1}(\gamma_{1n}) = g^{-1}(\gamma')$ with the properties of Lemma 4.2.B. Using induction, we may assume that $g^{-1}(\gamma') = h(\gamma_1)$ for some $h \in G_S$. Then $g \circ h(\gamma_1) = \gamma'$ and $g \circ h \in G_S$. This proves our claim.

So, let C' be a circle such that $C_1 \cap C' = \emptyset$ and $C_1 \cup C'$ does not separate S , and let $\gamma' = \langle C' \rangle$. By Lemma 4.1.E(i) we have $t_{\gamma_1} t_{\alpha_1}(\gamma_1) = \alpha_1$, $t_{\alpha_1} t_{\beta_1}(\alpha_1) = \beta_1$, $t_{\beta_1} t_{\alpha_2}(\beta_1) = \alpha_2$, and $t_{\alpha_2} t_{\gamma_2}(\alpha_2) = \gamma_2$. It follows that $g(\gamma_1) = \gamma_2$ for some $g \in G_S$ (namely, for the product of the above eight Dehn twists). Therefore, it is sufficient to prove that $f(\gamma_2) = \gamma'$ for some $f \in G_S$. Note that both C_2 and C' are disjoint from C_1 and both $C_1 \cup C_2$ and $C_1 \cup C'$ do not separate S .

Let R be the result of cutting S along C_1 , and let Q be the result of gluing two discs D_1, D_2 to the two boundary components of R . Clearly, both C_2 and C' are contained in Q (and even in R) and do not separate Q . Consider the subgroup G_Q of Mod_Q generated by Dehn twists about the circles $A_2, \dots, A_g, B_2, \dots, B_{g-1}, C_2, \dots, C_g$ (clearly, all these circles are contained in Q). Cf. Fig. 4.6. Since the genus of Q is equal to $g - 1$, we have $G_Q = \text{Mod}_Q$ by the inductive assumption.

Since $G_Q = \text{Mod}_Q$, we can find a diffeomorphism $F : Q \rightarrow Q$ such that $F(C_2)$ is isotopic to C' in Q and the isotopy class of F belongs to G_Q . Since G_Q is generated by the Dehn twists about the circles $A_2, \dots, A_g, B_2, \dots, B_{g-1}, C_2, \dots, C_g$, we may assume that F is equal to a composition of twist diffeomorphisms supported in some annuli with boundary components isotopic to these circles. Clearly, we may assume these annuli to be disjoint from the discs D_1, D_2 added to R . In particular, we may assume that F is fixed on D_1 and D_2 . Such a diffeomorphism F can be obtained by cutting a (unique) diffeomorphism $F^\sim : S \rightarrow S$ fixed on C_1 along C_1 and gluing in the identity diffeomorphisms $\text{id}_{D_1}, \text{id}_{D_2}$. Obviously, $F^\sim$ is equal to a composition of twist diffeomorphisms supported in the same annuli as F . In particular, its isotopy class $f^\sim$ belongs to G_S .

If $F(C_2)$ is not only isotopic, but equal to C' , then $F^\sim(C_2)$ is also equal to

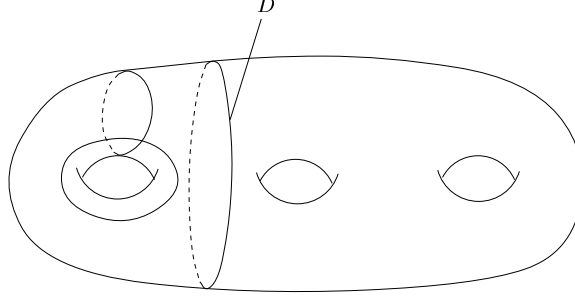


Fig. 4.7

C' , and hence $f^\sim(\gamma_2) = \gamma'$. This completes the proof in the case $F(C_2) = C'$, since $f^\sim \in G_Q$. In the rest of the proof we deal with the complications arising from the fact that in general we only know that $F(C_2)$ is isotopic to C' in Q .

Let $\{K_t : Q \rightarrow Q\}_{0 \leq t \leq 1}$ be an isotopy moving $F(C_2)$ into C' , so that $K_0 = \text{id}_Q$, $K_1(F(C_2)) = C'$. During this isotopy, the discs D_1, D_2 may be moved. Note that D_1 and D_2 are disjoint from $F(C_2)$ (because F is fixed on D_1 and D_2), and hence $K_1(D_1)$ and $K_1(D_2)$ are disjoint from C' . Since C' does not separate Q , we can move the discs $K_1(D_1), K_1(D_2)$ back to their original position D_1, D_2 without moving C' . More precisely, there is an isotopy $\{L_t : Q \rightarrow Q\}_{0 \leq t \leq 1}$ such that $L_0 = \text{id}_Q$, L_t is fixed on C' for all t , and $L_1 \circ K_1|_{D_1} = \text{id}_{D_1}$, $L_1 \circ K_1|_{D_2} = \text{id}_{D_2}$. Let $H = L_1 \circ K_1$. Clearly, $H(F(C_2)) = C'$, H is isotopic to id_Q and $H|_{D_1} = \text{id}_{D_1}$, $H|_{D_2} = \text{id}_{D_2}$. Like F , the diffeomorphism H can be obtained by cutting a (unique) diffeomorphism $H^\sim : S \rightarrow S$ fixed on C_1 along C_1 and gluing in the identity diffeomorphisms $\text{id}_{D_1}, \text{id}_{D_2}$. Let $h^\sim$ be the isotopy class of $H^\sim$.

Since $H \circ F(C_2) = C'$, we have $H^\sim \circ F^\sim(C_2) = C'$ and hence $h^\sim \circ f^\sim(\gamma_2) = \gamma'$. Recall that $f^\sim \in G_S$ and hence $h^\sim \circ f^\sim \in G_S$ if $h^\sim \in G_S$. Therefore, in order to complete the proof it is sufficient to show that $h^\sim \in G_S$. In view of the properties of H stated in the last paragraph, it is sufficient, in fact, to prove the following claim.

Claim. Let $H^\sim : S \rightarrow S$ be a diffeomorphism fixed on C_1 , preserving the sides of C_1 and such that the result H of cutting $H^\sim$ along C_1 and gluing in the identity diffeomorphisms $\text{id}_{D_1}, \text{id}_{D_2}$ is isotopic to id_Q . Then the isotopy class $h^\sim$ of $H^\sim$ belongs to G_S .

Let us now prove this claim. First, by Lemma 4.1.E(ii) some representative $I^\sim$ of the isotopy class $i^\sim = t_{\alpha_1} t_{\gamma_1}^2 t_{\alpha_1}$ maps C_1 to C_1 and interchanges the sides of C_1 . Let $I' : R \rightarrow R$ be the result of cutting $I^\sim$ along C_1 and let $i' \in \text{Mod}_R$ be its isotopy class. Next, by Lemma 4.1.G, $t^\sim = (t_{\gamma_1} t_{\alpha_1})^6$ is equal to t_{δ_0} ,

where δ_0 is the isotopy class of the circle D_0 in Fig. 4.7. In particular, $t^\sim$ can be represented by a twist diffeomorphism $T^\sim$ about D_0 . We may assume that $T^\sim$ is supported in an annulus disjoint from C_1 . Let $T' : R \rightarrow R$ be the result of cutting $T^\sim$ along C_1 and let $t' \in \text{Mod}_R$ be its isotopy class. Let G be the subgroup of Mod_R generated by i' , t' and the Dehn twists about $A_2, \dots, A_g, B_2, \dots, B_{g-1}, C_2, \dots, C_g$.

Let $H_C^\sim : R \rightarrow R$ be the result of cutting H along C (recall that $R = S_C$) and let $h_C^\sim \in \text{Mod}_R$ be its isotopy class. Suppose that $h_C^\sim \in G$, i.e. that $h_C^\sim$ can be represented as a product of i' , t' and the Dehn twists about $A_2, \dots, A_g, B_2, \dots, B_{g-1}, C_2, \dots, C_g$ on R . Then $h^\sim$ differs from the corresponding product of $i^\sim$, $t^\sim$ and the Dehn twists about $A_2, \dots, A_g, B_2, \dots, B_{g-1}, C_2, \dots, C_g$ (considered now on S) by some power of the Dehn twist t_{γ_1} about C_1 . Since $i^\sim = t_{\alpha_1} t_{\gamma_1}^2 t_{\alpha_1}$ and $t^\sim = (t_{\gamma_1} t_{\alpha_1})^6$ belong to G_S , it follows that in this case $h^\sim \in G_S$. Therefore, it is sufficient to prove that $h_C^\sim \in G$. By the assumption, $h_C^\sim$ belongs to the kernel of the natural map $\rho : \text{PMod}_R \rightarrow \text{PMod}_Q$ from 2.8. So, it is sufficient to prove that $\ker \rho \subset G$.

The kernel $\ker \rho$ is equal to the image of the canonical homomorphism $j : PB_2(Q) \rightarrow \text{Mod}_R$ from 2.8, and hence is generated by the images of the fundamental groups $\pi_1(Q_1)$, $\pi_1(Q_2)$, where Q_i is the result of gluing only one disc D_i to R (and $i = 1, 2$).

Consider first the element a of $\pi_1(Q_2)$ represented by the loop in Fig. 4.8(a). In view of Lemma 4.1.I, its image in Mod_R is equal to $t_\beta t_\gamma^{-1}$, where β, γ are the isotopy classes of circles B, C from Fig. 4.8(b). Clearly, B, C are isotopic in R to B_1, C_2 respectively, and hence this image is equal to $t_{\beta_1} t_{\gamma_2}^{-1} \in G$. Let us consider next the element b of $\pi_1(Q_2)$ represented by the loop in Fig. 4.8(c) (a loop going round the hole). In view of Lemma 4.1.I, its image in Mod_R is equal to $t_\delta t_\varepsilon^{-1}$, where δ, ε are the isotopy classes of circles D, E from Fig. 4.8(d). Clearly, D is isotopic to D_0 and E is a trivial circle (homotopic to a boundary component). It follows that $t_\delta = t_{\delta_0}$, $t_\varepsilon = 1$, and hence $t_\delta t_\varepsilon^{-1} = t_{\delta_0} = t^\sim$. Therefore, the image of b also belongs to G .

Recall that $fj(\sigma)f^{-1} = j(f_*(\sigma))$ for any $f \in \text{Mod}_R$, $\sigma \in B_2(Q)$ (cf. 2.8). It follows that $j(f_*(\sigma)) \in G$ if $f \in G$, $j(\sigma) \in G$. In particular, if the image of some element $c \in \pi_1(Q_2)$ belongs to G and $f \in G$, then the image of $f_*(c)$ also belongs to G (if f is not in PMod_R , i.e. if f interchanges the holes, then $f_*(c) \in \pi_1(Q_1)$, but actually we do not need this case). Now, Fig. 4.9(a)-(g) shows how to produce new elements of $\pi_1(Q_2)$ from the element a by applying consecutively various Dehn twists from G to a . Clearly, together with b these elements generate $\pi_1(Q_2)$. It follows that the image of $\pi_1(Q_2)$ in Mod_R is contained in G .

It remains to show the same for the image of $\pi_1(Q_1)$. This can be done by similar pictures. But it is easier to note that conjugation by the element i' , interchanging the holes of R , maps the image of $\pi_1(Q_2)$ into the image of $\pi_1(Q_1)$. It follows that (since $i' \in G$) the image of $\pi_1(Q_1)$ is also contained in G . And since the images of $\pi_1(Q_1)$, $\pi_1(Q_2)$ generate $\ker \rho$ by Corollary 2.8.B and

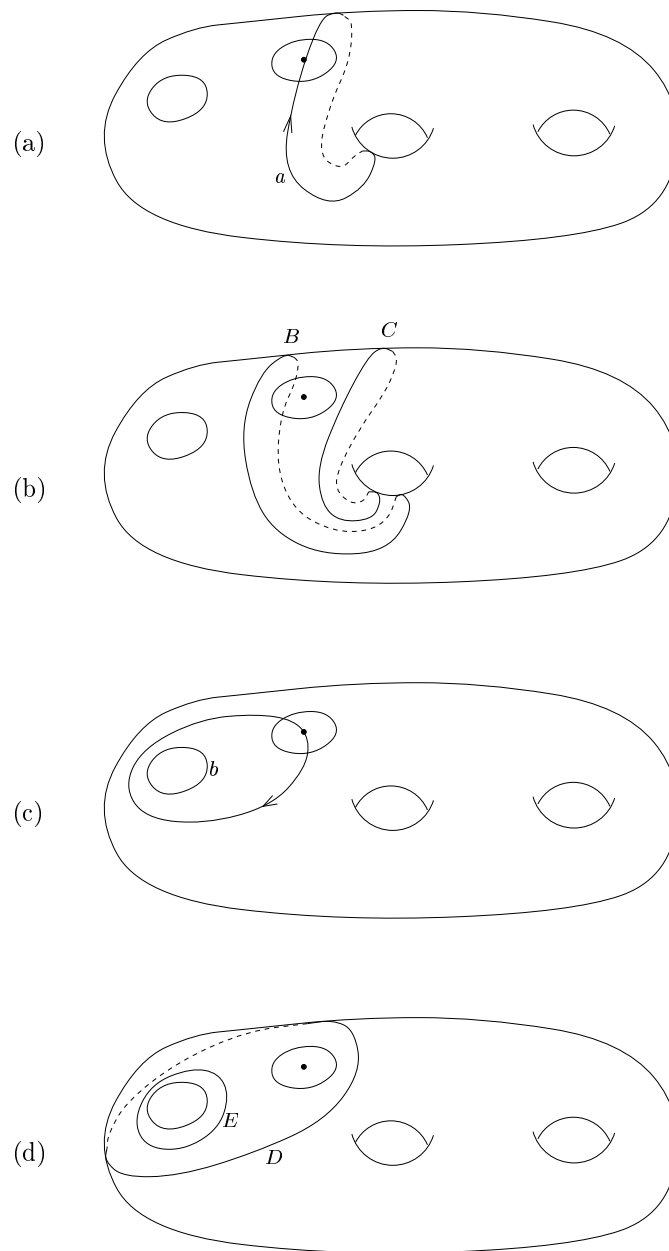


Fig. 4.8

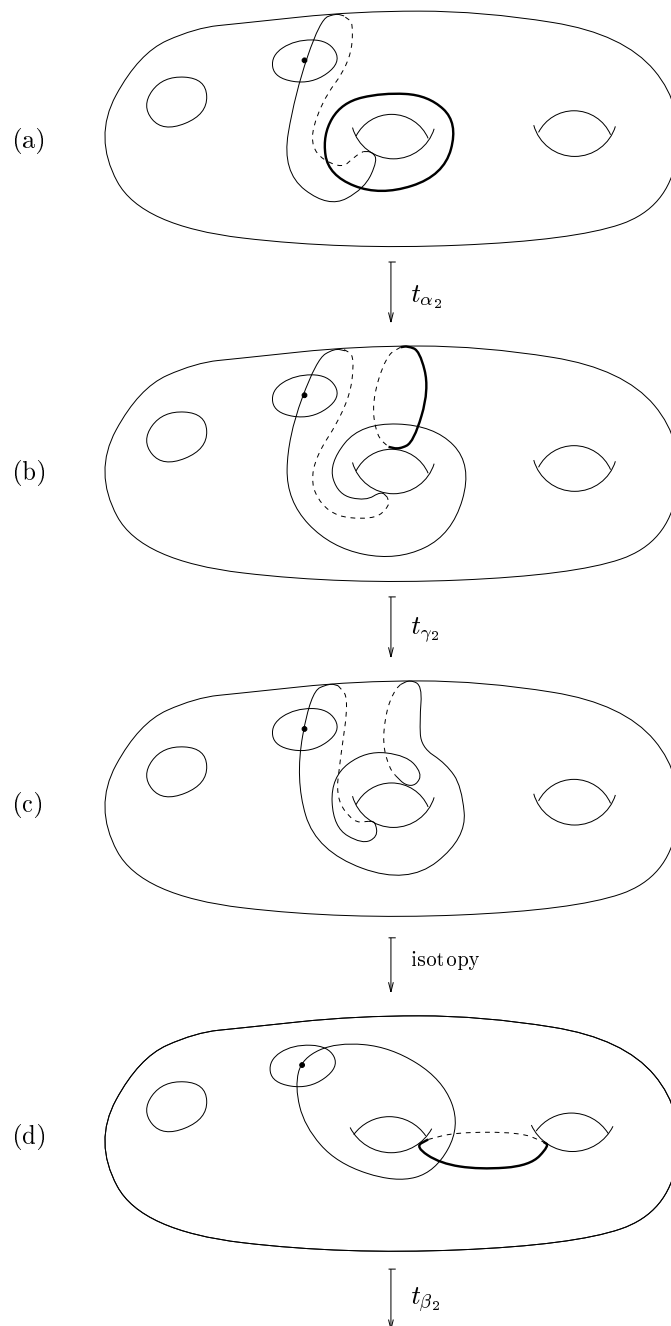


Fig. 4.9(a)–(d)

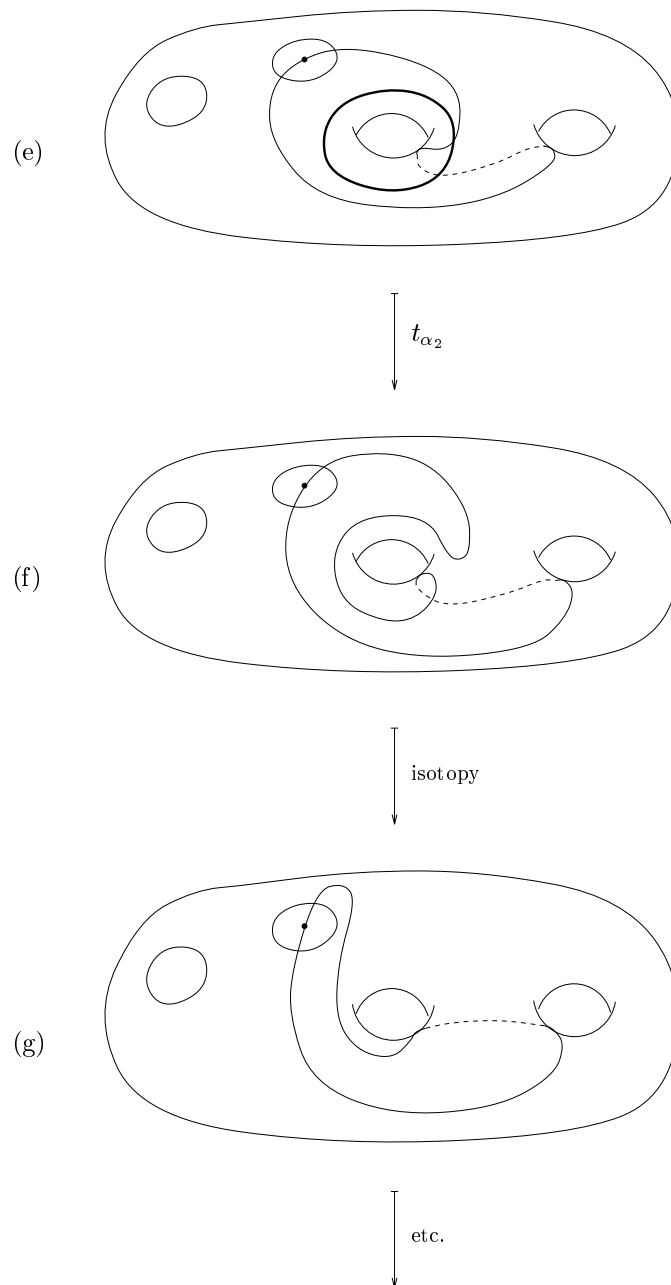


Fig. 4.9(e)–(g)

Theorem 2.8.C, we conclude that $\ker \rho \subset G$. As we already said, this completes the proof of the claim, and hence of the theorem. $\square$

M. Dehn [D2] was the first to prove that the groups Mod_S are finitely generated and to provide an explicit finite set of generators for them. His set of generators consisted of $2g(g-1)$ Dehn twists for a closed surface S of genus $g \geq 3$ and of 5 Dehn twists for a closed surface of genus 2. Much later, and independent of Dehn's work, Theorem 4.2.D was proved by W. B. R. Lickorish [Li2], [Li6]. An exposition of the work of Lickorish was given by J. Birman [Bir3]; see [Bir3], Chapter 4. Again, to a certain extent, our proof follows the simplified proof of J. Birman [Bir4] (cf. [Bir4], Theorem 4.7 and Corollary 4.8), and the use of $C(S)$ leads to further radical simplifications and is closer in spirit to Dehn's original approach. (We also added some details corresponding to the proof of Corollary 4.8 from [Bir4], omitted in [Bir4].)

Corollary 4.2.E *Mod_S is finitely generated for any compact (orientable) surface S .*

Proof. Since PMod_S is of finite index in Mod_S , it is sufficient to prove that the groups PMod_S are finitely generated. Let us glue discs to the boundary components of S one at a time. Using the second exact sequence of Theorem 2.8.C with $n = 1$ (recall that $PB_1(Q) = \pi_1(Q)$ for any surface Q), the fact that the fundamental groups of compact surfaces are finitely generated, and the induction on the number of boundary components, we can reduce our assertion to the case of closed surfaces S . But for closed S , Theorem 4.2.D provides an explicit finite system of generators. $\square$

Corollary 4.2.F *If S is a closed surface of genus g , then Mod_S is generated by Dehn twists about the $2g+1$ circles $A_1, \dots, A_g, B_1, \dots, B_{g-1}, C_1, C_2$ presented in Fig. 4.5.*

Proof. This result is due to S. Humphries [Hum], and we refer to [Hum] for a sequence of pictures showing how to transform C_i into a circle isotopic to C_{i+2} by a sequence of 16 twist diffeomorphisms about $A_i, B_i, C_{i+1}, A_{i+1}$, and B_{i+1} . In view of Lemma 4.1.C, this implies that one can express the Dehn twist about C_{i+2} in terms of Dehn twists about $C_i, A_i, B_i, C_{i+1}, A_{i+1}$, and B_{i+1} . It follows that we can consecutively eliminate Dehn twists about $C_g, C_{g-1}, \dots, C_3$ from our list of generators. $\square$

This result is complemented by a theorem of S. Humphries [Hum] to the effect that $2g+1$ is actually the minimal number of Dehn twist generators of Mod_S if S is a closed surface of genus g . If we do not require the generators to be Dehn twists, then one needs, in fact, only two generators. Namely, the following theorem holds.

Theorem 4.2.G *If S is a closed surface of genus g , then Mod_S is generated by the two elements*

$$t_{\gamma_1} t_{\alpha_1} t_{\beta_1} t_{\alpha_2} t_{\beta_2} \cdots t_{\beta_{g-1}} t_{\alpha_g},$$

$$t_{\gamma_{g-1}} t_{\gamma_g}^{-1},$$

where $\alpha_1, \beta_1, \dots, \alpha_g, \gamma_1, \gamma_{g-1}, \gamma_g$ are, respectively, the isotopy classes of the circles $A_1, B_1, \dots, A_g, C_1, C_{g-1}, C_g$ presented in Fig. 4.5.

This nice result is due to B. Wajnryb [W2]. He deduced it from the above Corollary 4.2.F by some ingenious computations. His proof also covers the case of surfaces with one boundary component (if we made a hole in the surface in Fig. 4.5 at the right end of it, then the Dehn twists about the same circles may serve as generators). The point of this theorem is in providing the *minimal* number of generators (clearly, Mod_S is not cyclic, and hence cannot be generated by one element!). As B. Wajnryb explains in [W2], it is quite easy to find a generating set consisting of three elements, and already W. B. R. Lickorish [Li3] noticed that four elements generate Mod_S . While the property of being generated by two elements has no immediate important consequences, it traditionally attracts the attention of group theorists.

4.3 Finite presentations

After we have proved that the groups Mod_S are finitely generated, it is only natural to ask if they are finitely presented (and to look for a finite presentation, if they are). In the spirit of our proof of the Dehn-Lickorish theorem (i.e., Theorem 4.2.D), we will base our approach to this question upon the action of Mod_S on $C(S)$. Such an approach is made possible by the following general theorem.

Theorem 4.3.A *Suppose that a group G acts on a CW-complex X permuting its cells and suppose that:*

- (i) *X is simply connected (in particular, X is connected);*
- (ii) *the isotropy group of every vertex of X is finitely presented;*
- (iii) *the isotropy group of every edge of X is finitely generated;*
- (iv) *the number of orbits of cells of dimension ≤ 2 is finite.*

Then G is finitely presented.

An elegant proof of this theorem, based on the Bass-Serre theory [Ser2], is given by K. Brown [Bro2]; cf. [Bro2], Theorem 4. But it can also be easily proved directly. In the special case when G acts transitively on the set of vertices, such a proof is contained in the report of F. Laudenbach [La] about the paper of A. Hatcher and W. Thurston [HatT]; cf. [La], §2. Implicitly this special case was proved and used already by A. Hatcher and W. Thurston [HatT]. Laudenbach's proof can be easily extended to the general case. K. Brown [Bro2] gives a long

list of precursors of Theorem 4.3.A, but the papers [HatT], [La] apparently were unknown to him. It is worth adding to his list also the results of H. Behr [Be], Section 1, and of J. L. Koszul [Kosz], Chapter III, §2. While H. Behr [Be] states his sufficient condition for the finite presentability in terms of actions on discrete metric spaces, J. L. Koszul [Kosz] reworked it in the language of actions on graphs, and obtained, in fact, a necessary and sufficient condition. Actually, as it is clear from his definition[†] of the homotopy of loops, J. L. Koszul [Kosz] works not with graphs, but with flag complexes (cf. 3.1). Compared with [Kosz], the main novelty of Theorem 4.3.A lies in the fact that locally infinite complexes are admitted. According to W. Harvey [Harv2], the fact that the complexes of curves are not locally finite also was an (at least a psychological) obstacle to overcome.

Note that all proofs of Theorem 4.3.A allow, in principle, to construct a finite presentation of G starting from finite presentations of the isotropy groups of vertices, finite sets of generators of the isotropy groups of edges, the combinatorial structure of X , and, of course, the action of G on X . This leads usually to a fairly complicated presentation, which one may try to simplify. For Mod_S , such an approach was successfully realized by B. Wajnryb in his work [W1] discussed below.

The following two lemmas admit direct and elementary proofs. But we prefer to deduce them from Theorem 4.3.A and the well known fact that for any finitely presented group there is a finite CW-complex having this group as its fundamental group. Our proof of the first lemma was suggested by [Kosz], §III.2, Remark 2.

Lemma 4.3.B *Let $1 \rightarrow H \rightarrow G \rightarrow F \rightarrow 1$ be an exact sequence of groups. If H and F are finitely presented, then G is also finitely presented.*

Proof. Let Y be a finite CW-complex with $\pi_1(Y) = F$, and let X be the universal cover of Y . So, X is simply connected and F freely acts on X . This action induces an action of G on X . Since F acts on X freely, the isotropy groups of vertices and edges under the action of G are all isomorphic to the group H , which is finitely presented. Since Y is finite, the number of orbits of cells (under the action of either F or G) is finite. Hence we may apply Theorem 4.3.A and conclude that G is finitely presented. $\square$

Lemma 4.3.C *Let H be a subgroup of finite index of a group G . Then H is finitely presented if and only if G is finitely presented.*

Proof. Suppose first that G is finitely presented. Let Y be a finite CW-complex with $\pi_1(Y) = G$, and let X be the universal cover of Y . Then H acts

[†]in [Kosz], p. 37, on the second line $(a_0, \dots, a_i, a_{i+1}, \dots, a_n)$ should be replaced by $(a_0, \dots, a_i, a_{i+2}, \dots, a_n)$

freely on X and the number of the orbits of the cells is finite because $Y = X/G$ is finite and the index $[G : H]$ is finite. Theorem 4.3.A now implies that H is finitely presented.

Suppose now that H is finitely presented. Let

$$H' = \bigcap_{g \in G} gHg^{-1}.$$

This intersection is actually finite, because $[G : H] < \infty$, and hence H' is of finite index in both G and H . By the already proved part of the lemma, H' is finitely presented. Since H' is obviously normal in G , we have an exact sequence of groups $1 \rightarrow H' \rightarrow G \rightarrow G/H' \rightarrow 1$. Since G/H' is finite, it is finitely presented. Now Lemma 4.3.B implies that G is finitely presented. $\square$

Theorem 4.3.D *Mod $_S$ is finitely presented for any compact (orientable) surface S .*

Proof. In view of Lemma 4.3.C it is sufficient to prove that the groups PMod_S are finitely presented. In order to deal with PMod_S we will use double induction on the genus and the number of boundary components, as in the proof of Theorem 4.2.C.

Suppose that $\partial S \neq \emptyset$. Let R be the result of gluing a disc to one of the boundary components of S . By the inductive assumption PMod_R is finitely presented. The kernel of the natural homomorphism $\rho : \text{PMod}_S \rightarrow \text{PMod}_R$ from 2.8 is finitely presented (by Theorem 2.8.C this kernel is isomorphic to $\pi_1(R)$ with few exceptions; these exceptions are easy to deal with directly), and hence Lemma 4.3.C implies that PMod_S is also finitely presented. This completes the induction step of the induction on the number of boundary components.

Next, suppose that S is closed. If S is a sphere, then $\text{PMod}_S = 1$. If S is a torus, then PMod_S is isomorphic to $SL_2(\mathbf{Z})$. It is well known that the latter group is finitely presented (cf. [Ser3], for example). So, we may assume that the genus of S is ≥ 2 . In this case we are going to apply Theorem 4.3.A to the action of PMod_S on the geometric realization of $C(S)$. We need to verify the assumptions of Theorem 4.3.A. Theorem 3.2.C implies (i) (cf. also the discussion preceding Theorem 3.2.C). Let us now check (iii). Suppose that $\langle C_1 \rangle, \langle C_2 \rangle$ are connected by an edge of $C(S)$, and let G be the isotropy group of (the geometric realization of) this edge. We may assume that $C_1 \cap C_2 = \emptyset$. Let G_0 be the subgroup of PMod_S consisting of the isotopy classes of diffeomorphisms $F : S \rightarrow S$, such that $F(C_1) = C_1$, $F(C_2) = C_2$ and F preserves the sides of both circles C_1, C_2 . Clearly, G_0 is a subgroup of finite index in G (in fact, the index is ≤ 4). Such diffeomorphisms F can be cut along $C_1 \cup C_2$, and this leads to a canonical homomorphism $G_0 \rightarrow \text{PMod}_Q$, where Q is the result of cutting S along $C_1 \cup C_2$. Clearly, this homomorphism is surjective, its kernel is generated by the Dehn twists about C_1 and C_2 (the latter assertion

follows from Corollary 4.1.B applied twice), and these Dehn twists commute (by Lemma 4.1.F(i)). Since S is closed, the genus of the components of Q is (strictly) less than the genus of S . So, the inductive assumption implies that PMod_Q is finitely presented. By Lemma 4.3.B, G_0 is also finitely presented. Finally, by Lemma 4.3.C the isotropy group G is finitely presented. So, we proved an assertion even stronger than (iii). The proof of (ii) is similar and simpler. In order to check (iv), notice that the number of orbits of cells of dimension ≤ 2 is nothing more than the number of arrangements of ≤ 3 disjoint circles on S considered up to a diffeomorphism. Clearly, this number is finite. So, we can apply Theorem 4.3.A and conclude that PMod_S is finitely presented. This completes the induction step of the induction on genus, and hence the proof. $\square$

Theorem 4.3.D was originally proved by J. McCool [McC] by methods of the combinatorial group theory. His approach allows, in principle, computation of an explicit finite presentation of Mod_S for a given S , but this turned out to be too difficult to be done for any S . Later on, A. Hatcher and W. Thurston [HatT] suggested a geometric approach to the finite presentability and to a construction of an explicit finite presentation. The above proof was suggested by the author in [I5]. It has some features in common with [HatT], such as the use of families of functions in order to prove the simply connectivity of an appropriate complex (cf. the proof of Theorems 3.2.B and 3.2.C above), and the use of Theorem 4.3.A (only implicit in [HatT]). In some sense, this proof can be considered as an ultimate simplification of the proof of Hatcher and Thurston (at least for the time being). One more approach to this theorem is indicated in 5.4.

The complex used in [HatT] is very complicated, and A. Hatcher and W. Thurston did not achieve their goal of writing down an explicit finite presentation (that does not diminishes the significance of their exceptionally beautiful paper!). Their complex was somewhat simplified by J. Harer [Har1], and using this simplified complex B. Wajnryb [W1] succeeded in writing down an explicit finite presentation for closed surfaces and for surfaces with one boundary component. The proof involves some extremely complicated computations. The reader should be warned that [W1] contains some mistakes, even in the statements of the main theorems, corrected much later in [BW]. Recently, B. Wajnryb [W3] provided a new exposition of his work. In addition, [W3] contains a new proof of the simply connectivity of the Hatcher-Thurston complex.

Let us briefly describe the Wajnryb presentation for closed surfaces. As generators he uses the Humphries generators of Corollary 4.2.F. Relations are the following. First, for any pair of disjoint circles from the list of Corollary 4.2.F, the Dehn twists about them are related by the commutation relation of Lemma 4.1.F(i). Second, for any pair of circles from this list intersecting (transversely) at one point, the Dehn twists about them are related by the Artin relation of Lemma 4.1.F(ii). There are three additional relations. In order to describe them, let us, as usual, denote the isotopy classes of the circles

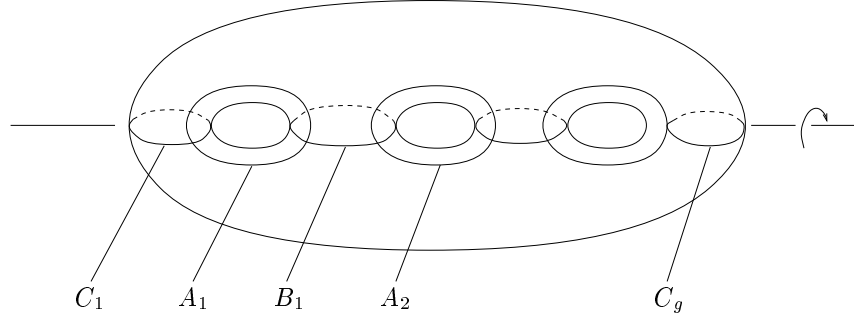


Fig. 4.10

$A_1, \dots, A_g, B_1, \dots, B_{g-1}, C_1, \dots, C_g$ in Fig. 4.5 by $\alpha_1, \dots, \alpha_g, \beta_1, \dots, \beta_{g-1}, \gamma_1, \dots, \gamma_g$, respectively.

In order to describe the first relation, we need a circle D_2 on S intersecting transversely A_2 at exactly one point, not intersecting any other circle in Fig. 4.5, and not isotopic to C_2 (up to isotopy D_2 is characterized by these properties; it is situated similarly to C_2 , but is contained in the lower part of S). Let $\delta_2 = \langle D_2 \rangle$. Then $\delta_2 = u(\gamma_2)$, where $u = t_{\alpha_2} t_{\beta_1} t_{\alpha_1} t_{\gamma_1}^2 t_{\alpha_1} t_{\beta_1} t_{\alpha_2}$, as one may check straightforwardly. Hence, $t_{\delta_2} = u t_{\gamma_2} u^{-1}$. It turns out that $(t_{\gamma_1} t_{\alpha_1} t_{\beta_1})^4 = t_{\gamma_2} t_{\delta_2}$, and using our formulas for t_{δ_2} and u , we can express this relation in terms of the Humphries generators. This is the first additional relation.

The next relation is, essentially, the lantern relation of Lemma 4.1.H, expressed in terms of the Humphries generators. To write it down, B. Wajnryb chose an embedding of S_0^4 in S such that the boundary circles C_0, C_1, C_2, C_3 of S_0^4 are mapped to the circles C_1, B_2, B_3, C_3 of Fig. 4.5. Then he expressed the Dehn twists about the images of the circles C_{12}, C_{13}, C_{23} , and the Dehn twist about C_3 in terms of the Humphries generators. After this he wrote the lantern relation in terms of these expressions. This is the second additional relation.

The last relation is concerned with the so-called *hyperelliptic involution* of S . It is an orientation-preserving diffeomorphism $H : S \rightarrow S$ of order 2 (i.e. $H \circ H = \text{id}_S$). Up to isotopy, it is characterized by the following properties: it maps each of the $2g + 1$ circles $C_1, A_1, B_1, A_2, B_2, \dots, B_{g-1}, A_g, C_g$ to itself, and reverses the orientation of all these circles (and hence interchanges the components of the complement in S of the union of these $2g + 1$ circles). It can be represented by a 180° rotation around the horizontal axis of symmetry in Fig. 4.10. The isotopy class h of H is also called the *hyperelliptic involution*. (Note that, as it is described here, the hyperelliptic involution depends on the choice of a collection of circles diffeomorphic to the one in Fig. 4.5, and different choices lead to different, but conjugate, hyperelliptic involutions. For the purposes of writing down the Wajnryb presentation, we need only one such collection of circles.) Since $H(C_g) = C_g$ and H is orientation-preserving, $h t_{\gamma_g} h^{-1} = t_{\gamma_g}$ by

Lemma 4.1.C, or, what is the same, $ht_{\gamma_g} = t_{\gamma_g}h$. The third additional relation is, essentially, this commutation relation. It turns out that $h = t_{\gamma_g}ft_{\gamma_g}$, where $f = t_{\alpha_g}t_{\beta_{g-1}} \cdots t_{\beta_1}t_{\alpha_1}t_{\gamma_1}^2t_{\alpha_1}t_{\beta_1} \cdots t_{\beta_{g-1}}t_{\alpha_g}$. Clearly, h commutes with t_{γ_g} if and only if f does. Wajnryb expressed t_{γ_g} in terms of the Humphries generators, and then, using this expression and the above formula for f , he wrote the commutation relation $ft_{\gamma_g} = t_{\gamma_g}f$ in terms of the Humphries generators. This is his third (and the last) additional relation.

The Wajnryb presentation for surfaces with one boundary component is obtained from the Wajnryb presentation for closed surfaces by omitting the third additional relation. (If we made a hole in the surface on Fig. 4.5 at the right end of it, the Dehn twist about the same circles serve as generators.)

The above description of the Wajnryb presentations in geometrical terms is, essentially, borrowed from J. Birman's survey [Bir5]. (Note that the relation (6) in [Bir5] is incorrect as stated. One should replace in it the element G by the above expression for the hyperelliptic involution h .)

Now, let S be a closed surface of genus 2. In this case the second additional relation follows from the other relations. The first finite presentation of Mod_S in this case was discovered by P. Bergau and J. Mennicke [BeM]. Their proof turned out to be incomplete, and their presentation was justified only about ten years later by J. Birman and H. Hilden [BirH]. Instead of the first two additional relations of the Wajnryb presentation, this presentation contains the following two relations:

$$(t_{\gamma_1}t_{\alpha_1}t_{\beta_1}t_{\alpha_2}t_{\gamma_2})^6 = 1;$$

$$(t_{\gamma_2}t_{\alpha_2}t_{\beta_1}t_{\alpha_1}t_{\gamma_1}^2t_{\alpha_1}t_{\beta_1}t_{\alpha_2}t_{\gamma_2})^2 = 1$$

(the second one expresses the fact that the hyperelliptic involution has order 2). A special feature of this case is that the hyperelliptic involution h commutes with all 5 generators of Theorem 4.2.D or Corollary 4.2.F (known already to Dehn [D2]), as is clear from the above description of h , and hence belongs to the center of Mod_S . In fact, it generates the center in this case. On the contrary, if the genus is ≥ 3 , then h does not commute with t_{γ_2} , and, in fact, the center is trivial. Cf. 7.5.

For surfaces S of genus 0 (i.e. a spheres with holes), Mod_S is closely related to the classical Artin braid groups. This allows to write down a presentation of Mod_S in this case. For the details, see [Bir3], Chapter 4.

Recently, S. Gervais [Ge2] used the Wajnryb presentation and induction on the number of the boundary components in order to derive an explicit (and very symmetric) presentation of Mod_S for surfaces S with an arbitrary number of boundary components.

Finally, let us mention a related work of S. Gervais [Ge1], where he derived some *infinite* presentations of mapping class groups, similar to the Wajnryb presentation, but having as generators all Dehn twists or all Dehn twists about nonseparating circles. His presentation with all Dehn twists as generators was recently simplified by F. Luo [Luo3].

5 Teichmüller spaces

Teichmüller spaces provide another class of fundamental geometric objects on which mapping class groups act. Our approach to Teichmüller spaces is closer in spirit to that of R. Fricke [FrK] than to the approach of O. Teichmüller himself. In particular, we define the Teichmüller spaces in terms of metrics of constant curvature, and not in terms of Riemann surfaces as Teichmüller did. In fact, the work of Fricke precedes the work of Teichmüller by about forty years. Probably, Fricke was also the first to define the mapping class groups.

In our discussion we stress the connection of Teichmüller spaces with complexes of curves (and especially the results of W. Harvey [Harv1], [Harv2]; cf. 5.4) and other combinatorial objects, namely, the so-called ideal triangulations (cf. 5.5). As the first application, we will indicate another approach to the finite presentability of the mapping class groups (cf. Corollary 5.4.B). In the next section we will present a much deeper application of Teichmüller spaces to the mapping class groups, namely, the computation of the virtual cohomological dimension of the mapping class groups (cf. Theorems 6.4.A, 6.4.B and 6.4.C).

5.1 Definitions

Let us assume that our (compact orientable) surface S admits a riemannian metric of constant curvature -1 with geodesic boundary. As is well known (see 2.5), such metrics exist if and only if S has negative Euler characteristic. Let us fix some number $l > 0$ and consider only metrics such that all components of the boundary ∂S have length l . The choice of l is largely irrelevant, but at a later point it will be convenient to have l small. Let us denote by H_S the set of such metrics. We can endow H_S with some convenient topology, say the C^∞ -topology, by considering riemannian metrics as tensor fields. Since diffeomorphisms act on tensor fields (by the push-forward), the group $\text{Diff}(S)$ acts on H_S . The quotient space $H_S/\text{Diff}_0(S)$ by the subgroup of diffeomorphisms isotopic to the identity is called the *Teichmüller space* of S and is denoted by T_S . Clearly, the quotient group $\text{Mod}_S = \text{Diff}(S)/\text{Diff}_0(S)$ acts on T_S and the quotient space T_S/Mod_S is equal to $H_S/\text{Diff}(S)$, which is called the *moduli space of hyperbolic structures* on S and is denoted by M_S . Hence $T_S/\text{Mod}_S = M_S$.

Sometimes (cf., for example, 5.5) it is more convenient to consider another version of Teichmüller spaces, namely the one related to so-called punctured surfaces. By a *punctured surface* R we understand a closed (orientable) surface *together* with a finite subset of it. The points of this finite subset are called *punctures* and we denote by R° the result of deleting all punctures from the surface. Let R be a punctured surface such that R° admits a complete riemannian metric of constant curvature -1 . Again, it is well known that such metrics exist if and only if R° has negative Euler characteristic. We denote by H_R the set of such metrics. This set also can be endowed with a natural topology. This time, the group $\text{Diff}(R^\circ)$ acts on H_R , and the quotient space

$H_S/\text{Diff}_0(R^\circ)$ by the subgroup of diffeomorphisms isotopic to the identity is called the *Teichmüller space* of R and is denoted by T_R . Naturally, the group $\text{Mod}_R = \text{Diff}(R^\circ)/\text{Diff}_0(R^\circ)$ is called the *mapping class group* of R . It naturally acts on T_R and the quotient space T_R/Mod_S is equal to $H_R/\text{Diff}(R^\circ)$, which is called the *moduli space of hyperbolic structures* on R and is denoted by M_R . Clearly, $T_R/\text{Mod}_R = M_R$.

Note that there is no real difference between mapping class groups of compact and punctured surfaces. If S is a compact surface, then we can collapse each component of ∂S into a point (one for each boundary component) and get a closed surface. Images of boundary components form a finite subset of this surface, and the resulting surface together with this finite subset is a punctured surface. Let us denote it by R . Clearly, $S \setminus \partial S = R^\circ$. This leads to a natural map $\text{Diff}(S) \rightarrow \text{Diff}(R^\circ)$. The homomorphism $\text{Mod}_S \rightarrow \text{Mod}_R$ induced by it is easily seen to be an isomorphism.

5.2 Length functions and Fenchel-Nielsen flows

Let α be a vertex of $C(S)$, i.e. the isotopy class of a nontrivial circle. For any metric $h \in H_S$ the isotopy class α contains exactly one geodesic with respect to the metric h . Let us denote by $l_\alpha(h)$ the length of this geodesic. Clearly, this length depends only on the image of h in T_S . Hence, we can consider l_α as a function $T_S \rightarrow \mathbf{R}_{>0}$. We call this function the *length function* corresponding to α . For a simplex $\sigma = \{\alpha_1, \dots, \alpha_m\}$ of $C(S)$ we may consider the map $L_\sigma : T_S \rightarrow \mathbf{R}_{>0}^m$ given by the formula $L_\sigma(x) = \{l_{\alpha_1}(x), \dots, l_{\alpha_m}(x)\}$. Clearly, it comes from a similar map $L_\sigma : H_S \rightarrow \mathbf{R}_{>0}^m$. The map L_σ is especially important when σ is a simplex of maximal dimension. In this case it turns out to be a principal bundle over $\mathbf{R}_{>0}^m$ having as the structure group the additive group $\mathbf{R}^m$.

Let us describe the corresponding action of $\mathbf{R}^m$ on T_S . Let $h \in H_S$ be a metric representing a point $x \in T_S$, and let C_i be the geodesic circle with respect to h in the isotopy class α_i ($1 \leq i \leq m$). If we cut S along this geodesic C_i , we get a new surface (with a metric of curvature -1 and geodesic boundary), which has two new boundary components resulting from C_i . In order to restore S from this new surface, we have to glue back these two boundary components. Clearly, these boundary components should be glued by an isometry, but this does not define the gluing uniquely. Two isometric gluings may differ by a rotation. It seems at first sight that the set of all possible isometric gluings is parameterized by a circle. In fact, since we are interested in the resulting point of T_S (and not only of M_S), i.e. we are interested in the resulting metric up to isotopy (i.e., up to the action of $\text{Diff}_0(S)$) and not only up to isometry, the set of all possible gluings is parameterized by the universal cover of this circle, i.e., by a copy of $\mathbf{R}$. More precisely, there are (at least) two natural identifications of this universal cover with $\mathbf{R}$: the one corresponding to the measuring of the amount of rotation by the angle, and the other corresponding to the measuring of the amount of rotation by the length (one can get from the first to the second

by multiplication by $l_{\alpha_i}(h)/2\pi$). Another ambiguity comes from the need to orient the circle C_i in order to get the identification, but none of these is of any importance for us now. To summarize, we may cut the original surface and then glue it back by a gluing differing from the one leading back to the original surface by a parameter $r \in \mathbf{R}$. This defines a map $\mathbf{R} \times T_S \rightarrow T_S$, and it is easy to see that this map is an action of the additive group $\mathbf{R}$. This action is usually called a *Fenchel-Nielsen flow*. Of course, this action depends on the vertex α_i . For any $h \in H_S$, the corresponding circles C_i , $1 \leq i \leq m$, are disjoint (because σ is a simplex), and this easily implies that the actions corresponding to different vertices of σ commute. Hence, they define an action of $\mathbf{R}^m$.

The fact that this action turns L_σ into a principal bundle amounts, essentially, to the following. In the notation of the previous paragraph, if the simplex σ is maximal and we cut S along all circles C_i , $1 \leq i \leq m$, we will get a collection of discs with two holes. A metric of the constant curvature -1 with geodesic boundary on a disc with two holes is uniquely determined, up to isotopy, by the lengths of its boundary components. Hence, the cut surface is determined by $L_\sigma(x)$. Clearly, if two surfaces (with metrics) lead to the same cut surface, then they differ only by the gluing, i.e. by the action of $\mathbf{R}^m$. It follows that the orbits of $\mathbf{R}^m$ are exactly the fibers of the map L_σ . One needs also to check that the action of $\mathbf{R}^m$ on T_S is free, and also the existence of local sections. We refer to [FLP], Exposé 7, for this.

5.3 The topology of Teichmüller spaces

As we just saw, the Teichmüller space T_S is the total space of a principal bundle over $\mathbf{R}_{>0}^m$ with the fiber $\mathbf{R}^m$. Since $\mathbf{R}_{>0}^m$ is contractible, it follows that T_S is homeomorphic to $\mathbf{R}_{>0}^m \times \mathbf{R}^m$, and hence to $\mathbf{R}^{2m}$. The number m is the maximal number of pairwise disjoint pairwise non isotopic non-trivial circles on S , hence $m = 3g - 3 + b$, where g is the genus and b is the number of boundary components of S . We conclude that T_S is homeomorphic to $\mathbf{R}^{6g-6+2b}$.

In addition, one can introduce a smooth structure on T_S in a number of ways, all leading to the same smooth structure. This smooth structure can be characterized by the fact that all length functions L_α are smooth. In fact, a finite number of them is sufficient to embed T_S in a Euclidean space $\mathbf{R}^n$ as a smooth submanifold. The straightforward approach requires $9g - 9 + 3b$ length functions (cf. [FLP], Exposés 7 and 11). But for a closed S it is sufficient to use only $\dim T_S + 1$ length functions. And if we consider another version of the Teichmüller space T_S , in which the lengths of the boundary components are not fixed as in 5.1, then for *non-closed* S it is sufficient to use only $\dim T_S$ length functions. See P. Schmutz [Sch] and the references there for these results.

No matter what definition of the smooth structure is used, it is easy to see that the principal bundle from 5.2 is smooth. It follows that T_S is not only homeomorphic, but also diffeomorphic to a Euclidean space (no results about topological manifolds need to be used!). Also, the Fenchel-Nielsen flows

are smooth. Moreover, we can replace *smooth* by *real analytic* everywhere. For closed S the Teichmüller space T_S even has a canonical complex analytic structure, but it plays no role in the sort of questions we are interested in (applications to the mapping class groups).

The discussion in 5.2 and in this section applies equally well to Teichmüller spaces of punctured surfaces. In particular, Teichmüller space T_R of a punctured surface R is diffeomorphic to $\mathbf{R}^{6g-6+2b}$, where g is the genus of R and b is the number of punctures, unless R is a torus without punctures, in which case T_R is diffeomorphic to $\mathbf{R}^2$.

5.4 Corners and the mapping class groups

As we noticed in 5.1, Mod_S acts on T_S . This action is not free, because Mod_S has torsion and T_S is homeomorphic to a Euclidean space (any element of finite order has a fixed point in T_S , say by the Smith theory — see, for example, [FLP], Exposé 11, end of § IV). But it is as close to being free as possible: any torsion free subgroup Γ of Mod_S acts freely on T_S (the argument goes as follows: if the isotopy class of a diffeomorphism F fixes a point of T_S represented by a metric $h \in H_S$, then F is isotopic to an isometry F' of the metric h ; it is well known that the group of isometries of a metric of constant curvature -1 on S is finite). It is well known that Mod_S contains torsion free subgroups, even of finite index. In fact, the kernel of the natural homomorphism $\text{Mod}_S \rightarrow \text{Aut}(H_1(S, \mathbf{Z}/m\mathbf{Z}))$ defined by the action of diffeomorphisms on homology is such a subgroup for any $m \geq 3$. This result is a combination of a theorem of J.-P. Serre (cf. Theorem 6.8.A) and a classical result of J. Nielsen (cf. Theorem 7.1.A) about realization of finite cyclic subgroups of Mod_S by subgroups of $\text{Diff}(S)$; see Corollary 7.1.B. In addition, this action is properly discontinuous. Hence, for torsion free Γ the quotient space T_S/Γ is a $K(\Gamma, 1)$ -space and, of course, a manifold. It turns out that this manifold is never compact, but one can construct a compact manifold from it if Γ is of finite index in Mod_S either by adding some boundary at infinity or by deleting a neighborhood of infinity. Moreover, one can do this in a fairly canonical way.

Before we describe these constructions, we note that the resulting compact manifolds with boundary will *not* be smooth manifolds in the usual sense, but smooth manifolds with corners. Recall that a *smooth manifold with corners* is simply a smooth manifold modeled on the products $\mathbf{R}_{\geq 0}^n \times \mathbf{R}^m$ for all n, m (and not only for $n = 1$ as manifolds with boundary). The simplest type of such manifolds is presented by the subsets of usual smooth manifolds locally described by inequalities of the form $x_1 \geq 0, \dots, x_n \geq 0$, where $(x_1, \dots, x_n, \dots, x_{n+m})$ is a chart.

Let us fix some small $\varepsilon > 0$ and let us assume (only for convenience) that the boundary length l fixed in 5.1 is equal to ε . Let $T_S(\varepsilon)$ be the subset of T_S defined by the inequalities $l_\alpha(x) \geq \varepsilon$ for all isotopy classes α of non-trivial circles. Clearly, $T_S(\varepsilon)$ is invariant under the action of Mod_S on T_S . The basic

properties of $T_S(\varepsilon)$ are summarized in the following theorem.

Theorem 5.4.A *If ε is sufficiently small, then the following holds.*

- (i) *The action of Mod_S on $T_S(\varepsilon)$ is properly discontinuous.*
- (ii) *Any torsion free subgroup Γ of Mod_S acts on $T_S(\varepsilon)$ freely, and its quotient space $T_S(\varepsilon)/\Gamma$ is a smooth manifold with corners.*
- (iii) *The quotient space $T_S(\varepsilon)/\text{Mod}_S$ is compact.*
- (iv) *$T_S(\varepsilon)$ is contractible.*
- (v) *The boundary $\partial T_S(\varepsilon)$ is homotopy equivalent to the (geometric realization of the) complex of curves $C(S)$.*

Proof. First, the ε should be chosen so small that the following holds: if h is a metric of constant curvature -1 on S and C_1, C_2 are two geodesic circles (with respect to h) on S of length $\leq \varepsilon$, then C_1, C_2 are either equal or disjoint. The existence of such an ε is well known; cf. [A], Chapter II, Corollary of Lemma 1 in Section 3.3; it also follows from [Bu], Corollary 4.1.2. The property (i) and the first part of the property (ii) immediately follow from the corresponding properties of T_S itself. Property (iii) follows from the so-called Mumford compactness theorem [Mumf2]; see [Bu], Theorem 6.6.5 or [A], Chapter II, Theorem in Section 3.4 for a closely related result. The proof of (iv) is the most technical part of the proof of the theorem; we refer to [I13], Theorem 3 for a proof. Next, we turn to the property (v). For any vertex α of $C(S)$, let us consider the subset $T_S(\varepsilon, \alpha) = \{x \in T_S(\varepsilon) \mid l_\alpha(x) = \varepsilon\}$ of $T_S(\varepsilon)$. This subset turns out to be closely related to $T_{S_\alpha}(\varepsilon)$, where S_α is the result of cutting S along a representative of α , and one can use this together with (iv) for S_α in order to prove that $T_S(\varepsilon, \alpha)$ is contractible. It is easy to see that the boundary of $T_S(\varepsilon)$ is equal to the union of subsets $T_S(\varepsilon, \alpha)$. By the choice of ε , any collection $T_S(\varepsilon, \alpha_i)$ of such subsets has a nonempty intersection if and only if the vertices α_i can be represented by disjoint circles. Hence, $C(S)$ is the nerve of a cover of $\partial T_S(\varepsilon)$ by contractible subsets. The property (v) follows. Finally, the second part of the property (ii) follows from the fact that $T_S(\varepsilon)$ itself is a manifold with corners. This follows, in turn, from the fact that in a neighborhood of a point $x \in \partial T_S(\varepsilon)$ the subset $T_S(\varepsilon)$ is defined by a *finite* number of the inequalities of the form $l_{\alpha_i} \geq \varepsilon$ and, moreover, the corresponding α_i form a simplex, and hence can be included as coordinate functions in a chart (by 5.2).

A more detailed outline of the proof is contained in [I5], Section 4, with complete details provided in [I13]. An outline of another proof is contained in [Har5], Section 3. $\square$

Corollary 5.4.B *Mod_S is finitely presented.*

Proof. Let Γ be a torsion free subgroup of finite index in Mod_S . It follows from (i) and (iv) that $T_S(\varepsilon)/\Gamma$ is a $K(\Gamma, 1)$ -space. Because Γ is of finite index in Mod_S , it follows from (iii) that $T_S(\varepsilon)/\Gamma$ is compact. In view of (ii), $T_S(\varepsilon)/\Gamma$

is a smooth manifold with corners. The last property implies that $T_S(\varepsilon)/\Gamma$ is homotopy equivalent to a finite CW-complex (moreover, it admits a finite triangulation). Hence, there exists a finite CW-complex which is a $K(\Gamma, 1)$ -space. It follows that Γ is finitely presented. By Lemma 4.3.C, Mod_S is also finitely presented. $\square$

The idea behind Theorem 5.4.A is due to W. Harvey [Harv1], [Harv2]. In [Harv2], W. Harvey announced a construction of a manifold with corners X_S , containing the Teichmüller space T_S as its interior and such that the properties (i)–(v) of Theorem 5.4.A hold with $T_S(\varepsilon)$ replaced by X_S . He was motivated by the analogy (first noticed also by him) between the mapping class groups and arithmetic groups (cf. 9.1) and by the work of A. Borel and J.-P. Serre [BorS]. Adding the corners to T_S in this manner turned out to be a more delicate matter than excising a part from T_S as in Theorem 5.4.A. A proof of the existence of such an X_S was provided in [I12], together with a geometric interpretation of the points in the complement $X_S \setminus T_S = \partial X_S$.

5.5 Ideal triangulations

Let R be a punctured surface in the sense of 5.1 with a *non empty* set of punctures. An *ideal triangulation* of R is a collection of (embedded) arcs with disjoint interiors connecting the punctures of R such that if we cut R along all these arcs, we will get a collection of triangles (with sides resulting from these arcs). The word *ideal* is justified by the standard understanding that the punctures are points at the infinity. In other words, we ideally triangulate R° rather than R . Note also that two endpoints of an arc are permitted to coincide and a triangle of an ideal triangulation is *not* determined by the set of its vertices (which are punctures) in general.

The ideal triangulations, considered up to isotopy, are exactly the top dimensional simplices of a simplicial complex $A(R)$, which we now define. The vertices of $A(R)$ are the isotopy classes of non-trivial arcs connecting punctures of R . Here an arc is called *trivial* if two its endpoints coincide and the resulting circle bounds a disc in R which does not contain other punctures. The isotopies are permitted, naturally, only in the class of arcs connecting punctures, so they are fixed on the endpoints. The simplices are defined in a manner similar to the simplices of complexes of curves: a collection of vertices forms a simplex if these vertices can be represented by arcs with disjoint interiors. Like complexes of curves, $A(R)$ is a *flag complex* (cf. 3.1). As in the case of complexes of curves, this can be established by using complete metrics of constant curvature -1, this time on R° , and geodesic arcs. Clearly, Mod_R acts on $A(R)$.

We also need a subcomplex $A_\infty(R)$ of $A(R)$. It has the same vertices as $A(R)$. A collection of vertices forms a simplex of $A_\infty(R)$ if those vertices can be represented by arcs with disjoint interiors such that the result of cutting R along these arcs has at least one component which is not a disc meeting the set of

punctures only in its boundary. In other words, simplices of $A(R)$ which are *not* simplices of $A_\infty(R)$ correspond to the isotopy classes of *ideal cell decompositions* of R in an obvious sense.

Theorem 5.5.A *Let Δ^{b-1} be the open standard $b - 1$ -dimensional simplex, where b is the number of punctures of R . Let us endow Δ^{b-1} with the trivial action of Mod_R , and the product $T_R \times \Delta^{b-1}$ with the product action. Then $T_R \times \Delta^{b-1}$ is Mod_R -equivariantly homeomorphic to the geometric realization of $A(R)$ with the geometric realization of $A_\infty(R)$ removed.*

Note that the homeomorphism implied in this theorem is *not* canonical: different proofs lead to apparently different homeomorphisms.

The early history of this theorem is not well documented in the literature. According to [Har5], Chapter 2, the idea of this triangulation is due to W. Thurston; his approach was based on hyperbolic geometry. The first published account is contained in [Har4], where an alternative proof, due to D. Mumford and based on work of K. Strebel on quadratic differentials on Riemann surfaces (cf. [Str1], [Str2]), was outlined. Proofs based on hyperbolic geometry were published by R. Penner [Pe1] and by B. Bowditch and D. B. A. Epstein [BowE]. Penner's approach seems to be more remote from the original ideas of W. Thurston in that the Lorentz model of the hyperbolic plane plays a crucial role in it. His ideas were extended, to some degree, to the higher dimensions [EpP] and to closed surfaces [NP], and in [Pe6] R. Penner applied them to the so-called *universal Teichmüller space*. Unfortunately, all this falls outside the scope of the present paper.

The complexes $A(R)$ and $A_\infty(R)$ are, together with complexes of curves, the basic combinatorial objects of the topology of surfaces. To a large extent they are simpler than complexes of curves, Theorem 5.5.A being the main reason, but they are not defined for *closed* surfaces. Among the main applications of the complexes $A(R)$ and Theorem 5.5.A are the computation of the virtual Euler characteristic of the mapping class group by (independently) J. Harer and D. Zagier [HarZ] and R. Penner [Pe2], [Pe3] (cf. 6.7), computations and estimates of the Weil–Petersson volumes of Teichmüller spaces by R. Penner [Pe5], Theorem 8.1.A about automorphisms of complexes of curves (cf. 8.4), the proof of E. Witten's conjecture [Wi] by M. Kontsevich [Ko] and the theorem of L. Mosher [Mos2], [Mos3] to the effect that the mapping class groups are *automatic* in the sense of [Ep]. For an exposition of the work of M. Kontsevich we refer to [Loo1], [Loo2], and for an introduction to L. Mosher's results we refer to his own survey [Mos4].

As an easy and useful corollary of Theorem 5.5.A we point out the following result.

Corollary 5.5.B *Every simplex of $A(R)$ is a face of a top dimensional simplex. Every codimension 1 simplex of $A(R)$ is a face of 1 or 2 top dimensional*

simplices. Any two top dimensional simplices Δ, Δ' of $A(R)$ can be connected by a chain of simplices $\Delta = \Delta_1, \dots, \Delta_m = \Delta'$ such that any two consecutive simplices Δ_i, Δ_{i+1} have a common codimension 1 face.

A much more elementary proof of this Corollary was provided by A. Hatcher [Hat3]; cf. [Hat3], Corollary.

6 Cohomological properties

This section is devoted to cohomological properties of mapping class groups. It is centered around the computation of the so-called virtual cohomological dimension of mapping class groups (see 6.4). As preparation for this computation, we present in 6.1 a very short and idiosyncratic introduction to the cohomology of groups. This introduction is heavily slanted toward phenomena crucial for an understanding of both mapping class groups and arithmetic groups (we will discuss the latter in 9.1) and known as the Bieri-Eckmann duality (cf. [BieE]). The latter is actually a manifestation of the Poincaré-Lefschetz duality, as we try to explain in 6.1. After this introduction, we prove in 6.2-6.4 J. Harer's theorem [Har2], [Har4] computing the virtual cohomological dimension of the mapping class groups. In 6.5-6.7 we review, mostly without proofs, some related cohomological properties of mapping class groups. Finally, in 6.8 we review some results about cohomology of the mapping class groups with finite coefficients.

6.1 Cohomology of groups and the Poincaré-Lefschetz duality

The goal of this section is to present an introduction to cohomology of groups and a fragment of the theory of R. Bieri and B. Eckmann [BieE] (see, especially, Theorem 6.1.H).

Let Γ be a discrete group and let X be a $K(\Gamma, 1)$ -space with a fixed isomorphism $\Gamma \rightarrow \pi_1(X)$, which will be used to identify these groups. The homology or cohomology of the group Γ can be defined as the homology or cohomology of the space X . In order to capture the essential properties of Γ , one has to consider also homology and cohomology with so-called *twisted* coefficients. Let M be a Γ -module (in other words, M is a module over the group ring $\mathbf{Z}[\Gamma]$). Then the homology $H_*(\Gamma, M)$ and cohomology $H^*(\Gamma, M)$ of Γ with (twisted) coefficients M can be defined as follows.

For a space Y let us denote by $C_*(Y)$ its singular complex, or, if Y is a CW-complex, its complex of cellular chains. Let $X^\sim \rightarrow X$ be the universal covering of X . The fundamental group $\Gamma = \pi_1(X)$ acts naturally on $X^\sim$, turning $C_*(X^\sim)$ into a complex of Γ -modules. By definition,

$$H_*(\Gamma, M) = H_*(X, M) = H^*(C_*(X^\sim) \otimes_\Gamma M),$$

where $\otimes_\Gamma$ means the tensor product over the group ring $\mathbf{Z}[\Gamma]$. If $M = \mathbf{Z}$ with the trivial structure of Γ -module (i.e., $\gamma m = m$ for all $\gamma \in \Gamma$, $m \in M$), then

$$C_*(X^\sim) \otimes_\Gamma M = C_*(X^\sim) \otimes_\Gamma \mathbf{Z} = C_*(X)$$

and we recover the usual homology of Γ and X . (In fact, if two singular simplices or cells c_1, c_2 in $X^\sim$ are projected to the same simplex or cell in X , then $c_1 = \gamma c_2$ for some $\gamma \in \Gamma$, and hence $c_1 \otimes m = \gamma c_2 \otimes m = c_2 \otimes \gamma m = c_2 \otimes m$ for any $m \in M$.) Cohomology is defined in a similar way:

$$H^*(\Gamma, M) = H^*(X, M) = H^*(\text{Hom}_\Gamma(C_*(X^\sim), M)),$$

where $\text{Hom}_\Gamma(C_*(X^\sim), M) \subset C^*(X^\sim, M)$ is the group of *equivariant* M -valued cochains, i.e. the group of cochains f such that $f(\gamma c) = \gamma f(c)$ for any $\gamma \in \Gamma$ and any chain c . As above, if $M = \mathbf{Z}$ with the trivial structure of Γ -module, we recover the usual cohomology of Γ and X .

If we choose a lift to $X^\sim$ of any singular simplex or cell in X , then any equivariant cochain will be defined by its values on these lifts. It follows that $\text{Hom}_\Gamma(C_*(X^\sim), M)$ and $C^*(X, M)$ are isomorphic as graded abelian groups. But the differentials are different (in the first complex, the differential is *twisted*).

Now, let us define the *cohomological dimension* $\text{cd } \Gamma$ of Γ as the supremum (which can be ∞) of the numbers n such that $H^n(\Gamma, M) \neq 0$ for some Γ -module M .

Theorem 6.1.A *If X is a closed topological manifold of dimension n , then $\text{cd } \Gamma = n$.*

Proof. First, note that $H^m(X, M) = 0$ for $m > n$ by same reasons as for trivial coefficients. Thus we only need to find a Γ -module M such that $H^n(X, M) \neq 0$. If X is orientable, then we can take $M = \mathbf{Z}$ with the trivial structure of Γ -module, otherwise we should take the *orientation module* $\mathbf{Z}_{\text{or}}$. It is isomorphic to $\mathbf{Z}$ as an abelian group, but $\gamma \in \Gamma$ acts on it by multiplication by -1 if γ reverses the orientation of $X^\sim$, and by 1 otherwise. $\square$

This proof at least partially motivates the need for twisted coefficients in the definition of $\text{cd } \Gamma$.

As usual, a short exact sequence of coefficients

$$0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$$

leads to a long exact sequence of cohomology groups:

$$\dots \rightarrow H^n(\Gamma, M') \rightarrow H^n(\Gamma, M) \rightarrow H^n(\Gamma, M'') \rightarrow H^{n+1}(\Gamma, M') \rightarrow \dots$$

It is the key tool in the proof of the following lemma.

Lemma 6.1.B *If $\text{cd } \Gamma < \infty$, then $\text{cd } \Gamma$ is equal to the supremum of numbers n such that $H^n(\Gamma, M) \neq 0$ for some free $\mathbf{Z}[\Gamma]$ -module M .*

Proof. Let $n = \text{cd } \Gamma$. Choose M such that $H^n(\Gamma, M) \neq 0$ and choose a short exact sequence

$$0 \rightarrow M' \rightarrow F \rightarrow M \rightarrow 0$$

with F free. Then we have an exact sequence

$$H^n(\Gamma, F) \rightarrow H^n(\Gamma, M) \rightarrow H^{n+1}(\Gamma, M'),$$

where $H^n(\Gamma, M) \neq 0$ and $H^{n+1}(\Gamma, M') = 0$ because $n + 1 > \text{cd } \Gamma$. It follows that $H^n(\Gamma, F) \neq 0$. $\square$

Lemma 6.1.C *If X is a finite CW-complex, then $H^n(\Gamma, F) = 0$ for all free $\mathbf{Z}[\Gamma]$ -modules F if and only if $H^n(\Gamma, \mathbf{Z}[\Gamma]) = 0$.*

Proof. The “only if part” is trivial. In order to prove the “if” part, suppose that $H^n(\Gamma, \mathbf{Z}[\Gamma]) = 0$. Then, for any finitely generated free module $F = (\mathbf{Z}[\Gamma])^m$, we have $H^n(\Gamma, F) = H^n(\Gamma, (\mathbf{Z}[\Gamma])^m) = H^n(\Gamma, \mathbf{Z}[\Gamma])^m = 0^m = 0$.

Now, any free module is equal to the direct limit of its finitely generated free submodules. As we will explain in a moment, the fact that X is a finite CW-complex implies that the functor $H^n(\Gamma, \cdot) = H^n(X, \cdot)$ commutes with the direct limits. Hence, the vanishing of the cohomology with coefficients in finitely generated free modules implies the vanishing of the cohomology with coefficients in an arbitrary free module. This completes the proof, modulo the claim about commuting with the direct limits.

It is well known that for any ring R (we are interested in the case $R = \mathbf{Z}[\Gamma]$) and any projective R -module Q the functor $Q \otimes_R \cdot$ commutes with the direct limits. In addition, if P is a *finitely generated* projective R -module, then $\text{Hom}_R(P, M) = P^* \otimes_R M$ for any R -module, where $P^* = \text{Hom}(P, R)$. It follows that for a finitely generated projective R -module P the functor $\text{Hom}_R(P, \cdot)$ commutes with the direct limits.

Returning to our situation, notice that if X is a finite CW-complex and we consider cellular chains, then $C_k(X^\sim)$ are finitely generated free (and hence projective) $\mathbf{Z}[\Gamma]$ -modules. Thus the result of the previous paragraph implies that the functor $\text{Hom}_\Gamma(C_k(X^\sim), \cdot)$ commutes with the direct limits. It remains to note that taking the (co)homology groups of a complex always commutes with the direct limits. $\square$

Corollary 6.1.D *If X is a finite CW-complex, then $\text{cd } \Gamma$ is equal to the maximum of numbers n such that $H^n(\Gamma, \mathbf{Z}[\Gamma]) \neq 0$.*

Proof. Since X is a finite CW-complex, $\dim X < \infty$. If we use cellular chains, then $C_k(X^\sim) = 0$ for $k > \dim X$, and hence $H^k(\Gamma, M) = H^n(X, M) = 0$ for $k > \dim X$. It follows that $\text{cd } \Gamma < \infty$. It remains to apply Lemmas 6.1.B and 6.1.C. $\square$

The last corollary shows the importance of the groups $H^n(\Gamma, \mathbf{Z}[\Gamma])$. Our next goal is to provide a geometric interpretation of these cohomology groups.

Lemma 6.1.E *For any Γ -module M the group $\text{Hom}_\Gamma(M, \mathbf{Z}[\Gamma])$ is naturally isomorphic to the group $\text{Hom}_c(M, \mathbf{Z})$ of all homomorphisms of abelian groups $f : M \rightarrow \mathbf{Z}$ such that for any $m \in M$ the image $f(\gamma m)$ is equal to 0 for almost all $\gamma \in \Gamma$ (i.e., for all γ with the exception of finitely many of them).*

Proof. Any homomorphism $F : M \rightarrow \mathbf{Z}[\Gamma]$ of Γ -modules can be written in the form

$$F(m) = \sum_{\gamma \in \Gamma} f_\gamma(m) \gamma$$

for some homomorphisms $f_\gamma : M \rightarrow \mathbf{Z}$. Clearly, for any m the image $f_\gamma(m)$ is equal to 0 for almost all $\gamma \in \Gamma$ (by the definition of $\mathbf{Z}[\Gamma]$). Since F is a homomorphism of Γ -modules, we have $F(\beta m) = \beta F(m)$ for any $m \in M$, $\beta \in \Gamma$. It follows that

$$\sum_{\gamma \in \Gamma} f_\gamma(\beta m) \gamma = \sum_{\gamma \in \Gamma} f_\gamma(m) \beta \gamma,$$

and hence $f_{\beta\gamma}(\beta m) = f_\gamma(m)$ for all $\beta, \gamma \in \Gamma$ and $m \in M$. In particular, $f_\gamma(m) = f_1(\gamma^{-1}m)$ for all $\gamma \in \Gamma$, $m \in M$, where $1 \in \Gamma$ is the unit element of Γ . It follows that F is determined by f_1 . Moreover, because for any m the image $f_1(\gamma m) = f_{\gamma\gamma^{-1}}(\gamma m) = f_{\gamma^{-1}}(m)$ is equal to 0 for almost all γ , the homomorphism f_1 belongs to $\text{Hom}_c(M, \mathbf{Z})$.

Conversely, for a homomorphism $f_1 : M \rightarrow \mathbf{Z}$ belonging to $\text{Hom}_c(M, \mathbf{Z})$, we can define a homomorphism $F : M \rightarrow \mathbf{Z}[\Gamma]$ of Γ -modules by the formula

$$F(m) = \sum_{\gamma \in \Gamma} f_1(\gamma^{-1}m) \gamma.$$

Thus $F \mapsto f_1$ establishes a natural bijection $\text{Hom}_\Gamma(M, \mathbf{Z}[\Gamma]) \rightarrow \text{Hom}_c(M, \mathbf{Z})$. $\square$

Lemma 6.1.F *If X is a finite CW-complex, then $H^*(X, \mathbf{Z}[\Gamma])$ is naturally isomorphic to the cohomology group with compact support $H_c^*(X^\sim)$ of $X^\sim$.*

Proof. By Lemma 6.1.E,

$$\begin{aligned} H^*(\Gamma, \mathbf{Z}[\Gamma]) &= H^*(X, \mathbf{Z}[\Gamma]) = H^*(\text{Hom}_\Gamma(C_*(X^\sim), \mathbf{Z}[\Gamma])) = \\ &= H^*(\text{Hom}_c(C_*(X^\sim), \mathbf{Z})). \end{aligned}$$

If we use the cellular chains, then $\text{Hom}_c(C_*(X^\sim), \mathbf{Z})$ consists of cochains which are equal to 0 on almost all cells in the preimage of any given cell in X . Because X has only a finite number of cells, these are exactly the cochains equal to 0 on almost all cells in $X^\sim$, i.e., the cochains with compact support. It follows that the last cohomology group is nothing other than the cohomology groups with compact support of $X^\sim$. $\square$

Corollary 6.1.G *If X is a finite CW-complex, then $\text{cd } \Gamma$ is equal to the maximum number n such that $H_c^n(X^\sim, \mathbf{Z}) \neq 0$. $\square$*

Now, it is the time to make a crucial step and apply the Poincaré-Lefschetz duality.

Theorem 6.1.H *Suppose that X is a compact topological manifold of dimension d with boundary (may be, empty) and, simultaneously, is a finite CW-complex. Let m be the minimum number such that $\tilde{H}_m(\partial X^\sim) \neq 0$, where $\tilde{H}_*(\cdot)$, as usual, denotes the reduced homology. Then $\text{cd } \Gamma = d - m - 1$. (If there is no such m , then $\text{cd } \Gamma = 0$.)*

Proof. By the Poincaré-Lefschetz duality $H_c^n(X^\sim) = H_{d-n}(X^\sim, \partial X^\sim)$. Because X is a $K(\Gamma, 1)$ -space, $X^\sim$ is contractible, and hence $H_{d-n}(X^\sim, \partial X^\sim) = \tilde{H}_{d-n-1}(\partial X^\sim)$. It remains to apply Corollary 6.1.G (note that $m = d - n - 1$ if and only if $n = d - m - 1$). $\square$

The assumptions of this theorem are satisfied, for example, when X is a smooth manifold with corners (as in the applications we have in mind) and, hence, can be triangulated.

6.2 Classifying spaces and universal bundles

Here we discuss some auxiliary results about classifying spaces and universal bundles which will be needed in 6.3 and 6.4.

Let G be a topological group acting on a topological space X . Let $EG \rightarrow BG$ be the Milnor universal principal G -bundle. In particular, $EG \rightarrow BG$ is a locally trivial bundle and its total space EG is contractible. We will denote by $EX \rightarrow BG$ the associated bundle with the fiber X . Recall that $EX = EG \times_G X$. A choice of a base point $x \in X$ defines a map $G \rightarrow X$ given by the formula $g \mapsto gx$ and a map $EG \rightarrow EX$ given by the formula $y \mapsto y \times_G x$.

Lemma 6.2.A *If $G \rightarrow X$ is a Serre fibration, then the map $EG \rightarrow EX$ is also a Serre fibration.*

Proof. Locally, over any sufficiently small open set $U \subset BG$, the bundle $EG \rightarrow BG$ is isomorphic to the trivial bundle $U \times G \rightarrow U$, the associated bundle is isomorphic to $U \times X \rightarrow U$ and the map $EG \rightarrow EX$ is the map $U \times G \rightarrow U \times X$ induced by $G \rightarrow X$. If $G \rightarrow X$ is a Serre fibration, then, obviously, $U \times G \rightarrow U \times X$ is a Serre fibration also. It follows that $EG \rightarrow EX$ is a Serre fibration locally (over EX). By a well known theorem (see, for example, [Bre], Theorem VII.6.11), this implies that $EG \rightarrow EX$ is a Serre fibration. $\square$

Lemma 6.2.B *Let H be the stabilizer of the base point x in G . Suppose that the above map $G \rightarrow X$ is a Serre fibration. Then the part*

$$\pi_1(X) \longrightarrow \pi_1(EX) \longrightarrow \pi_1(BG)$$

of the homotopy sequence of the bundle $EX \rightarrow BG$ can be canonically identified with the part

$$\pi_1(X) \longrightarrow \pi_0(H) \longrightarrow \pi_0(G)$$

of the homotopy sequence of the fibration $G \rightarrow X$.

Proof. Let $\pi_1(BG) \rightarrow \pi_0(G)$ be the boundary map from the homotopy sequence of the bundle $EG \rightarrow BG$. Since EG is contractible, it is an isomorphism. By Lemma 6.2.A, the map $EG \rightarrow EX$ is a Serre fibration; obviously, its fiber is H . Let $\pi_1(EX) \rightarrow \pi_0(H)$ be the boundary map from the homotopy sequence of this fibration. Again, since EG is contractible, it is an isomorphism. Direct check shows that the following diagram

$$\begin{array}{ccccc} \pi_1(X) & \longrightarrow & \pi_1(EX) & \longrightarrow & \pi_1(BG) \\ \downarrow & & \downarrow & & \downarrow \\ \pi_1(X) & \longrightarrow & \pi_0(H) & \longrightarrow & \pi_0(G) \end{array}$$

is commutative, and hence provides the required identification. $\square$

If G acts on another topological space X' and a G -equivariant map $X \rightarrow X'$ is given, then there is a natural map $EX \rightarrow EX'$ commuting with the projections $EX \rightarrow BG$, $EX' \rightarrow BG$. If $X \rightarrow X'$ is a (weak) homotopy equivalence, then $EX \rightarrow EX'$ is also a (weak) homotopy equivalence, and we can identify the homotopy sequences of fibrations $EX \rightarrow BG$ and $EX' \rightarrow BG$.

6.3 Mess subgroups

In this section we discuss some subgroups of the mapping class groups discovered by G. Mess [Me]. These subgroups will be our main tool for estimating the virtual cohomological dimension of the mapping class groups from *below* (cf. the proof of Theorem 6.4.A).

Let S_g be a closed orientable surface of genus g , and let $\text{Mod}_g = \text{Mod}_{S_g}$. Let S_g^1 be an orientable surface of genus g with 1 boundary component, and let $\mathcal{M}_g^1 = \mathcal{M}_{S_g^1}$. This section is devoted to some subgroups B_g , B_g^1 of Mod_g , $\mathcal{M}_g^1$ respectively, where $g \geq 2$. They were introduced by G. Mess [Me], and we call them *Mess subgroups*. Although the construction of these involves some choices, it is essentially canonical.

The construction of the Mess subgroups has a recursive character. We start with any subgroup B_2 of Mod_2 generated by Dehn twists about any three pairwise disjoint pairwise nonisotopic circles on S_2 . Thus B_2 is isomorphic to $\mathbf{Z}^3$. Suppose that B_g , $g \geq 2$ is already defined.

We may assume that S_g is obtained from S_g^1 by gluing a disc D^2 to the boundary of S_g^1 . The extension of diffeomorphisms of S_g^1 fixed on ∂S_g^1 by the identity across the disc D^2 defines a homomorphism $\mathcal{M}_g^1 \rightarrow \text{Mod}_g$ as in 2.8. Let B_g^1 be the preimage of B_g under this homomorphism.

In order to define B_{g+1} , consider some embedding $S_g^1 \rightarrow S_{g+1}$ and identify S_g^1 with its image. The extension of diffeomorphisms of S_g^1 fixed on ∂S_g^1 by the identity across the complement of S_g^1 in S_{g+1} defines a homomorphism $i : \mathcal{M}_g^1 \rightarrow \text{Mod}_{g+1}$. Note that the closure of the complement of S_g^1 in S_{g+1} is a torus with one hole. Let us choose some nontrivial circle in this torus with one hole and consider the Dehn twist $t \in \text{Mod}_{g+1}$ about this circle. Let T be the infinite cyclic group generated by t . We define B_{g+1} as the group generated by $i(B_g^1)$ and T .

This completes the construction of the Mess subgroups. A couple of remarks are in order. First, recall (see 2.6) that the restriction of diffeomorphisms of S_g on D^2 defines a fibration $\text{Diff}(S_g) \rightarrow \text{Emb}(D^2, S_g)$. As explained in 2.6, its fiber can be identified with $\text{Diff}(S_g^1 \text{ fix } \partial)$. Since $\text{Diff}(S_g)$ is simply connected (by Theorem 2.7.G) and $\text{Emb}(D^2, S_g)$ is connected, the homotopy sequence of this fibration ends with

$$1 \longrightarrow \pi_1(\text{Emb}(D^2, S_g)) \longrightarrow \pi_0(\text{Diff}(S_g^1 \text{ fix } \partial)) \longrightarrow \pi_0(\text{Diff}(S_g)) \longrightarrow 1,$$

i.e., with

$$1 \longrightarrow \pi_1(\text{Emb}(D^2, S_g)) \longrightarrow \mathcal{M}_g^1 \longrightarrow \text{Mod}_g \longrightarrow 1,$$

where the homomorphism $\mathcal{M}_g^1 \rightarrow \text{Mod}_g$ is exactly the one used above. Hence, we have a short exact sequence

$$1 \longrightarrow \pi_1(\text{Emb}(D^2, S_g)) \longrightarrow B_g^1 \longrightarrow B_g \longrightarrow 1.$$

Since $\text{Emb}(D^2, S_g)$ is weakly homotopy equivalent to the unit tangent bundle $UT(S_g)$ (by Theorem 2.6.C), we can also write this short exact sequence as

$$1 \longrightarrow \pi_1(UT(S_g)) \longrightarrow B_g^1 \longrightarrow B_g \longrightarrow 1.$$

Note also that the homomorphism $i : \mathcal{M}_g^1 \rightarrow \text{Mod}_{g+1}$ is injective by Theorem 2.7.I and that the twist t obviously commutes with the image of i . It follows that B_{g+1} is isomorphic to $B_g^1 \times \mathbf{Z}$. Now we come to the main property of the Mess subgroups (cf. [Me], Proposition 1).

Theorem 6.3.A *There exists a closed topological manifold of dimension $4g-5$ which is a $K(B_g, 1)$ -space. Similarly, there exists a closed topological manifold of dimension $4g-2$ which is a $K(B_g^1, 1)$ -space. These topological manifolds can be chosen to be simultaneously CW-complexes.*

Proof. The idea is to start with the 3-dimensional torus, which can be taken as a $K(B_2, 1)$ -space and to construct a $K(B_g^1, 1)$ -space as the total space of a bundle with the fiber $UT(S_g)$ over an already constructed $K(B_g, 1)$ -space, and then get a $K(B_{g+1}, 1)$ -space by multiplying this total space by the circle S^1 . We would like this bundle to realize the sequence $\pi_1(UT(S_g)) \rightarrow B_g^1 \rightarrow B_g$ as the π_1 -part of its homotopy sequence. Note that $UT(S_g)$ is a $K(\pi, 1)$ -space, because it is the total space of a circle bundle over S_g , which is a $K(\pi, 1)$ -space. Suppose that a $K(B_g, 1)$ -space K_g is already constructed.

Let $G = \text{Diff}(S_g)$, $X = \text{Emb}(D^2, S_g)$ with the obvious action of $\text{Diff}(S_g)$, and let X' be the unit tangent bundle $UT(S_g)$ with the following action of $\text{Diff}(S_g)$: a diffeomorphism acts on a unit vector by its differential and the subsequent normalization. Let $X \rightarrow X'$ be the map $u : \text{Emb}(D^2, S_g) \rightarrow UT(S_g)$ from 2.6. It is obviously G -equivariant. By Theorem 2.6.C, it is a weak homotopy equivalence. Hence the induced map $EX \rightarrow EX'$ is also a weak homotopy equivalence. It follows that we can naturally identify the homotopy sequences of bundles $EX \rightarrow BG$ and $EX' \rightarrow BG$.

Now, since components of $\text{Diff}(S_g)$ are weakly contractible by Theorem 2.7.G, the classifying space $BG = B\text{Diff}(S_g)$ is a $K(\text{Mod}_g, 1)$ -space. Hence, the inclusion $B_g \rightarrow \text{Mod}_g$ leads to a map $K_g \rightarrow BG$ from our $K(B_g, 1)$ -space K_g to $BG = B\text{Diff}(S_g)$. Consider the bundle $K_g^1 \rightarrow K_g$ with the fiber $X' = UT(S_g)$ induced from the bundle $EX' \rightarrow BG$ by this map. Since both $UT(S_g)$ and K_g are $K(\pi, 1)$ -spaces, the homotopy sequence of the induced bundle shows that K_g^1 is also a $K(\pi, 1)$ -space. In addition, this homotopy sequence ends with the short exact sequence

$$1 \longrightarrow \pi_1(UT(S_g)) \longrightarrow \pi_1(K_g^1) \longrightarrow \pi_1(K_g) \longrightarrow 1.$$

This short exact sequence naturally maps to the end

$$1 \longrightarrow \pi_1(UT(S_g)) \longrightarrow \pi_1(EX') \longrightarrow \pi_1(BG) \longrightarrow 1$$

(recall $X' = UT(S_g)$) of the homotopy sequence of the bundle $EX' \rightarrow BG$. Moreover, $\pi_1(K_g^1)$ is obviously isomorphic to the preimage of $\pi_1(K_g) = B_g$ under the map $\pi_1(EX') \rightarrow \pi_1(BG)$. By the previous paragraph, this short exact sequence is naturally isomorphic to the end

$$1 \longrightarrow \pi_1(X) \longrightarrow \pi_1(EX) \longrightarrow \pi_1(BG) \longrightarrow 1$$

of the homotopy sequence of the bundle $EX \rightarrow BG$. Now, consider the fibration $G = \text{Diff}(S_g) \rightarrow X = \text{Emb}(D^2, S_g)$. Clearly, $H = \text{Diff}(S_g^1 \text{ fix } \partial)$ is the stabilizer of the embedding $D^2 \rightarrow S_g$ and, hence, is the fiber of this fibration. By Lemma 6.2.B, the last short exact sequence is naturally isomorphic to the end

$$1 \longrightarrow \pi_1(X) \longrightarrow \pi_0(H) \longrightarrow \pi_0(G) \longrightarrow 1$$

of the homotopy sequence of the fibration $G \rightarrow X$. But this short exact sequence is nothing other than our sequence

$$1 \longrightarrow \pi_1(\text{Emb}(D^2, S_g)) \longrightarrow \mathcal{M}_g^1 \longrightarrow \text{Mod}_g \longrightarrow 1.$$

It follows that $\pi_1(K_g^1)$ is isomorphic to the preimage of B_g under the map $i : \mathcal{M}_g^1 \rightarrow \text{Mod}_g$, i.e. to B_g^1 . Hence, we can take K_g^1 as our $K(B_g^1, 1)$ -space, and then put $K_{g+1} = K_g^1 \times S^1$.

This completes our construction of $K(B_g, 1)$ - and $K(B_g^1, 1)$ -spaces. Since they are constructed by consecutively taking the total spaces of locally trivial bundles with 3-manifold fibers and multiplying by S^1 (and starting with the 3-dimensional torus), these spaces are, obviously, topological manifolds. With some additional care, one can turn these spaces into CW-complexes (or even smooth manifolds, and then triangulate them). This proves all statements of the theorem with the exception of the correctness of the dimension values.

Clearly, $\dim K_g^1 = \dim K_g + 3$, $\dim K_{g+1} = \dim K_g^1 + 1$ and $\dim K_2 = 3$. Therefore, $\dim K_g = 4g - 5$ and $\dim K_g^1 = 4g - 2$. This completes the proof. $\square$

Corollary 6.3.B $\text{cd } B_g = 4g - 5$, $\text{cd } B_g^1 = 4g - 2$.

Proof. Combine Theorem 6.3.A with Theorem 6.1.A. $\square$

6.4 Virtual cohomological dimension

One of the main properties of the cohomological dimension is the inequality $\text{cd } \Gamma' \leq \text{cd } \Gamma$, which holds for $\Gamma' \subset \Gamma$. We refer to [Bro1] for a proof and only note that it uses in a crucial way the fact that all *twisted* coefficients are allowed; cf. [Bro1], Chapter VIII, Proposition 2.4. It is well known that $\text{cd } A = \infty$ for a finite cyclic group A ; it follows that any group with torsion has infinite cohomological dimension. In order to get an interesting invariant for groups with torsion, we modify the definition in the way suggested by J.-P. Serre [Ser2].

If Γ is *virtually torsion free*, i.e. contains a subgroup of finite index which is torsion free, then, by a fundamental theorem of J.-P. Serre [Ser2] (cf. also [Bro1], Chapter VIII, Theorem 3.1 or [BeK], Section 3) all torsion free subgroups of finite index of Γ have the same cohomological dimension. The common value of the cohomology dimension of such subgroups is called the *virtual cohomological dimension* of Γ and is denoted by $\text{vcd } \Gamma$.

Since Mod_S always has torsion, $\text{cd } \text{Mod}_S = \infty$ and the interesting invariant is $\text{vcd } \text{Mod}_S$. The results of 6.1 and 6.3 allow us to compute it. We start with closed surfaces.

Theorem 6.4.A *If S is a closed surface of genus $g \geq 2$, then $\text{vcd } \text{Mod}_S = 4g - 5$. If S is a sphere or a torus, then $\text{vcd } \text{Mod}_S = 0$ or 1 respectively.*

Proof. If S is a sphere, then $\text{Mod}_S = 1$ and the result is trivial. If S is a torus, then Mod_S is isomorphic to $SL_2(\mathbf{Z})$. The last group, as is well known, contains a finitely generated free group as a subgroup of finite index. It is easy to see that the cohomological dimension of a free group is 1. Hence, $\text{vcd Mod}_S = 1$ if S is a torus.

In the rest of the proof we assume that S is a closed surface of genus $g \geq 2$. Consider a subgroup Γ of finite index in Mod_S acting freely on the Teichmüller space T_S (such subgroups do exist by 5.4). It is sufficient to show that $\text{cd } \Gamma = 4g - 5$ for any such Γ (we do not need to know in advance that such a group Γ is torsion free, since $\text{cd } \Gamma < \infty$ implies this).

Consider the intersection $\Gamma' = \Gamma \cap B_g$ of Γ with the Mess subgroup B_g . It is obviously of finite index in B_g . Hence, we can take as a $K(\Gamma', 1)$ -space a finite sheeted covering space of the $K(B_g, 1)$ -space provided by Theorem 6.3.A. Clearly, this $K(\Gamma', 1)$ -space is a closed manifold of dimension $4g - 5$. By Theorem 6.1.A, $\text{cd } \Gamma' = 4g - 5$. It follows that $\text{cd } \Gamma \geq 4g - 5$.

In order to prove the opposite inequality, consider the manifold with corners $T_S(\varepsilon)$ from 5.4 and put $X = T_S(\varepsilon)/\Gamma$. Since the action of Γ on T_S is free, X is also a manifold with corners, and since $T_S(\varepsilon)$ is contractible (cf. 5.4), we can take $T_S(\varepsilon)$ as the universal cover $X^\sim$ of X . By 5.4, the boundary $\partial T_S(\varepsilon)$ is homotopy equivalent to (the geometric realization of) the complex of curves $C(S)$. Hence, Theorem 3.2.C implies that $\partial X^\sim = \partial T_S(\varepsilon)$ is $2g - 3$ -connected. Now we apply Theorem 6.1.H. Since $\partial X^\sim$ is $2g - 3$ -connected, the number m from this theorem is $\geq 2g - 2$ (such an m exist, because we already saw that $\text{cd } \Gamma \neq 0$). Finally, $d = \dim X^\sim = \dim T_S(\varepsilon) = \dim T_S = 6g - 6$, and hence Theorem 6.1.H implies that $\text{cd } \Gamma \leq (6g - 6) - (2g - 2) - 1 = 4g - 5$. This completes the proof. $\square$

Theorem 6.4.B *If S is a surface of genus $g \geq 2$ with $b \geq 1$ boundary components, then $\text{vcd Mod}_S = 4g - 4 + b$.*

Proof. Let $S_{g,b}$ be a surface of genus g with b boundary components, and let $\text{Mod}_{g,b} = \text{Mod}_{S_{g,b}}$. In order to compute $\text{vcd Mod}_{g,b}$, we will construct a subgroup $\Gamma_{g,b}$ of finite index in $\text{Mod}_{g,b}$ and a $K(\Gamma_{g,b}, 1)$ -space. Since the dimension of this space will be $< \infty$, and hence $\text{cd } \Gamma_{g,b}$ will be $< \infty$, the subgroup $\Gamma_{g,b}$ will be automatically torsion free, and therefore we will have $\text{vcd Mod}_{g,b} = \text{cd } \Gamma_{g,b}$. The cohomological dimension $\text{cd } \Gamma_{g,b}$ will be computed with the help of Theorem 6.1.H using our $K(\Gamma_{g,b}, 1)$ -spaces.

The construction of subgroups $\Gamma_{g,b}$ has a recursive character. We start with any subgroup $\Gamma_{g,0}$ of finite index in $\text{Mod}_{g,0}$ acting freely on the Teichmüller space $T_{S_{g,0}}$. Let $L_{g,0} = T_{S_{g,0}}(\varepsilon)/\Gamma_{g,0}$, where $T_{S_{g,0}}(\varepsilon)$ is the manifold with corners from 5.4; $L_{g,0}$ will be our $K(\Gamma_{g,0}, 1)$ -space. Suppose that a subgroup $\Gamma_{g,b}$ of finite index in $\text{Mod}_{g,b}$ and a $K(\Gamma_{g,b}, 1)$ -space $L_{g,b}$ are already constructed.

Let us fix a point x in the interior of $S_{g,b}$. Let $* = \{x\}$; it is a 0-dimensional submanifold of $S_{g,b}$. Clearly, we can identify $\text{Emb}^\diamond(*, S_{g,b})$ with the interior

$\text{int } S_{g,b}$ of $S_{g,b}$ (recall that the embeddings in $\text{Emb}^\diamond(*, S_{g,b})$ can be extended to diffeomorphisms of $S_{g,b}$; see 2.6).

Now, consider the group $\text{Mod}_{g,(b,1)} = \pi_0(\text{Diff}(S_{g,b} \text{ fix } *))$. In view of the last paragraph of 2.8, the group $\text{Mod}_{g,(b,1)}$ is isomorphic to the subgroup of the group $\text{Mod}_{g,b+1}$ consisting of the isotopy classes of diffeomorphisms $S_{g,b+1} \rightarrow S_{g,b+1}$ preserving (setwise) one fixed boundary component. In particular, $\text{Mod}_{g,(b,1)}$ is isomorphic to a subgroup of finite index (actually, of index $b+1$) of $\text{Mod}_{g,b+1}$.

Let $G = \text{Diff}(S_{g,b})$ and let $X = \text{int } S_{g,b}$ with the obvious action of G . The map $G \rightarrow X$ given by $g \mapsto gx$ is nothing but the evaluation of diffeomorphisms at x ; it can be identified with the map $\text{Diff}(S_{g,b}) \rightarrow \text{Emb}^\diamond(*, S_{g,b})$ from 2.6. In particular, Theorem 2.6.A implies that $G \rightarrow X$ is a Serre fibration. The stabilizer H of the point $x \in X$ (i.e., the fiber of $G \rightarrow X$) is equal to $\text{Diff}(S_{g,b} \text{ fix } *)$. The homotopy sequence of this fibration ends with

$$1 \rightarrow \pi_1(\text{int } S_{g,b}) \rightarrow \pi_0(\text{Diff}(S_{g,b} \text{ fix } *)) \rightarrow \pi_0(\text{Diff}(S_{g,b})) \rightarrow 1,$$

i.e. with

$$1 \rightarrow \pi_1(\text{Int } S_{g,b}) \rightarrow \text{Mod}_{g,(b,1)} \rightarrow \text{Mod}_{g,b} \rightarrow 1.$$

By Lemma 6.2.B, this short exact sequence can be naturally identified with

$$1 \rightarrow \pi_1(X) \rightarrow \pi_1(EX) \rightarrow \pi_1(BG) \rightarrow 1$$

(where $X = \text{int } S_{g,b}$).

Next, we would like to replace the open surface $\text{int } S_{g,b}$ by the compact surface $S_{g,b}$. Let X' be the surface $S_{g,b}$ with the natural action of $G = \text{Diff}(S_{g,b})$. The inclusion $X \rightarrow X'$ is, obviously, G -equivariant and is a homotopy equivalence. In view of the remarks at the end of 6.2, we can identify the last exact sequence with

$$1 \rightarrow \pi_1(X') \rightarrow \pi_1(EX') \rightarrow \pi_1(BG) \rightarrow 1$$

(where $X' = S_{g,b}$). We can write this short exact sequence more explicitly as

$$1 \rightarrow \pi_1(S_{g,b}) \rightarrow \pi_1(ES_{g,b}) \rightarrow \pi_1(B\text{Diff}(S_{g,b})) \rightarrow 1.$$

Now, as in the proof of Theorem 6.3.A, we use the fact that $B\text{Diff}(S_{g,b})$ is a $K(\text{Mod}_{g,b}, 1)$ -space, and hence the inclusion $\Gamma_{g,b} \rightarrow \text{Mod}_{g,b}$ leads to a map $L_{g,b} \rightarrow \text{Diff}(S_{g,b})$ from our $K(\Gamma_{g,b}, 1)$ -space $L_{g,b}$ to $B\text{Diff}(S_{g,b})$. Consider the bundle $L_{g,b+1} \rightarrow L_{g,b}$ with the fiber $S_{g,b}$ induced from the bundle $EX' = ES_{g,b} \rightarrow BG = B\text{Diff}(S_{g,b})$ by this map. Since both $S_{g,b}$ and $L_{g,b}$ are $K(\pi, 1)$ -spaces, the homotopy sequence of this bundle shows that $L_{g,b+1}$ is also a $K(\pi, 1)$ -space. In addition, this homotopy sequence ends with the short exact sequence

$$1 \rightarrow \pi_1(S_{g,b}) \rightarrow \pi_1(L_{g,b+1}) \rightarrow \pi_1(L_{g,b}) \rightarrow 1.$$

The latter naturally maps to the last short exact sequence of the previous paragraph, and a trivial diagram chase shows that $\pi_1(L_{g,b+1})$ is isomorphic to the

preimage of $\Gamma_{g,b}$ under the map $\pi_1(ES_{g,b}) \rightarrow \pi_1(B\text{Diff}(S_{g,b}))$. In view of the above, this preimage is naturally isomorphic to the preimage of $\Gamma_{g,b}$ under the map $\text{Mod}_{g,(b,1)} \rightarrow \text{Mod}_{g,b}$. We define $\Gamma_{g,b+1}$ to be equal to this preimage (obviously, it is of finite index in $\text{Mod}_{g,b+1}$) and take $L_{g,b+1}$ as our $K(\Gamma_{g,b+1}, 1)$ -space.

This completes the construction of our subgroups $\Gamma_{g,b}$ and $K(\Gamma_{g,b}, 1)$ -spaces $L_{g,b}$. Since the latter are constructed by consecutively taking the total spaces of locally trivial bundles with surface (in general, with boundary) fibers starting with a manifold with corners, the spaces $L_{g,b}$ are topological manifolds with boundary and we also can turn them into CW-complexes (compare the proof of Theorem 6.3.A). Obviously, $\dim L_{g,b+1} = \dim L_{g,b} + 2$ for all b .

In order to apply Theorem 6.1.H, we need to analyze the universal covers $L_{g,b}^\sim$ of manifolds $L_{g,b}$ and their boundaries $\partial L_{g,b}^\sim$. If the universal cover $L_{g,b}^\sim$ is already constructed, we can construct $L_{g,b+1}^\sim$ in two steps. First, take the bundle $M_{g,b} \rightarrow L_{g,b}^\sim$ with fiber $S_{g,b}$ induced from the bundle $L_{g,b+1} \rightarrow L_{g,b}$ by the map $L_{g,b}^\sim \rightarrow L_{g,b}$. Since the base $L_{g,b}^\sim$ of this bundle is contractible, this bundle is actually trivial, and hence $M_{g,b}$ is homeomorphic to $S_{g,b} \times L_{g,b}^\sim$. As the second step, we take the universal cover of $M_{g,b}$. This leads to $S_{g,b}^\sim \times L_{g,b}^\sim$, where $S_{g,b}^\sim$ is the universal cover of the surface $S_{g,b}$. Thus, $L_{g,b+1}^\sim$ is homeomorphic to $S_{g,b}^\sim \times L_{g,b}^\sim$, and hence the boundary $\partial L_{g,b+1}^\sim$ is homeomorphic to

$$\partial S_{g,b}^\sim \times L_{g,b}^\sim \cup S_{g,b}^\sim \times \partial L_{g,b}^\sim.$$

Now, there are two different cases to consider: $b = 0$ and $b \geq 1$.

In the first case, $S_{g,0}^\sim$ has no boundary and $\partial L_{g,1}^\sim$ is homeomorphic to $S_{g,0}^\sim \times \partial L_{g,0}^\sim$. Since $S_{g,0}^\sim$ is contractible, it follows that $\partial L_{g,1}^\sim$ is homotopy equivalent to $\partial L_{g,0}^\sim$.

In the second case, $\partial S_{g,b}^\sim$ is non empty and actually consists of a countable (infinite) number of components homeomorphic to the real line. Since $S_{g,b}^\sim$ is contractible, it follows that the pair $(S_{g,b}^\sim, \partial S_{g,b}^\sim)$ is homotopy equivalent to the pair (CZ, Z) , where Z is a discrete space consisting of a countable number of points and CZ is the cone of Z . The last pair, in turn, is homotopy equivalent to the pair $(\mathbf{R}, \mathbf{Z})$. It follows that $\partial L_{g,b+1}^\sim$ is homotopy equivalent to

$$\mathbf{Z} \times L_{g,b}^\sim \cup \mathbf{R} \times \partial L_{g,b}^\sim.$$

Since the universal covers $L_{g,b}^\sim$ are contractible, the last space is homotopy equivalent to a bouquet of an infinite number of suspensions $\Sigma \partial L_{g,b}^\sim$.

Now, let $d(b) = \dim L_{g,b}$ and let $m(b)$ be the minimum number m such that $\tilde{H}_m(\partial L_{g,b}^\sim) \neq 0$ (we assume that the genus g is fixed). As we saw above, $d(b+1) = d(b) + 2$ for all b . The results of the two previous paragraphs imply that $m(1) = m(0)$ and $m(b+1) = m(b) + 1$ for $b \geq 1$. Combining these remarks with Theorem 6.1.H, we see that $\text{cd } \Gamma_{g,1} = \text{cd } \Gamma_{g,0} + 2$ and $\text{cd } \Gamma_{g,b+1} = \text{cd } \Gamma_{g,b} + 1$ for $b \geq 1$. Since $\text{cd } \Gamma_{g,0} = \text{vcd } \text{Mod}_{g,0} = 4g - 5$ by Theorem 6.4.A, the theorem follows. $\square$

Theorem 6.4.C *If S is a sphere with b holes, then $\mathrm{vcd} \mathrm{Mod}_S = 0$ for $b \leq 3$ and $\mathrm{vcd} \mathrm{Mod}_S = b - 3$ for $b \geq 3$. If S is a torus with b holes, then $\mathrm{vcd} \mathrm{Mod}_S = 1$ for $b = 0$ and $\mathrm{vcd} \mathrm{Mod}_S = b$ for $b \geq 1$.*

Proof. The proof is similar to the proof of the previous theorem. Since for a sphere S with ≤ 2 holes or a torus S without holes the components of $\mathrm{Diff}(S)$ are not contractible (and hence $B\mathrm{Diff}(S)$ is not a $K(\mathrm{Mod}_S, 1)$ -space), the inductive argument should start with a sphere with 3 holes and a torus with 1 hole. We leave the details to the reader. $\square$

The first nontrivial estimate $\mathrm{vcd} \mathrm{Mod}_S \leq 6g - 9$ of $\mathrm{vcd} \mathrm{Mod}_S$ for a closed surface of genus g was proved in [I2], [I3] (the trivial estimate is $\mathrm{vcd} \mathrm{Mod}_S \leq \dim T_S = 6g - 6$). It was deduced from the simply connectivity of $C(S)$, which, in turn, was deduced from the simply connectivity of the Hatcher-Thurston complex [HatT]. Theorems 6.4.A–6.4.C are due to J. Harer [Har2], [Har4]. Another proof was provided by the author [I5]. The paper [I5] contains also a computation of the virtual cohomological dimension of the mapping class groups of nonorientable surfaces. Both papers [Har4] and [I5] actually prove a stronger result than Theorems 6.4.A–6.4.C. Namely, they prove that Mod_S contains a subgroup of finite index which is a *group with duality* of some dimension in the sense of Bieri–Eckmann [BieE] and compute this dimension. The *dimension* of a group with duality is built into its definition (like the dimension of a smooth manifold or a Poincaré space); it is always $< \infty$ and turns out to be equal to the cohomological dimension. The last property allows the use of this result for the computation of $\mathrm{vcd} \mathrm{Mod}_S$. We refer to [I5], Section 6 for an expository account of this stronger result, and to [Bro1], Chapter VIII, or [BeK] for more details about groups with duality. The proof of the lower estimate for the cohomological dimension in our proof of Theorem 6.4.A follows the ideas of G. Mess [Me], while the proof of the upper estimate (which turned out to be equal to the lower one) follows the standard approach of [I2], [I3], [Har2], [Har4], [I5]. The above proof of Theorem 6.4.B seems to be new. Both of these proofs avoid the full power of the theory of groups with duality (a fragment of this theory was included in 6.1) and also do not use the complete description of the homotopy type of $C(S)$ (they use only Theorem 3.2.C, and only this for closed surfaces).

6.5 Homology stability

In its simplest form, the homology stability for the mapping class groups asserts that the homology group $H_n(\mathrm{Mod}_S)$ (with trivial coefficients $\mathbf{Z}$) does not depend on S provided S is closed and the genus g of S is sufficiently large compared to n . The implied restriction on g is usually called *the domain of stability*. The first result of this sort, with the domain of stability $g \geq 3n + 1$, was proved by J. Harer [Har2], [Har3]. Such results are similar to, and motivated by, the classical homology stability theorems in the algebraic K -theory. The simplest

of those asserts that the homology group $H_n(SL_g(\mathbf{Z}))$ does not depend on g provided g is sufficiently large compared to n . The more advanced versions of the homology stability theorems in algebraic K -theory cover some other natural series of groups, like the special linear groups over rings or the symplectic groups, and usually provide an explicit domain of stability (the role of g is played by some natural parameter). A common feature of the proofs of the homology stability theorems in algebraic K -theory is the need to include into the picture a wider class of groups than the original infinite series: this is required by the way the induction is arranged in the proofs. The situation for the mapping class groups is similar: one needs to consider the groups $\mathcal{M}_S$ for non-closed surfaces S , in addition to the groups Mod_S for closed S . The groups $\mathcal{M}_S$ have a natural advantage over our usual version Mod_S , namely the existence of a natural map $\mathcal{M}_R \rightarrow \mathcal{M}_S$ when R is a subsurface of S . This map is given by the extension of diffeomorphisms of R fixed on ∂R by the identity to diffeomorphisms of S . In fact, we can completely switch to the groups $\mathcal{M}_S$ in this context, because $\mathcal{M}_S = \text{Mod}_S$ for closed S (and the homology stability does not hold for the groups Mod_S for non-closed surfaces S ; for the second homology group this follows from Theorem 6.6.C).

Theorem 6.5.A *Let R be a connected subsurface of a connected surface S . Let g_R be the genus of R . The map $H_m(\mathcal{M}_R) \rightarrow H_m(\mathcal{M}_S)$ induced by the natural map $\mathcal{M}_R \rightarrow \mathcal{M}_S$ is an epimorphism if*

$$g_R \geq 2n + 1$$

and is an isomorphism if

$$g_R \geq 2n + 2.$$

If, in addition, S is not closed, then the map $H_m(\mathcal{M}_R) \rightarrow H_m(\mathcal{M}_S)$ is an epimorphism if

$$g_R \geq 2n$$

and is an isomorphism if

$$g_R \geq 2n + 1.$$

The proof of Theorem 6.5.A is fairly technical. In the outline, it follows the Maazen–van der Kallen [vdKal] approach to the homology stability theorems in algebraic K -theory. On the geometric side, one of the basic tools is the action of the groups $\mathcal{M}_S$ on some versions of the complexes of curves $\mathcal{C}(S)$. The higher connectivity of these versions plays a crucial role. On the algebraic side, the central role is played by some spectral sequences associated with an action of a group on a simplicial complex and a carefully arranged induction on the genus and the number of boundary components of S . For an introduction to the ideas of the proof of Theorem 6.5.A, we refer to [I5], Section 7 and to [I16], Section 1. The details omitted in [I5] are provided in [I11]. The case of closed S requires an additional argument and is dealt with in [I16].

Corollary 6.5.B *If S is a closed surface of genus $\geq 2n + 2$, then up to isomorphism the homology group $H_n(\text{Mod}_S)$ does not depend on S .*

Proof. First, recall that $\text{Mod}_S = \mathcal{M}_S$ for closed S .

If S, S' are two closed surfaces of different genera, there is no natural homomorphism between the groups $\mathcal{M}_S, \mathcal{M}_{S'}$. In order to compare the homology of $\mathcal{M}_S, \mathcal{M}_{S'}$, we can choose two diffeomorphic subsurfaces R, R' of S, S' respectively and apply Theorem 6.5.A to the maps $\mathcal{M}_R \rightarrow \mathcal{M}_S, \mathcal{M}_{R'} \rightarrow \mathcal{M}_{S'}$. If we choose subsurfaces R, R' such that their genus is equal to the lesser of the genera of S, S' , the corollary will follow immediately. $\square$

The domain of stability of Theorem 6.5.A and Corollary 6.5.B is substantially better than the J. Harer's original domain of stability $g \geq 3n + 1$ from [Har3]. For non-closed surfaces, the domain of stability $g \geq 2n + 1$ of Theorem 6.5.A is exactly the same as the best known domain of stability for $H_n(SL_g(\mathbf{Z}))$. On the other hand, if we consider homology with rational (as opposed to the integer) coefficients, then the domain of stability can be further improved. (A similar phenomenon is well known in the algebraic K -theory.)

Theorem 6.5.C *Let R be a connected subsurface of a connected surface S with non empty boundary. Let g_R be the genus of R . The map $H_m(\mathcal{M}_R, \mathbf{Q}) \rightarrow H_m(\mathcal{M}_S, \mathbf{Q})$ induced by the natural map $\mathcal{M}_R \rightarrow \mathcal{M}_S$ is an isomorphism if*

$$g_R \geq \frac{3n}{2}.$$

For odd n , this map is surjective if

$$g_R \geq \frac{3n}{2} - 2.$$

This theorem is due to J. Harer [Har10], who also proved that the domain of stability of this theorem cannot be improved. The question about the exact domain of stability for integral coefficients remains open.

Finally, we mention two other stability theorems. In [Har6], J. Harer proved a homology stability theorem for subgroups of the mapping class groups consisting of the isotopy classes of diffeomorphisms preserving some spin structure on the surface (these subgroups obviously have finite index in the corresponding mapping class groups). The main result of [I16] is a homology stability theorem for so-called *twisted* coefficients, i.e. for coefficients in a non-trivial module. Actually, more important than the fact that the coefficients are twisted is that the natural examples of twisted coefficients substantially depend on the surface in question, i.e. are not *constant*. As the simplest (and typical) example one may consider $H_n(\mathcal{M}_S, H_1(S))$, where $\mathcal{M}_S$ acts on $H_1(S)$ in the natural way. In this special case, we have the following result.

Theorem 6.5.D *Let R be a connected subsurface of a connected surface S . Let g_R be the genus of R . Suppose that both R and S have exactly 1 boundary component. Then the map $H_m(\mathcal{M}_R, H_1(R)) \rightarrow H_m(\mathcal{M}_S, H_1(S))$ induced by the natural maps $\mathcal{M}_R \rightarrow \mathcal{M}_S$, $H_1(R) \rightarrow H_1(S)$ is an epimorphism if*

$$g_R \geq 2n + 1,$$

and is an isomorphism if

$$g_R \geq 2n + 3.$$

This is a special case of Corollary 4.9 from [I16] (the *coefficient system* $S \mapsto H_1(S)$ is of *degree* 1 in the sense of [I16]). E. Looijenga [Loo3] provided a new proof of a special case of the stability theorem for twisted coefficients of [I16]. Namely, he considers only coefficients which factor through the natural representation $\mathcal{M}_S \rightarrow Sp_g(\mathbf{Z})$, where g is the genus of S , and which are *rational* vector spaces. In this situation he extends the results of [I16] to closed surfaces and, more importantly, explicitly computes the stable homology (i.e., the homology in the domain of stability) with such twisted coefficients in terms of the stable homology with trivial coefficients $\mathbf{Q}$. Looijenga's methods are largely algebro-geometric, but he uses also the homology stability theorem for *constant* coefficients.

Surprisingly, homology stability with twisted coefficients does not hold for closed surfaces, even for the first homology group, as the following theorem implies.

Theorem 6.5.E *If S is a closed surface of genus $g \geq 2$, then $H_1(\mathcal{M}_S, H_1(S)) = \mathbf{Z}/(2g - 2)\mathbf{Z}$.*

This result is due to Sh. Morita [Mor4] (cf. [Mor4], Corollary 5.4).

6.6 The low-dimensional homology groups

We start by computing the first homology group of PMod_S .

Lemma 6.6.A *$H_1(\text{PMod}_S)$ depends only on genus of S .*

Proof. It is sufficient to show that $H_1(\text{PMod}_S)$ does not change if we make a hole in S . So, let R be the result of removing from S the interior of a disc embedded in S . Consider the exact sequence

$$\pi_1(S) \rightarrow \text{PMod}_R \rightarrow \text{PMod}_S \rightarrow 1$$

from 2.8. If $a \in \pi_1(S)$ is the homotopy class of an embedded loop in S , then the image of a in PMod_R has the form $t_\alpha t_\beta^{-1}$ for some circles α, β in R . Moreover, if a is the homotopy class of an embedded non separating loop, then both circles

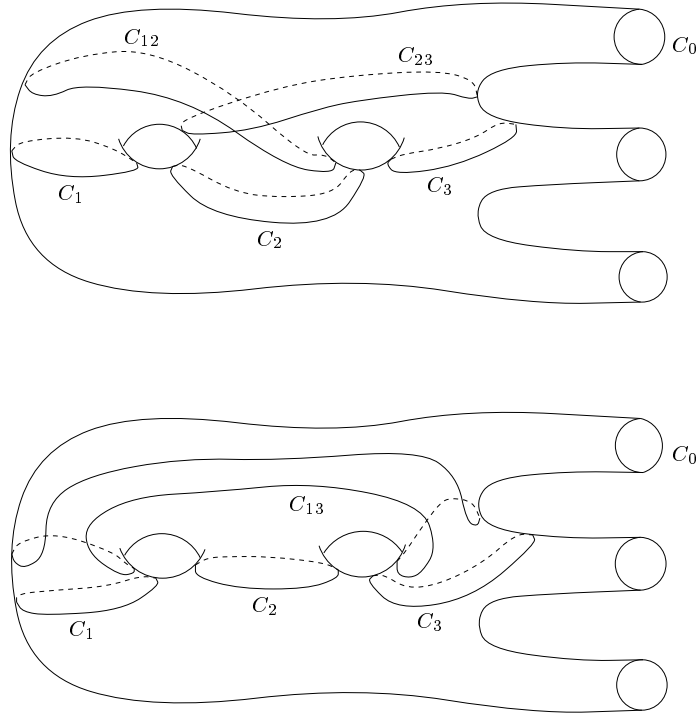


Fig. 6.1

α and β are non separating; see Lemma 4.1.I. It follows that t_α is conjugate to t_β , and hence their images in $H_1(\text{PMod}_R)$ are equal. Therefore, the image of a in $H_1(\text{PMod}_R)$ is 0. Since $\pi_1(S)$ is generated by the homotopy classes of embedded non separating loops (if S has non empty boundary, then some of the standard generators are separating, but they can easily be replaced by two non separating generators), the image of $\pi_1(S)$ in $H_1(\text{PMod}_R)$ is 0. Using the above exact sequence, we see that $H_1(\text{PMod}_R)$ is isomorphic to $H_1(\text{PMod}_S)$. $\square$

Theorem 6.6.B *Let S be a surface of genus g . The first homology group $H_1(\text{PMod}_S)$ is equal to $\mathbf{Z}/12\mathbf{Z}$ if $g = 1$, to $\mathbf{Z}/10\mathbf{Z}$ if $g = 2$, and to 0 if $g \geq 3$.*

Proof. If S is a closed surface of genus 1, (i.e., a torus), then PMod_S is isomorphic to $SL_2(\mathbf{Z})$ and the result is well known (to algebraists, at least). If S is a closed surface of genus 2, then, probably, the best way to compute $H_1(\text{PMod}_S)$ is to use the presentation of Mod_S due to Bergau-Mennicke [BeM] and Birman-Hilden [BirH]; cf. end of 4.3. Given the presentation, the computation of $H_1(\text{PMod}_S)$ is routine and can be left to the reader. This proves the

theorem for closed surfaces of genus 1 and 2, and then Lemma 6.6.A implies it for all surfaces of genus 1 and 2.

In the generic case $g \geq 3$ we actually do not need Lemma 6.6.A. In this case, we can embed the sphere with 4 holes S_0^4 from Lemma 4.1.H in S in such a way that all circles C_i , $0 \leq i \leq 3$ and C_{ij} , $1 \leq i < j \leq 3$ will be non separating in S ; see Fig. 6.1. For such an embedding all Dehn twists t_i , t_{ij} from the lantern relation are conjugate, and hence represent the same element of $H_1(\text{PMod}_S)$. The lantern relation $t_0 t_1 t_2 t_3 = t_{12} t_{13} t_{23}$ implies that this element is equal to 0. On the other hand, by Theorem 4.2.C, PMod_S is generated by Dehn twists about non separating circles. All such Dehn twists are conjugate to t_0 , and hence represent 0 in $H_1(\text{Mod}_S)$. It follows that $H_1(\text{PMod}_S) = 0$. $\square$

The path to this nice argument for the genus ≥ 3 case, which is due to J. Harer [Har1], was fairly long. First, D. Mumford [Mumf1] proved that $H_1(\text{Mod}_S)$ is cyclic of order dividing 10 for closed surfaces S of genus ≥ 2 . Then J. Birman [Bir2] computed $H_1(\text{Mod}_S)$ for closed surfaces S of genus 1 and 2, and proved that for genus ≥ 3 the order of this group is 1 or 2. Finally, J. Powell [Pow] proved that this order is actually 1 by means of some complicated computations.

Theorem 6.6.C *Let S be a surface of genus g with b boundary components. If $g \geq 5$, then $H_2(\text{PMod}_S) = \mathbf{Z}^{b+1}$.*

This result is due to J. Harer [Har1]. One should warn the reader that [Har1] contains some mistakes, even in the statement of the main theorem (it is asserted in [Har1] that $H_2(\text{PMod}_S) = \mathbf{Z} \oplus \mathbf{Z}/(2g-2)\mathbf{Z}$ for closed surfaces S of genus g). But the general agreement seems to be that Theorem 6.6.C is correct. The main tool in [Har1] is an action of PMod_S on a modification of the Hatcher-Thurston complex [HatT] and the spectral sequences associated with this action. Later on, Harer [Har7] suggested an alternative method to compute the second homology group with *rational* coefficients, and also extended this new method to the third homology group. The approach of [Har7] is close in spirit to the proofs of the homology stability theorems 6.5, and uses the actions of the mapping class groups on complexes similar to those used to prove the stability theorems. The spectral sequences associated with these actions relate the homology groups of a mapping class group with the homology groups of the stabilizers of the simplices, which are, in fact, mapping class groups of smaller surfaces. This allows use of an inductive process for the computations. Let us state the main result of [Har7].

Theorem 6.6.D *Let S be a surface of genus g .*

$H_2(\text{PMod}_S, \mathbf{Q})$ is equal to 0 if $g = 2$, and to $\mathbf{Q}$ if $g \geq 3$.

$H_3(\text{PMod}_S, \mathbf{Q}) = 0$ if $g \geq 6$.

Using the same approach, in [Har8] J. Harer computed $H_2(\cdot, \mathbf{Q})$ for subgroups of the mapping class groups consisting of the isotopy classes of diffeomorphisms preserving some spin structure on the surface, and in [Har9] he also computed the fourth homology group of PMod_S . More precisely, the main result of [Har9] is the following theorem.

Theorem 6.6.E *Let S be a surface of genus g with b boundary components. $H_4(\text{PMod}_S, \mathbf{Q}) = \mathbf{Q}^2$ if $g \geq 6$, $b \geq 1$ or if $g \geq 10$.*

The computations involved in the proof of this theorem are extremely complicated. Finally, in [HL] R. Hain and E. Looijenga report that recently J. Harer has extended his computations to $H_5(\text{PMod}_S, \mathbf{Q})$. This homology group turns out to be 0 for sufficiently high genus.

6.7 Virtual Euler characteristic

In the spirit of Serre's approach [Ser2], the usual Euler characteristic $\text{ch}(\cdot) = \sum_{i \geq 0} \text{rank } H^i(\cdot)$ is not the right invariant for discrete groups Γ with torsion, even when it is well defined. One should consider instead the so-called *virtual Euler characteristic* $\chi(\cdot)$. If Γ contains a subgroup Γ' of finite index admitting a finite CW-complex as a $K(\Gamma', 1)$ -space (this implies that Γ' is torsion free), then, by definition,

$$\chi(\Gamma) = \frac{1}{[\Gamma : \Gamma']} \text{ch}(\Gamma').$$

It turns out that this number does not depend on the choice of Γ' . In general and in the most interesting cases, $\chi(\Gamma)$ is not an integer, but only a rational number.

For the virtual Euler characteristic there is an analogue of the usual formula computing the Euler characteristic of a CW-complex by counting the number of cells of various dimensions. Suppose that Γ acts on a CW-complex T preserving its CW-structure. Suppose that, in addition, T is contractible, some subgroup Γ' of finite index acts freely on T (and hence T/Γ' is a $K(\Gamma', 1)$ -space), the number of orbits of cells is finite and the isotropy groups of cells are finite. Then

$$\chi(\Gamma) = \sum_{\sigma} (-1)^{\dim \sigma} \frac{1}{[\Gamma_{\sigma} : 1]},$$

where Γ_{σ} is the isotropy group of the cell σ , $[\Gamma_{\sigma} : 1]$ is its order and σ runs over some set of representatives of the orbits of cells. If Γ acts on T freely, this formula reduces to the usual formula for the Euler characteristic of the $K(\Gamma, 1)$ -space T/Γ . For more details about the virtual Euler characteristic we refer to [Ser2] and to [Bro1], Section IX.7.

Theorem 6.7.A *If S has one boundary component, then $\chi(\text{Mod}_S) = -B_{2g}/2g = \zeta(1 - 2g)$, where g is the genus of S , B_{2g} is the $2g$ -th Bernoulli number and ζ is the Riemann ζ -function.*

Proof. Let R be a once-punctured surface of the same genus as S . By 5.1, Mod_S is isomorphic to Mod_R . Hence, we may consider Mod_R instead of Mod_S .

We will use the ideal triangulation of the Teichmüller space T_R introduced in 5.5. Let us consider the (first) barycentric subdivision of the complex $A(R)$, and take the union $\Sigma(R)$ of the barycentric stars of those simplices of $A(R)$ which are *not* simplices of $A_\infty(R)$. This union has a natural structure of a CW-complex with the barycentric stars as cells. One may say that $\Sigma(R)$ is the complex dual to the ideal triangulation of T_R from 5.5. It is not hard to see that $\Sigma(R)$ is a Mod_R -equivariant deformation retract of T_R . In particular, $\Sigma(R)$ is contractible. Cells of $\Sigma(R)$ are in one-to-one correspondence with simplices of $A(R)$ which are not simplices of $A_\infty(R)$, i.e., with ideal cell decompositions of R . Under this correspondence, the orbits of cells correspond to the *topological types* of ideal cell decompositions. Since the number of these topological types is obviously finite, the number of orbits is also finite. The isotropy group of a cell corresponding to an ideal cell decomposition is, clearly, equal to the group of isotopy classes of diffeomorphisms preserving this ideal cell decomposition, and hence is finite. (On the contrary, the isotropy groups of simplices in $A_\infty(R)$ are infinite.) As in the case of Teichmüller spaces T_S of surfaces with boundary (see 5.4), there are subgroups of finite index in Mod_R acting freely on T_R , and hence on $\Sigma(R)$. It follows that we can use the CW-complex $\Sigma(R)$ with its natural action of Mod_R in order to compute $\chi(\text{Mod}_R)$ by the above formula.

The application of this formula to this situation is not a simple matter at all. The problem is essentially a combinatorial one, since an ideal cell decomposition of R is an essentially combinatorial object, and the corresponding isotropy group is the group of symmetries of this combinatorial object. But there is no simple way to list all ideal cell decompositions (up to homeomorphism, of course) and to find the orders of their groups of symmetries. By some breathtaking combinatorial arguments, looking like sort of a miracle for an uninitiated topologist or algebraic geometer, one can compute exactly the combination of these orders leading to $\chi(\text{Mod}_R)$. We refer to the original papers of J. Harer and D. Zagier [HarZ] and R. Penner [Pe3] for the details. While in [HarZ] the needed technique (integration over the space of hermitian matrices!) appears as a *deus ex machine*, R. Penner [Pe3] motivates it by some ideas from quantum physics. A simplified proof was provided by M. Kontsevich [Ko]; see [Ko], Appendix D. More recently, D. Zagier [Z] provided a considerably simpler proof of the key combinatorial fact needed for the calculation of $\chi(\text{Mod}_R)$. This new proof is based on the theory of characters of symmetric groups (and avoids the integration over the space of hermitian matrices). Despite this dramatic simplification, the original proofs retain, in the author's view, considerable interest. $\square$

Corollary 6.7.B *If S is a closed surface of genus $g \geq 2$, then $\chi(\text{Mod}_S) = B_{2g}/4g(g-1)$.*

Proof. The corollary immediately follows from the theorem if we use the short exact sequence

$$1 \rightarrow \pi_1(S) \rightarrow \text{Mod}_{S_1} \rightarrow \text{Mod}_S \rightarrow 1,$$

where S_1 is the result of making one hole in S (cf. 2.8), the fact that virtual Euler characteristic is multiplicative in short exact sequences (cf. [Bro1], Chapter IX, Proposition 7.3), and the easy fact that $\chi(\pi_1(S))$ is equal to the Euler characteristic of S , i.e. to $2 - 2g$. $\square$

Finally, we note that [HarZ] also contains some information about the usual Euler characteristic $\sum (-1)^i \dim H_i(\text{Mod}_S, \mathbf{Q})$.

6.8 Torsion in mapping class groups, torsion in their cohomology and related topics

Mapping class groups contain many torsion elements, which can be exploited to detect torsion in their cohomology. Clearly, any finite group Γ acting on a surface S by orientation-preserving diffeomorphisms may be considered as a finite subgroup of $\text{Diff}(S)$, and hence leads to a finite subgroup of Mod_S . In fact, the latter subgroup of Mod_S is isomorphic to Γ , as immediately follows from the next theorem.

Theorem 6.8.A *Let S be an orientable surface of negative Euler characteristic. If Γ is a finite subgroup of $\text{Diff}(S)$, then the canonical homomorphism*

$$\Gamma \rightarrow \text{Aut}(H_1(S, \mathbf{Z}/m\mathbf{Z}))$$

defined by the action of diffeomorphisms on homology is injective for any $m \geq 3$.

Note that the homomorphism $\Gamma \rightarrow \text{Aut}(H_1(S, \mathbf{Z}/m\mathbf{Z}))$ obviously factors through Mod_S . This theorem is essentially due to J.-P. Serre [Ser1]; for an elementary proof, see [I15], Theorem 1.3 and Supplement 1.4.

According to a famous theorem of S. Kerckhoff [Ke2], [Ke3], every finite subgroup of Mod_S can be obtained as the image of a finite subgroup of $\text{Diff}(S)$; see Theorem 7.4.B. In practice finite subgroups of Mod_S are always obtained from an action of a finite group on S , i.e. as images of finite subgroups of $\text{Diff}(S)$. Kerckhoff's theorem tells us that we will not miss any finite subgroup by restricting ourselves to this method. Note that every finite group admits an embedding into some mapping class group because, as is well known, every finite group occurs as a group of symmetries of a Riemann surface. Moreover, L. Greenberg [Gre] proved that every finite group is isomorphic to the group of all automorphisms of some Riemann surface.

Finite subgroups of Mod_S are used to detect nontrivial cohomology classes of Mod_S by restricting cohomology classes to appropriate finite subgroups. Of course, one can detect in this way only torsion cohomology classes, because

cohomology groups of finite groups always consist of torsion classes. The first result of this sort was obtained by R. Charney and R. Lee [ChL1]. It was improved by H. Glover and G. Mislin [GIM1], [GIM2] and by R. Charney and R. Lee [ChL2]. It is convenient to state these results in terms of the stable cohomology groups in the sense of homology stability discussed in 6.5. Note that there are cohomology stability theorems exactly similar to the homology stability theorems of 6.5. In fact, stability theorems for cohomology follow from the corresponding stability theorems for homology. In particular, $H^m(\mathcal{M}_S)$ does not depend on S if the genus of S is sufficiently large compared to m . Let us denote by $H^m(\mathcal{M})$ the group $H^m(\mathcal{M}_S)$ for surfaces S of sufficiently large genus. Similar notation will be used for (untwisted) coefficients other than $\mathbf{Z}$.

Theorem 6.8.B *For all k the stable cohomology group $H^{4k}(\mathcal{M})$ contains an element of order equal to the denominator of the rational number $B_{2k}/2k$, where B_{2k} is the $2k$ -th Bernoulli number.*

This theorem is due to H. Glover and G. Mislin [GIM1], [GIM2]. The numbers appearing in this theorem are the same as in the computation of the virtual Euler characteristic (cf. Theorem 6.7.A). This remarkable fact apparently has no (known) explanation. In order to construct elements of $H^*(\mathcal{M})$, Glover and Mislin consider closed surfaces S and use the canonical representation $\mathcal{M}_S \rightarrow \text{Aut}(H_1(S, \mathbf{C}))$ defined by the action of diffeomorphisms on homology. This leads to flat complex vector bundle (with the fiber $H_1(S, \mathbf{C})$) over $K(\mathcal{M}_S, 1)$, and the Chern classes of this bundle can be considered as elements of $H^*(\mathcal{M}_S)$. Since the bundle is flat, they are torsion elements. One needs to know that these Chern classes are non-zero and, moreover, have sufficiently high order. This is checked by restricting (pulling back) them to specially constructed finite cyclic subgroups of $\mathcal{M}_S$. The elements alluded to in the theorem are some combinations of these Chern classes.

The cyclic subgroups used to detect the order of cohomology classes from Theorem 6.8.B are constructed only for surfaces of fairly high genus (compared to the dimension of the classes). The (co)homology stability theorems imply that these cohomology classes survive till a much lower genus. It happens that sometimes these cyclic groups cannot act nontrivially on surfaces of such a low genus, and hence cannot be subgroups of the mapping class groups of these low genus surfaces. In this way, one gets examples of p -torsion cohomology classes of mapping class groups having no p -torsion elements (where p is some prime number). Such torsion elements in cohomology are somewhat unusual and are called *exotic* torsion elements. The existence of exotic torsion elements was observed in [GIM2].

Theorem 6.8.C *The stable cohomology $H^*(\mathcal{M}, \mathbf{Z}[1/2])$ with coefficients $\mathbf{Z}[1/2]$ contains as a direct summand the cohomology $H^*(\text{Im } J, \mathbf{Z}[1/2])$ of the space $\text{Im } J$ with the same coefficients.*

This theorem is due to R. Charney and R. Lee [ChL2]. The space $\mathrm{Im} J$ is a well known character from homotopy theory. It is known that its cohomology groups consist of torsion elements; replacing the coefficients $\mathbf{Z}$ by $\mathbf{Z}[1/2]$ just kills the 2-torsion. R. Charney and R. Lee use some cyclic subgroups in a manner similar to that of H. Glover and G. Mislin [G1M2] (but the subgroups of Charney and Lee are smaller and simpler to construct) together with some deep ideas from homotopy theory and algebraic K -theory (the methods of H. Glover and G. Mislin are more elementary). Another difference is that R. Charney and R. Lee use the canonical homomorphisms $\mathcal{M}_S \rightarrow \mathrm{Aut}(H_1(S, \mathbf{Z}/p\mathbf{Z}))$ (their images are contained in the symplectic groups over the field $\mathbf{F}_p = \mathbf{Z}/p\mathbf{Z}$, and a lot is known about the cohomology of these groups from the work of Quillen and others) instead of the homomorphisms $\mathcal{M}_S \rightarrow \mathrm{Aut}(H_1(S, \mathbf{C}))$ used by H. Glover and G. Mislin. Theorem 6.8.C can be used to produce some torsion classes in addition to those constructed by H. Glover and G. Mislin [G1M2].

One can easily realize $H^*(\mathcal{M})$ as the cohomology of some space. Let S_g^1 be a surface of genus g with one boundary component. We may assume that S_g^1 is contained in S_{g+1}^1 (and hence $S_{g+1}^1 \setminus \mathrm{int} S_g^1$ is a torus with two holes) for all $g \geq 1$. Then we have a sequence of inclusions $S_1^1 \rightarrow \dots \rightarrow S_g^1 \rightarrow S_{g+1}^1 \rightarrow \dots$, inducing inclusions of the corresponding mapping class groups $\mathcal{M}_1^1 \rightarrow \dots \rightarrow \mathcal{M}_g^1 \rightarrow \mathcal{M}_{g+1}^1 \rightarrow \dots$, where $\mathcal{M}_n^1 = \mathcal{M}_{S_n^1}$. Let $\mathcal{M}$ be the direct limit (i.e., the union) of these groups. Clearly, the cohomology $H^*(\mathcal{M})$ of this group is exactly what was denoted by $H^*(\mathcal{M})$ above. By the definition, $H^*(\mathcal{M})$ is the cohomology $H^*(K(\mathcal{M}, 1))$ of the space $K(\mathcal{M}, 1)$. A natural problem is to refine Theorem 6.8.C and produce a splitting on the level of spaces, not just cohomology groups. In fact, it is natural to replace $K(\mathcal{M}, 1)$ by the simply connected space $K(\mathcal{M}, 1)^+$ having the same cohomology groups as $K(\mathcal{M}, 1)$. Here $+$ denotes Quillen's plus-construction, which is well-defined for $K(\mathcal{M}, 1)$ because $H_1(K(\mathcal{M}, 1)) = 0$ as a corollary of Theorem 6.6.B. Charney and Cohen [ChC] proved that it is possible to find such a splitting in the stable homotopy category. Note that the difference between $K(\mathcal{M}, 1)$ and $K(\mathcal{M}, 1)^+$ disappears in the stable homotopy category.

Theorem 6.8.D ($\mathrm{Im} J$) $\otimes \mathbf{Z}[1/2]$, i.e. the space $\mathrm{Im} J$ localized away from 2, is a stable retract of $K(\mathcal{M}, 1)$.

The proof uses some noncyclic finite groups π_k (they are iterated wreath products of cyclic groups). But R. Charney and F. Cohen do not embed them in any $\mathcal{M}_S$. Instead of this, they construct maps $K(\pi_k, 1) \rightarrow K(\mathcal{M}, 1)^+$ (in fact, only over finite skeletons) using the homology stability theorems for mapping class groups in order to split natural maps of the form $K(\mathcal{M}_S, 1)^+ \rightarrow K(\mathcal{M}_R, 1)^+$, where S is a surface with one boundary component and R is the result of gluing a disc to this boundary component.

Recently, U. Tillmann [Til2] proved that the splitting of Theorem 6.8.C indeed can be realized on the level of spaces without passing to the stable homotopy category. Namely, she proved the following theorem.

Theorem 6.8.E *There is a space Y such that $K(\mathcal{M}, 1)^+$ is homotopy equivalent to $((\operatorname{Im} J) \otimes \mathbf{Z}[1/2]) \times Y$.*

The proof is based on Theorem 6.8.D and the following fundamental theorem of U. Tillmann [Til2].

Theorem 6.8.F *$K(\mathcal{M}, 1)^+$ has the homotopy type of an infinite loop space.*

This theorem is obviously of great interest independent of its applications to Theorem 6.8.E. The proof is based on a construction of suitable categories (using the results of a previous paper of U. Tillman [Til1]) which lead to an acceptable input for an infinite loop space machine. Then a new group completion theorem and Harer's homology stability theorem (see 6.5) are used to identify the resulting infinite loop space with $K(\mathcal{M}, 1)^+$. Tillmann's infinite loop space structure on $K(\mathcal{M}, 1)^+$ extends the loop space structure on $K(\mathcal{M}, 1)^+$ constructed previously by more elementary means by E. Miller [Mi] and C.-F. Bödigheimer [Bö]. But it is not clear yet if it extends the double loop space structure of [Mi] and [Bö]. See also [CoT] for related results about the homology operations in $H^*(\mathcal{M})$.

Now we turn to the mapping class group $\operatorname{Mod}_2 = \operatorname{Mod}_{S_2}$ of a closed surface S_2 of genus 2. R. Lee and S. Weintraub [LeW] and F. Cohen [Co1] proved that $H^*(\operatorname{Mod}_2, \mathbf{Z})$ is a torsion group, having nontrivial p -torsion (for prime p) only for $p = 2, 3$ or 5 . F. Cohen [Co1], [Co2] showed that the 5-torsion is completely captured by a cyclic subgroup of order 5. More precisely, the following theorem holds.

Theorem 6.8.G *There is a homomorphism $\mathbf{Z}/5\mathbf{Z} \rightarrow \operatorname{Mod}_2$ inducing an isomorphism on 5-primary components in homology and on homology with coefficients $\mathbf{Z}/5\mathbf{Z}$. In particular, $H_n(\operatorname{Mod}_2, \mathbf{Z})$ is equal to $\mathbf{Z}/5\mathbf{Z}$ plus some 2- and 3-torsion groups for odd n and is a sum of 2- and 3-torsion groups for even n .*

The cohomology of Mod_2 with coefficients $\mathbf{Z}/2\mathbf{Z}$, $\mathbf{Z}/3\mathbf{Z}$, and $\mathbf{Z}/5\mathbf{Z}$ was essentially completely described by F. Cohen and D. Benson [Co2], [Ben], [BeCo]. For the coefficients $\mathbf{Z}/3\mathbf{Z}$ and $\mathbf{Z}/5\mathbf{Z}$, even the ring structure of the cohomology was determined, while for the coefficients $\mathbf{Z}/2\mathbf{Z}$ the ranks of the cohomology groups were computed, but the question of which of two possible ring structures is correct was left open. These papers also contain some information about the integral cohomology of Mod_2 , the structure of which was further elucidated by F. Cohen [Co3]. We state only the result for the coefficients $\mathbf{Z}[1/2]$, which is especially simple.

Theorem 6.8.H *The ring $H^*(\operatorname{Mod}_2, \mathbf{Z}[1/2])$ is isomorphic to*

$$\mathbf{Z}[1/2][x, y, z]/(5x, 3y, 3z, z^2),$$

where x, y, z have degrees 2, 4, 5 respectively.

The proof invokes a connection between Mod_2 and the mapping class group $\text{Mod}_0^6 = \text{Mod}_{S_0^6}$ of a sphere with six holes S_0^6 due to Birman and Hilden [BirH]. Namely, there is a short exact sequence

$$1 \rightarrow \mathbf{Z}/2\mathbf{Z} \rightarrow \text{Mod}_2 \rightarrow \text{Mod}_0^6 \rightarrow 1,$$

with the image of $\mathbf{Z}/2\mathbf{Z}$ generated by the hyperelliptic involution (cf. end of 4.3 for the latter). In addition to this, a close relation between Mod_0^6 and the Artin braid group on six strings is exploited. This geometric input leads to cohomological computations by means of some powerful homotopy-theoretic techniques. For further results obtained by this approach we refer to [Co4], [Co5], where some noncyclic finite subgroups of $\mathcal{M}_S$, namely quaternion and dihedral groups in $\mathcal{M}_S$, play an essential role.

For further information about many of the topics discussed in this section, and for other related results, we refer to an excellent expository paper [Mis] by G. Mislin. In particular, it contains a discussion of results of H. Glover, G. Mislin and Y. Xia [GIMX1], [GIMX2], and of related results of Y. Xia [X1]–[X5]; see also [X6]–[X8] and [PriX] for further results.

7 Thurston's theory and its applications

This section is devoted to an overview of Thurston's theory of surface diffeomorphisms and some of its applications. For more details, we refer to Thurston's updated announcement of the main results [Th1], and for a detailed exposition to the proceedings of the Orsay seminar [FLP]. In the case of closed surfaces, a nice introduction from the point of view of hyperbolic geometry is provided by A. Casson and S. Bleiler [CB]. See also A. Hatcher [Hat2].

7.1 Classification of mapping classes

An element $f \in \text{Mod}_S$ is called *reducible* if it fixes some simplex of $C(S)$ (but, perhaps, permutes its vertices), and *irreducible* otherwise. As usual, we say that an element f is of *finite order* if $f^n = 1$ for some $n \neq 0$. If an element f is not of finite order and is irreducible, then it is called *pseudo-Anosov*. It turns out that elements in all three classes (reducible, finite order, pseudo-Anosov) can be represented by diffeomorphisms having some special properties. These special representatives play a crucial role in the understanding of corresponding mapping classes.

First of all, for a reducible element f fixing a simplex σ of $C(S)$, let us consider the union C of some disjoint circles representing the vertices of σ . We will call such a union a *realization* of σ . In this case there is a diffeomorphism $F : S \rightarrow S$ representing the isotopy class f and leaving C invariant; $F(C) = C$. Such a diffeomorphism F induces a diffeomorphism $F_C : S_C \rightarrow S_C$ of the result S_C of cutting S along C . The components of S_C are, in a definite sense,

simpler than S (they have larger Euler characteristic) and this allows the use of this procedure in inductive arguments. One minor difficulty comes from the fact that S_C has, in general, several components, and F_C may permute them in a nontrivial way. The standard way to deal with this is to replace F_C by some power of it not permuting the components. Another, more serious, problem comes from the fact that F_C may be isotopic to the identity, and then we lose all the information after the cutting. This happens exactly when f is a product of Dehn twists around the components of C . This case needs to be dealt with separately. Anyhow, any reducible mapping class can be represented by a *reducible* diffeomorphism, i.e. a diffeomorphism preserving the union of several pairwise disjoint and non-isotopic nontrivial circles.

While the results outlined in the previous paragraph are essentially elementary, the next case, namely the case of elements of finite order, is much deeper. Special representatives of such elements are provided by the following theorem of J. Nielsen [Ni4].

Theorem 7.1.A *If $f \in \text{Mod}_S$ is an element of finite order $n \neq 0$, then f can be represented by a diffeomorphism $F : S \rightarrow S$ of the same order n . Moreover, one can choose F to be an isometry of a metric of constant curvature on S with geodesic boundary.*

See [FLP], Exposé 11, § 4 for a proof of this theorem in the context of Thurston's theory and an outline of another proof due to W. Fenchel [Fe]. As a simple application of this theorem, consider the kernel $\Gamma_S(m)$ of the natural homomorphism

$$\text{Mod}_S \rightarrow \text{Aut}(H_1(S, \mathbf{Z}/m\mathbf{Z}))$$

defined by the action of diffeomorphisms on homology. Clearly, $\Gamma_S(m)$ has finite index in Mod_S .

Corollary 7.1.B *If $m \geq 3$, then $\Gamma_S(m)$ is torsion free.*

Proof. If the Euler characteristic of S is negative, it is sufficient to combine Nielsen's Theorem 7.1.A with Serre's Theorem 6.8.A. The remaining cases are elementary and left to the reader. $\square$

Finally, the remaining elements, which we called pseudo-Anosov, also can be represented by some very special diffeomorphisms, namely by *pseudo-Anosov diffeomorphisms*. (To be honest, they are only homeomorphisms, but their non-smoothness is very mild and harmless.) We will not give their definition here, which generalizes (the two-dimensional case of) the definition of *Anosov diffeomorphisms*, introduced by D. V. Anosov [An] and playing a central role in the theory of dynamical systems; see [FLP] or [Th1]. The existence of pseudo-Anosov diffeomorphisms is one of the main achievements of Thurston's theory of the diffeomorphisms of surfaces.

While the existence of nontrivial reducible mapping classes or mapping classes of finite order is quite obvious, it is not so for pseudo-Anosov mapping classes. Let us indicate a couple of ways to get such mapping classes. We need the following notion: we say that a set A of vertices of $C(S)$ (i.e., a set of isotopy classes of nontrivial circles on S) *fills* S if for every vertex γ there is a vertex $\alpha \in A$ such that any two circles representing α, γ have a non empty intersection. For finite sets A , this is equivalent to the following property: for any collection of circles representing all isotopy classes from A , all components of the complement of the union of these circles in S are either (open) discs or (half-open) annuli, and each of the latter contains a component of ∂S .

Theorem 7.1.C *If α, β are two isotopy classes of nontrivial circles such that $\{\alpha, \beta\}$ fills S , then all elements $t_\alpha^n t_\beta^{-m}$ for $m, n \geq 1$ are pseudo-Anosov.*

See [FLP], Exposé 13, Remark 2 after theorem III.3. Theorem III.3 itself shows that under the assumptions of Theorem 7.1.C, many other combinations of t_α, t_β are also pseudo-Anosov. These results were generalized by D. Long [Lo2] and then by R. Penner [Pe5] to the situations when more than two Dehn twists are involved. Further examples were provided by M. Bauer [Bau]. The following remarkable result, due to A. Fathi [Fathi], while not giving completely explicit examples of pseudo-Anosov elements, shows that they are in a very strong sense generic.

Theorem 7.1.D *Let $f \in \text{Mod}_S$ and α be a vertex of $C(S)$. If the set $\{f^n(\alpha) : n \in \mathbf{Z}\}$ fills S , then all elements $t_\alpha^n f$ are pseudo-Anosov except for at most 7 consecutive values of n .*

For example, if $\{\alpha, \beta\}$ fills S , then $\{t_\beta^n(\alpha) : n \in \mathbf{Z}\}$ also fills S . The paper [Fathi] includes some other results of this sort, but the following question remains open: is it possible to replace the Dehn twist t_α in this theorem by an arbitrary composition of Dehn twists about several disjoint circles, possibly at the cost of replacing the number 7 by some, may be undetermined, but *universal* number? The results of A. Fathi were preceded by a theorem of D. Long and H. Morton [LoM] in which only the finiteness of the number of exceptions (i.e., number of n such that $t_\alpha^n f$ is not pseudo-Anosov) was asserted, instead of the existence of a universal constant (≤ 7 by Theorem 7.1.D).

When one deals with individual mapping classes, one can usually reduce the problem to the case of irreducible mapping classes by the method of cutting outlined above. When one deals with the whole mapping class group or a general subgroup of it, usually there is no way to avoid reducible elements. But one can often restrict attention to following, in some sense simplest, type of reducible elements. Let us call a mapping class $f \in \text{Mod}_S$ *pure* if it can be represented by a diffeomorphism $F : S \rightarrow S$ fixing (pointwise) some union C of disjoint and pairwise non-isotopic nontrivial circles on S and such that F does not permute the components of $S \setminus C$ and induces on each component of the

cut surface S_C a diffeomorphism isotopic either to a pseudo-Anosov or to the identity diffeomorphism. For example, every pseudo-Anosov element is pure, as is also any product of Dehn twists about several disjoint circles. In the latter case the induced diffeomorphism of S_C is isotopic to the identity; in all other cases F induces a diffeomorphism isotopic to a pseudo-Anosov diffeomorphism on at least one component of S_C .

Theorem 7.1.E *If $m \geq 3$, then $\Gamma_S(m)$ consists of pure elements.*

This theorem strengthens Corollary 7.1.B, and is a natural sharpening of it in the framework of Thurston's theory. For a proof, see [I15], Corollary 1.8. This theorem often allows the immediate elimination of all difficulties related to elements of finite order and to possible permutations of components (of the cut surface) by reducible diffeomorphisms. For example, if one considers a subgroup G of Mod_S , it is often possible to replace it by $G \cap \Gamma_S(m)$, $m \geq 3$.

Another important tool for dealing with reducible elements is the notion of essential reduction classes introduced, in a slightly different form, by J. Birman, A. Lubotzky and J. McCarthy [BLM]. For a pure element $f \in \text{Mod}_S$ we say that a vertex α of $C(S)$ is an *essential reduction class* of f if $f(\alpha) = \alpha$ and $f(\beta) \neq \beta$ for all β such that α, β cannot be represented by disjoint circles. For a general element $f \in \text{Mod}_S$, we say that a vertex α of $C(S)$ is an *essential reduction class* of f if it is an essential reduction class of some nontrivial (and then, as one may prove, of any nontrivial) power f^n ($n \neq 0$), which is pure. Note that f has pure powers in view of Theorem 7.1.E and the fact that $\Gamma_S(m)$ has finite index in Mod_S . It turns out that the set of all essential reduction classes of f is a simplex of $C(S)$, which we call the *canonical reduction system* of f and denote by $\sigma(f)$. If f is pure, it fixes all vertices of $\sigma(f)$. In general f leaves $\sigma(f)$ invariant. The crucial fact is that $\sigma(f)$ is non empty if f is a reducible element of infinite order ($\sigma(f)$ is empty for any element of finite order f , reducible or not). In particular, cutting along some realization of $\sigma(f)$ provides a canonical way to simplify a reducible element f . If f is pure, then this leads to a diffeomorphism (of the cut surface) which leaves all components invariant and is isotopic on any component either to the identity, or to a pseudo-Anosov diffeomorphism.

7.2 Measuring against circles and its applications

In this section we discuss one of the basic technical tools of Thurston's theory: measuring various geometric objects on a surface S against the isotopy classes of nontrivial circles, i.e. the vertices of $C(S)$. We will denote by $V(S)$ the set of all vertices of $C(S)$, i.e. the set of all isotopy classes of nontrivial circles. One of the fundamental ideas of Thurston was to consider measurements with respect to all vertices simultaneously. The natural place to record all such measurements is the set $\mathcal{R}(S)$ of all functions $V(S) \rightarrow \mathbf{R}_{\geq 0}$, endowed with the product topology. The mapping class group Mod_S acts on $V(S)$, and this action induces an action

of Mod_S on $\mathcal{R}(S)$. The next important idea is to pass from the space $\mathcal{R}(S)$ to a projectivization of it, namely to the quotient $\mathcal{PR}(S)$ of $\mathcal{R}(S) \setminus \{0\}$ by the natural (diagonal) action of the multiplicative group $\mathbf{R}_{>0}$ on it. Clearly, the action of Mod_S on $\mathcal{R}(S)$ induces an action of Mod_S on $\mathcal{PR}(S)$.

Let us illustrate this approach by the example of the Teichmüller space T_S . A natural (but not the only possible) way to measure the points of T_S (i.e., the classes of hyperbolic metrics on S) with respect to the vertices $\alpha \in V(S)$ is to use the length functions $l_\alpha : T_S \rightarrow \mathbf{R}_{>0}$ from 5.2. The collection of all length functions defines an embedding $T_S \rightarrow \mathcal{R}(S)$. Clearly, this embedding is Mod_S -equivariant. Its image is contained, obviously, in $\mathcal{R}(S) \setminus \{0\}$. The composition $T_S \rightarrow \mathcal{PR}(S)$ of this embedding with the canonical projection $\mathcal{R}(S) \setminus \{0\} \rightarrow \mathcal{PR}(S)$ is still an embedding, which is also Mod_S -equivariant. It turns out that the embedding $T_S \rightarrow \mathcal{PR}(S)$ is a homeomorphism onto its image, this image has a compact closure in $\mathcal{PR}(S)$ and, moreover, this closure is homeomorphic to a closed disc (of the dimension equal to the dimension of T_S) under a homeomorphism taking the image of T_S into the interior of this disc. In this way, we achieve a natural compactification of T_S by a sphere called *Thurston's boundary* of Teichmüller space. The compactification itself is called *Thurston's compactification* of Teichmüller space. An advantage of this compactification is that the action of Mod_S extends to it by continuity (as is clear from the construction). S. Kerckhoff [Kel] proved that this desirable property does not hold for the classical compactifications of Teichmüller space by a sphere, the so-called *Teichmüller compactifications*. A Teichmüller compactification depends on a choice of a base point in T_S , and Kerckhoff's result holds for any choice of the base point. In fact, Kerckhoff proved that compactifications corresponding to different base points are often different, and this is closely related to the impossibility to extend the action of Mod_S to any of them by continuity.

Another example of the idea of measuring geometric objects against the isotopy classes of circles is provided by measuring the isotopy classes of circles themselves. For $\alpha, \beta \in V(S)$ let us define the *geometric intersection number* $i(\alpha, \beta)$ as the minimal number of points in the intersections $A \cap B$, where A, B run over all circle representatives of the isotopy classes α, β respectively. For any $\alpha \in V(S)$ the function $I_\alpha : \gamma \mapsto i(\gamma, \alpha)$ belongs to $\mathcal{R}(S)$. Moreover, it is easy to see that for any $\alpha \in V(S)$ there is some $\gamma \in V(S)$ such that $i(\alpha, \gamma) \neq 0$. Hence, $I_\alpha \in \mathcal{R}(S) \setminus \{0\}$ for every $\alpha \in V(S)$. By assigning I_α to α , we get a map $V(S) \rightarrow \mathcal{R}(S) \setminus \{0\}$. It is easy to see that this map is injective. Moreover, its composition with the projection $\mathcal{R}(S) \setminus \{0\} \rightarrow \mathcal{PR}(S)$ is still injective (this is a little more difficult). The remarkable fact is that the image of $V(S)$ under this composition $V(S) \rightarrow \mathcal{PR}(S)$ is contained in Thurston's boundary of T_S and is dense in this boundary. In some sense, this image plays the role of rational points of Thurston's boundary.

It turns out that other points of Thurston's boundary also admit a geometric interpretation. The relevant geometric object is a *measured foliation with singularities*, or, more precisely, the *Whitehead equivalence class* of such a foliation

on S . Usually one speaks simply about *measured foliations* on S . The set of all (Whitehead equivalence classes) of measured foliations on S is denoted by $\mathcal{MF}(S)$. We omit the definitions, referring to [FLP] and [Th1], and restrict ourselves to the following remarks. A measured foliation on S is a codimension 1 foliation on S (so the leaves have dimension 1). Since the Euler characteristic of S is almost never zero, S usually does not admit honest foliations, and we have to allow some singularities in them. The term *measured* means that our foliation is equipped with a so-called *transverse measure*, which allows us to measure locally (say, within a flow box of the foliation) the distance between the leaves. The transverse measure allows us to define the *transverse length* of curves and, in particular, of circles (if a curve is contained in a leaf, its transverse length is 0).

The notion of the transverse length of a circle allows us to deal with the measured foliations in a way similar to the way we dealt with hyperbolic metrics (or, rather, points of T_S) and circles (or, rather, vertices of $C(S)$) above. Namely, given a vertex $\gamma \in V(S)$ and a measured foliation μ , we define $l_\gamma(\mu)$ to be the infimum of transverse lengths with respect to μ of all circles in S representing the isotopy class γ . In contrast with the hyperbolic metrics (compare 5.2), this infimum is not always achieved (but it is achieved on a not necessarily embedded curve homotopic to circles representing γ). For any measured foliation μ , the function $L_\mu : \gamma \mapsto l_\gamma(\mu)$ belongs to $\mathcal{R}(S)$. It turns out that for every measured foliation μ there exists some $\gamma \in V(S)$ such that $l_\gamma(\mu) \neq 0$. Hence, $L_\mu \in \mathcal{R}(S) \setminus \{0\}$ for every $\mu \in \mathcal{MF}(S)$. By assigning L_μ to μ , we get a map $\mathcal{MF}(S) \rightarrow \mathcal{R}(S) \setminus \{0\}$. It is not very hard to see that this map is injective. In particular, we can equip $\mathcal{MF}(S)$ with the topology induced from $\mathcal{R}(S)$ by this map. The composition of this map with the projection $\mathcal{R}(S) \setminus \{0\} \rightarrow \mathcal{PR}(S)$ turns out to be not injective. The reason is very simple: the multiplication of transverse measures by positive constants defines a natural action of the multiplicative group $\mathbf{R}_{>0}$ on $\mathcal{MF}(S)$, and our map $\mathcal{MF}(S) \rightarrow \mathcal{R}(S) \setminus \{0\}$ is obviously $\mathbf{R}_{>0}$ -equivariant. So, it is only natural to consider the quotient $\mathcal{PMF}(S)$ of $\mathcal{MF}(S)$ by this action of $\mathbf{R}_{>0}$. Clearly, our map $\mathcal{MF}(S) \rightarrow \mathcal{R}(S) \setminus \{0\}$ leads to a map $\mathcal{PMF}(S) \rightarrow \mathcal{PR}(S)$, which turns out to be injective. The image of this map is equal to Thurston's boundary of T_S . The interpretation of points of Thurston's boundary in terms of measured foliations plays a crucial role in the proof of the fact that Thurston's boundary is a sphere forming a ball together with the Teichmüller space.

Note that any isotopy class of a nontrivial circle naturally leads to an equivalence class of measured foliations, and the images in $\mathcal{R}(S)$ of an isotopy class of a circle and of the corresponding measured foliation are the same. So, a measured foliation turns out to be, in some sense, a generalized circle.

It is worth pointing out that Thurston's boundary can be constructed and its main properties can be established, using only the notion of a measured foliation without any reference to Teichmüller spaces, and for many applications it is sufficient to use only Thurston's boundary itself (together with the action

of Mod_S on it).

7.3 Action of the mapping classes on Thurston's compactification of Teichmüller space

Let us denote by $\overline{T}_S$ Thurston's compactification of Teichmüller space, i.e. the union of T_S with its Thurston's boundary. In this section we describe the main features of the action of various classes of elements of Mod_S introduced in 7.1 on $\overline{T}_S$.

First of all, the elements of finite order can be characterized by the fact that they have a fixed point in the Teichmüller space T_S itself. The set of fixed points in T_S can be identified with some other Teichmüller space. A similar result probably holds for the set of fixed points in $\overline{T}_S$.

A pseudo-Anosov element f has exactly two fixed points in $\overline{T}_S$ and both of them belong to Thurston's boundary; there are no fixed points in T_S itself. One of the fixed points is in a natural sense *attracting*, and the other one is *repelling*. In the following theorem a denotes the attracting point and r denotes the repelling point.

Theorem 7.3.A *Let f be a pseudo-Anosov element. There exist two points a , r in Thurston's boundary of T_S , such that if U is any neighborhood of a in $\overline{T}_S$ and K is any compact set in $\overline{T}_S \setminus \{r\}$, then $f^n(K) \subset U$ for all sufficiently large n .*

So, all points in $\overline{T}_S \setminus \{r\}$ converge under the action of the iterations of f to a , uniformly on compact sets. This justifies the terms “attracting” and “repelling”. Since both T_S and its boundary are obviously invariant, a similar result holds for Thurston's boundary instead of $\overline{T}_S$. One can find a somewhat weaker result (not asserting the uniformity of the convergence) in [FLP] (cf. [FLP], Exposé 12, Corollaire II.1); Theorem 7.3.A itself easily follows, for example, from Theorem 7.3.B below.

The action of reducible elements is more complicated. If a reducible pure element f is not a composition of Dehn twists about several disjoint circles (so, if we cut S along a realization of $\sigma(f)$, then the induced diffeomorphism of the cut surface is isotopic to a pseudo-Anosov diffeomorphism on at least one component), then we have a picture similar to the above description of the action of a pseudo-Anosov element. We state the result only for the action on Thurston's boundary. Its set of fixed points consists of two components, one of which is in a natural sense *attracting*, while the other one is *repelling*. In the following theorem, A denotes the attracting set, and R denotes the repelling set.

Theorem 7.3.B *Let f be a pure element of Mod_S which is not a composition of Dehn twists about several disjoint circles. There exist two disjoint subsets*

A, R of Thurston's boundary of T_S , such that if U is any neighborhood of A in Thurston's boundary and K is any compact set in the complement of R , then $f^n(K) \subset U$ for all sufficiently large n .

This result is proved in [I15], Appendix, where also the components of the fixed point set are completely described. Theorem 7.3.A is a special case of Theorem 7.3.B and this description. Note that a weaker result, in which the attracting and the repelling sets are not disjoint in general (cf. [I15], Theorem 3.5), is sufficient for most applications. Another version of this weaker result was proved by J. McCarthy [Mc2]; cf. [Mc2], Uniform Convergence Lemma.

7.4 Some applications

We start with a famous result of Thurston about 3-manifolds, which in fact served as the motivation for Thurston's theory of surface diffeomorphisms. Recall that the mapping torus M_f of map $f : X \rightarrow X$ is the quotient space $X \times [0, 1] / (x, 0) \sim (f(x), 1)$. If X is a manifold and f is diffeomorphism, then M_f is also a manifold.

Theorem 7.4.A *Let $f : S \rightarrow S$ be a diffeomorphism of a closed surface S . The mapping torus M_f admits a riemannian metric of constant negative curvature if and only if the isotopy class of f is pseudo-Anosov.*

A slightly more complicated result holds for surfaces with boundary as well. This theorem is the most difficult special case of Thurston's famous geometrization theorem for Haken manifolds. Of course, the proof involves much more than Thurston's theory of surface diffeomorphisms. The first account of this theorem is due to D. Sullivan [Su]; the exposition in [Su] is very demanding. Recently, new expositions of this theorem were published by C. McMullen [McM] (see [McM], Chapter 3) and J.-P. Otal [O]. C. McMullen's approach is close to the original ideas of Thurston (and refers to some still unpublished preprints of Thurston). The approach of J.-P. Otal follows the one of Thurston in the general outline, but provides a completely new proof of one of the key steps (the so-called double limit theorem); his exposition is essentially self-contained. Hopefully, Thurston's own proof [Th2] will also appear soon. See also notes by M. Kapovich [Kap].

The next theorem, due to S. Kerckhoff [Ke2], [Ke3], provides a solution of the so-called Nielsen realization problem.

Theorem 7.4.B *Let G be a finite subgroup of Mod_S . Then there is a finite subgroup $G^\sim$ of $\text{Diff}(S)$ which maps isomorphically onto G under the natural map $\text{Diff}(S) \rightarrow \text{Mod}_S$. Moreover, one can choose $G^\sim$ to be a subgroup of the group of isometries of some metric of constant curvature on S with geodesic boundary.*

One says that $G^\sim$ *realizes* G . Note that this theorem generalizes Theorem 7.1.A from cyclic to arbitrary subgroups of Mod_S . The key ingredient of the proof is provided by the so-called *earthquake flows* on the Teichmüller space T_S . These flows are related to measured foliations in the same manner as the Fenchel-Nielsen flows from 5.2 are related to circles. In fact, earthquake flows are the limits of (suitably chosen and parameterized) Fenchel-Nielsen flows. The main properties of the earthquake flows used in the proof of Theorem 7.4.B are the convexity of the length functions from 5.2 along the flow lines of earthquake flows and the fact that every two points can be connected by the flow line of an earthquake flow. We refer to [Ke3] for the details; see also [Ke4]. Some further developments resulting from this solution of the Nielsen realization problem are presented in [Ke5]. After Kerckhoff's work, S. Wolpert [Wo] and A. Tromba [Tr1] (see [Tr1], Section 6.4), [Tr2] found other proofs of Theorem 7.4.B, independent of Thurston's theory (but still based on convexity properties of some functions; in fact, the same length functions in Wolpert's proof, but along different paths in T_S).

The first applications of Thurston's theory to the algebraic properties of mapping class groups are due to J. McCarthy [Mc1], who noticed that Thurston's theory provides a good description of normalizers and centralizers of pseudo-Anosov elements, and to J. Birman, A. Lubotzky and J. McCarthy [BLM], who proved, in particular, the following theorem.

Theorem 7.4.C *If a subgroup of Mod_S contains a solvable subgroup of finite index, then it contains an abelian subgroup of finite index.*

Soon after [BLM], the author proved [I2] the following more strong theorem.

Theorem 7.4.D *Let G be a subgroup of Mod_S , and let $m \geq 3$. Then either G contains a free (nonabelian) group with two generators, or $G \cap \Gamma_S(m)$ is abelian.*

Since $\Gamma_S(m)$ is of finite index in Mod_S and a solvable group cannot contain a free group with two generators, Theorem 7.4.D implies Theorem 7.4.C. The following corollary of Theorem 7.4.D, which is only slightly weaker than Theorem 7.4.D itself, was simultaneously and independently proved by J. McCarthy [Mc2].

Theorem 7.4.E *Let G be a subgroup of Mod_S . Then either G contains a free group with two generators, or G contains an abelian subgroup of finite index.*

A similar property, with “abelian” replaced by “solvable”, holds for arbitrary linear groups (i.e., subgroups of $GL_n(k)$ for a field k) according to a famous theorem of J. Tits [Tits1]. In fact, the search for an analogue of Tits' theorem for subgroups of the mapping class groups led to Theorems 7.4.D and 7.4.E. Later on, the author [I2], [I6] discovered that some other fundamental theorems about the linear groups have analogues for subgroups of mapping class groups.

In order to state them, recall that a subgroup H of a group G is called *maximal* if for every subgroup K of G such that $H \subset K$, either $H = K$ or $K = G$. Recall also that the *Frattini subgroup* $F(G)$ of a group G is defined as the intersection of all maximal subgroups of G . An intuitive characterization of $F(G)$ is provided by the following elementary result: $F(G)$ is the set of all *nongenerating* elements of G , i.e. such elements $g \in G$ such that for every generating set X of G , the set $X \setminus \{g\}$ is also generating.

Theorem 7.4.F *Let G be a finitely generated subgroup of Mod_S . Then either G contains a maximal subgroup of infinite index, or G contains an abelian subgroup of finite index.*

It is easy to see that if a group G contains an abelian subgroup of finite index, then all maximal subgroups have finite index. While usually it is very easy to find a maximal subgroup of finite index (at least when there are nontrivial subgroups of finite index: take a subgroup of minimal index $\neq 1$), it is hard to prove the existence of maximal subgroups of infinite index. For linear groups, a theorem similar to Theorem 7.4.F (with “solvable” instead of “abelian”) was proved by G. A. Margulis and G. A. Soifer [MaSo]; their result motivated Theorem 7.4.F.

Theorem 7.4.G *Let G be a finitely generated subgroup of Mod_S . Then the Frattini subgroup $F(G)$ is nilpotent.*

This is an analogue of a theorem of V. P. Platonov [Pl] to the effect that Frattini subgroups of finitely generated linear groups are nilpotent. It was proved by the author [I6] after some partial results of D. Long [Lo1].

The proofs of Theorems 7.4.C–7.4.G are based on the properties of the action of the elements of Mod_S on Thurston’s boundary of the Teichmüller space T_S and, in particular, on the results discussed in 7.3. A crucial role is also played by the following result, in which we call a subgroup G of Mod_S *irreducible* if there is no nonempty simplex σ of $C(S)$ such that $g(\sigma) = \sigma$ for all $g \in G$.

Theorem 7.4.H *If G is an infinite irreducible subgroup of Mod_S , then G contains a pseudo-Anosov (in particular, an irreducible) element.*

Note that a similar statement for finite subgroups is false: according to J. Gilman [Gi], there are finite irreducible subgroups of Mod_S consisting entirely of reducible elements. In view of Theorems 7.4.B and 7.4.H, it is natural to classify the subgroups of Mod_S in a manner similar to Thurston’s classification of elements of Mod_S . Namely, a subgroup of Mod_S can be *reducible*, (i.e., not irreducible), finite, or infinite irreducible. (Note that these classes are not completely mutually exclusive: a subgroup can be finite and reducible.) The following theorem provides a useful complement to Theorem 7.4.H and this classification.

Theorem 7.4.I *If a subgroup G of Mod_S contains a pseudo-Anosov element, then either G contains an infinite cyclic subgroup of finite index generated by a pseudo-Anosov element, or G contains a free group freely generated by two pseudo-Anosov elements.*

Theorems 7.4.C–7.4.G are proved using the same general outline. First, we replace our group G by $G \cap \Gamma_S(m)$ for some $m \geq 3$. If the resulting subgroup (still denoted by G) is reducible, then one may cut S along some realization of a nonempty simplex σ of $C(S)$ such that $g(\sigma) = \sigma$ for all $g \in G$. This reduces the problem to a similar problem on the cut surface (which has simpler components). If a subgroup G is irreducible and infinite, then Theorems 7.4.H and 7.4.I provide us with pseudo-Anosov elements in G , which play the key role in the rest of the proof. (Note that finite groups present no problems in these theorems.) For the details, see [I15].

7.5 Commuting elements and Dehn twists

Thurston's theory allows us to formulate a very useful criterion for two elements of Mod_S to commute. We state it only for the pure elements, this case usually being sufficient for the applications.

Theorem 7.5.A *Let f, g be two pure elements of Mod_S . These elements commute (i.e. $fg = gf$) if and only if there is simplex σ of $C(S)$, its realization $C \subset S$ and representatives $F, G : S \rightarrow S$ of f, g respectively such that:*

- (i) *F and G preserve (setwise) every component of C and every component of $S \setminus C$;*
- (ii) *for every component R of the result S_C of cutting S along C , the diffeomorphisms $F_R, G_R : R \rightarrow R$ induced on R by F, G respectively are isotopic to some powers of the same pseudo-Anosov diffeomorphism of R or to the identity.*

The proof is based on the results of 7.3; see [I15], Section 8.12. As an application, we give a purely algebraic characterization of powers of Dehn twists inside subgroups of pure elements. Recall that the centralizer $C_G(g)$ of an element g of a group G is defined as the subgroup $\{f \in G : fg = gf\}$ and that the center $C(G)$ of a group G is defined as the subgroup $\{f \in G : fg = gf \text{ for all } g \in G\}$.

Theorem 7.5.B *Let G be a subgroup of Mod_S consisting entirely of pure elements (for example, let $G \subset \Gamma_S(m)$ for some $m \geq 3$). An element $g \in G$ is a nontrivial power of a Dehn twist if and only if the center $C(C_G(g))$ of the centralizer $C_G(g)$ of g is isomorphic to $\mathbf{Z}$ and is not equal to the centralizer $C_G(g)$ of g .*

In fact, $C(C_G(g))$ is isomorphic to $\mathbf{Z}$ only for nontrivial powers of Dehn twists and for pseudo-Anosov elements. The second condition serves to exclude

pseudo-Anosov elements. This theorem is a version of a characterization of Dehn twists in [18].

Since powers of Dehn twists obviously are pure elements, one can apply Theorem 7.5.A to them and get the following result (admittedly, known before Thurston's theory).

Theorem 7.5.C *Let t_α, t_β be two Dehn twists and let m, n be two nonzero integers. Then $t_\alpha^n t_\beta^m = t_\beta^m t_\alpha^n$ if and only if α and β can be represented by disjoint circles.*

Theorem 7.5.A also allows us to easily find centers of the mapping class groups.

Theorem 7.5.D *If S is neither a sphere with ≤ 4 holes, nor a torus with ≤ 2 holes, nor a closed surface of genus 2, then the center $C(\text{Mod}_S)$ is trivial (i.e. $C(\text{Mod}_S) = 1$). If S is a closed surface of genus 2, then the center $C(\text{Mod}_S)$ is isomorphic to $\mathbf{Z}/2\mathbf{Z}$ and is generated by the hyperelliptic involution from 4.3.*

The corresponding result for the cases excluded in this theorem is only slightly more complicated. Theorem 7.5.D is well known, but apparently there is no good reference for it.

Corollary 7.5.E *If S is neither a sphere with ≤ 4 holes, nor a torus with ≤ 2 holes, nor a closed surface of genus 2, then the canonical action of $\text{Mod}_S^\diamond$ on $C(S)$ is effective. If S is a closed surface of genus 2, the kernel of the canonical action of $\text{Mod}_S^\diamond$ on $C(S)$ is equal to the subgroup of order 2 generated by the hyperelliptic involution from 4.3.*

This easily follows from Theorem 7.5.D, if one uses the fact that Dehn twists generate PMod_S (by Theorem 4.2.C).

8 Automorphisms of complexes of curves

8.1 The theorem about automorphisms

The action of $\text{Mod}_S^\diamond$ on $C(S)$ induces a homomorphism $\text{Mod}_S^\diamond \rightarrow \text{Aut}(C(S))$. This homomorphism is injective if S is neither a sphere with ≤ 4 holes, nor a torus with ≤ 2 holes, nor a closed surface of genus 2; cf. Corollary 7.5.E. It turns out that this homomorphism is also surjective in these cases, and also for closed surfaces of genus 2.

Theorem 8.1.A *Suppose that S is not a sphere with ≤ 4 holes and not a torus with ≤ 2 holes. Then all automorphisms of $C(S)$ are induced by the elements of $\text{Mod}_S^\diamond$. In particular, if, in addition, S is not a closed surface of genus 2, then $\text{Aut}(C(S))$ is canonically isomorphic to $\text{Mod}_S^\diamond$. If S is a closed surface of genus 2, then $\text{Aut}(C(S))$ is canonically isomorphic to the quotient of the group $\text{Mod}_S^\diamond$ by the subgroup of order 2 generated by the hyperelliptic involution.*

This theorem is due to the author [I10] for surfaces S of genus ≥ 2 , and to M. Korkmaz [Kor] in the remaining cases. See also [I17] for an expanded and updated version of [I10]. Most of the proof (namely, the ideas discussed in 8.4) for the genus ≥ 2 case works also in the genus 0 and 1 cases. The main difficulty in treating these low genus cases, overcome by M. Korkmaz, consisted in finding and proving an analogue of Lemma 8.2.A below.

If S is either a sphere with 4 holes, or a torus with ≤ 1 holes, then $C(S)$ is an infinite set of vertices without any edges (i.e., $\dim C(S) = 0$) and, hence, $\text{Aut}(C(S))$ is an infinite symmetric group. Obviously, diffeomorphisms of S cannot permute vertices of $C(S)$ arbitrarily, even in these cases, and thus the conclusion of the theorem is not true for such S . If S is a sphere with at most 3 holes, then $C(S)$ is empty, but Mod_S° is not trivial (it includes at least one orientation-reversing mapping class). Hence, the conclusion of the theorem is not true, by trivial reasons, for such S also. F. Luo [Luo2] observed that $\text{Aut}(C(S))$ is not equal to Mod_S if S is a torus with 2 holes. The reason is very simple: If $S_{1,2}$ is a torus with 2 holes and $S_{0,5}$ is a sphere with 5 holes, then $C(S_{1,2})$ is isomorphic to $C(S_{0,5})$, but $\text{Mod}_{S_{1,2}}$ is not isomorphic to $\text{Mod}_{S_{0,5}}$; in fact, $\text{Mod}_{S_{0,5}}$ acts transitively on the set of vertices of $C(S_{0,5})$ and the action of $\text{Mod}_{S_{1,2}}$ on the set of vertices of $C(S_{1,2})$ has two orbits, corresponding to separating and nonseparating circles (cf. 8.6 for a similar phenomenon). F. Luo [Luo2] also suggested another approach to Theorem 8.1.A, still based on the ideas of [I10] (outlined below) and also on a multiplicative structure on the set of vertices of $C(S)$ introduced in [Luo1].

Theorem 8.1.A is similar to, and was motivated by, a well known theorem of J. Tits [Tits2] to the effect that all automorphisms of a Tits building (with some exceptions, as in our case) stem from automorphisms of the corresponding algebraic group; see [Tits2], Introduction, Problem (B). The theorem of Tits, in turn, has its roots in the *Fundamental Theorem of Projective Geometry*, asserting that maps of a projective space to itself which map lines to lines, planes to planes etc., are, in fact, (projectively) linear.

In 8.2-8.4 we will outline the main ideas of the proof of Theorem 8.1.A. After this, we will discuss applications of this result to mapping class groups and Teichmüller spaces.

8.2 Intersection number 1 property

We start by introducing some terminology. Two isotopy classes α, β of non-trivial circles on S (i.e., two vertices of $C(S)$) are said to have *geometric intersection number* 1 (respectively, 0) if they can be represented by two circles intersecting transversely in exactly one point (respectively, by two disjoint circles). If this is the case, we write $i(\alpha, \beta) = 1$ (respectively, $i(\alpha, \beta) = 0$). (These definitions form a special case of a general notion of geometric intersection number from 7.2.) Clearly, two isotopy classes have geometric intersection number 0 if and only if they are connected by an edge in $C(S)$. Obviously, these intersection

numbers are preserved by the action of $\text{Mod}_S^\diamond$. The starting point of the proof of Theorem 8.1.A (at least for surfaces of genus ≥ 1) is the fact that the property of having geometric intersection number 1 is preserved by all automorphisms of $C(S)$ (which a priori preserve only the property of having geometric intersection number 0). The key part of the proof of this fact is contained in the following lemma, at least when the genus is ≥ 2 .

Lemma 8.2.A *Suppose that the genus of S is at least 2. Let α_1, α_2 be isotopy classes of two nontrivial circles on S . Then the geometric intersection number $i(\alpha_1, \alpha_2) = 1$ if and only if there exist isotopy classes $\alpha_3, \alpha_4, \alpha_5$ of nontrivial circles having the following two properties:*

- (i) $i(\alpha_i, \alpha_j) = 0$ if and only if the i -th and j -th circles in Fig. 8.1 are disjoint;
- (ii) if α_4 is the isotopy class of a circle C_4 , then C_4 divides S into two parts, one of which is a torus with one hole containing some representatives of the isotopy classes α_1, α_2 .

Proof. Fig. 8.1 provides a proof of the “only if” part. In view of the property (ii), the proof of the “if” part can be reduced to an examination of a torus with one hole, which is not difficult. We omit the details. $\square$

Corollary 8.2.B *Suppose that the genus of S is at least 2. Every automorphism of $C(S)$ maps pairs of vertices having geometric intersection number 1 into pairs of vertices also having this property.*

Proof. It is sufficient to show that the properties (i) and (ii) of Lemma 8.2.A are preserved by every automorphism of $C(S)$. As we noted before the lemma, two vertices have geometric intersection number 0 if and only if they are connected by an edge in $C(S)$. It follows that the property (i) is preserved by all automorphisms. In order to prove that the property (ii) is also preserved, we will show how to express it in terms of the structure of $C(S)$ only.

Let us consider a vertex $\alpha = \langle C \rangle$ of $C(S)$. Let L_α be the link of α in $C(S)$, i.e. the set of all vertices of $C(S)$ connected by an edge with α (with the induced structure of a simplicial complex). Consider the graph L_α^* , having the same vertices as the link L_α and having as edges exactly those pairs of vertices which are *not* edges of L_α . It is easy to see that the connected components of L_α^* correspond exactly to the connected components of the result S_C of cutting S along C (at least if none of them is a sphere with 3 holes). After detecting the components of S_C with the help of L_α^* (when one of the components is a sphere with 3 holes, we can detect only the other, but this turns out to be sufficient), we can return to the link L_α itself and try use Corollary 3.3.B, or the table preceding it, to recognize the topological type of these components.

Let us apply these remarks to α_5 . The part of S *not* containing representatives of α_1, α_2 should have genus one less than the genus of S and one more

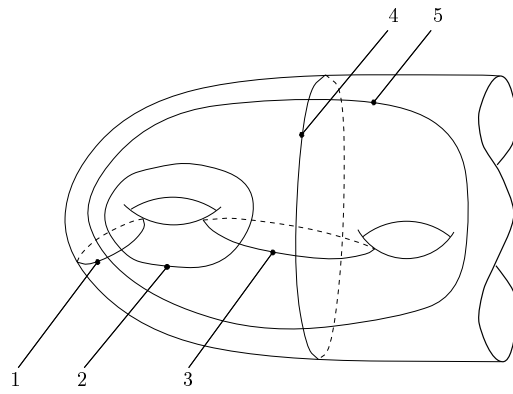


Fig. 8.1

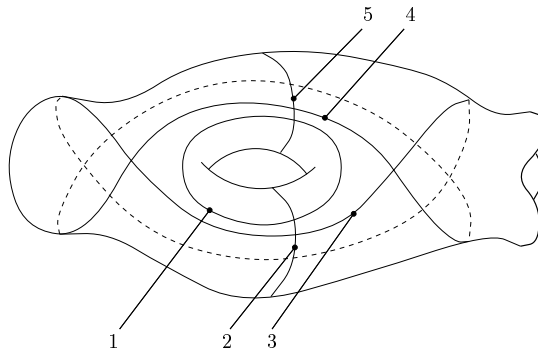


Fig. 8.2

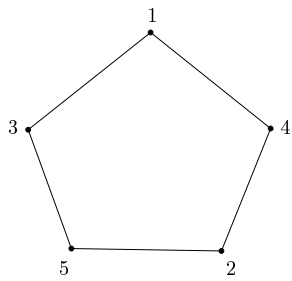


Fig. 8.3

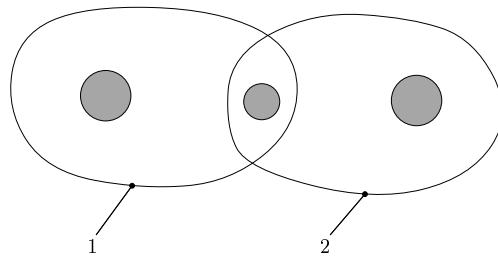


Fig. 8.4

boundary component. If the genus of S is ≥ 3 , these properties can be recognized with the help of Corollary 3.3.B. The case of genus 2 is only a little more difficult; one should use not only this Corollary, but also the table preceding it. It follows that the property (ii) can be expressed in terms of the structure of $C(S)$ alone and, hence, is preserved by all automorphisms of $C(S)$. This completes the proof. $\square$

Now, we turn to the genus 1 case. The following analogue of Lemma 8.2.A was proved by M. Korkmaz [Kor].

Lemma 8.2.C *Suppose that S is a torus with at least two holes. Let α_1, α_2 be isotopy classes of two nontrivial circles on S . Then the geometric intersection number $i(\alpha_1, \alpha_2) = 1$ if and only if there exist isotopy classes $\alpha_3, \alpha_4, \alpha_5$ having the following two properties:*

(i) $i(\alpha_i, \alpha_j) = 0$ if and only if the i -th and j -th circles in Fig. 8.2 are disjoint;

(ii) if α_i is the isotopy class of a circle C_i , then the circles C_1, C_2 and C_3 are non separating, and both C_4 and C_5 divide S into a torus with one hole and a sphere with $b + 1$ holes, where b is the number of boundary components of S .

Proof. Similarly to the proof of Lemma 8.2, the proof of the “only if” part is provided by the Fig. 8.2. The proof of the “if” part is a little more difficult than the proof of the “if” part of Lemma 8.2, but we again omit the details. $\square$

Corollary 8.2.D *Suppose that S is a torus with at least two holes. Every automorphism of $C(S)$ maps pairs of vertices having geometric intersection number 1 into pairs of vertices also having this property.*

The proof is more difficult than that of Corollary 8.2.B, partly because the table preceding Corollary 3.3.B is not sufficient to distinguish between the surfaces of genus 0 and 1. We omit fairly complicated details.

Surprisingly, the vertices $\alpha_1, \dots, \alpha_5$ of both Lemmas 8.2.A and 8.2.C with the edges connecting them form a pentagon, presented in Fig. 8.3, and a similar pentagon plays a role in the case of genus 0.

8.3 The case of a sphere with holes

At first sight, none of the above ideas can be applied when S is a sphere with holes: all circles on a sphere with holes are separating and, hence, two isotopy classes never have geometric intersection number 1. It turns out that instead of non separating circles, one can use circles bounding a disc with two holes in S . Then the role of pairs of isotopy classes having geometric intersection number 1 is taken over by the pairs of isotopy classes of such circles having the simplest possible non-trivial intersection, namely, intersecting as in Fig. 8.4,

where the shadowed discs represent holes. Such pairs of isotopy classes can be characterized in a manner similar to Lemmas 8.2.A and 8.2.C. A pentagon as in Fig. 8.3 appears again, but its five vertices are not sufficient now. We refer to [Kor] for the details.

Note that the circles bounding discs with two holes and pairs illustrated in Fig. 8.4 are exactly what is needed to make the machinery of the next section work (cf. the discussion of codings in 8.4).

8.4 Complexes of curves and ideal triangulations

Now we describe the ideas involved in the remaining part of the proof of Theorem 8.1.A. These ideas work (almost) equally well for surfaces of any genus, we restrict ourselves to the genus ≥ 2 case.

We need a minor modification $A(S)$ of the ideal triangulations of the Teichmüller spaces from 5.5. The vertices of the complex $A(S)$ are the isotopy classes $\langle I \rangle$ of *properly embedded non-trivial* (cf. 2.3) arcs I in S . The isotopies are not required to be fixed at the ends, but are required to run in the class of properly embedded arcs. A set of vertices $\{\gamma_0, \dots, \gamma_n\}$ forms a simplex of $A(S)$ if and only if the isotopy classes $\gamma_0, \dots, \gamma_n$ can be represented by pairwise disjoint arcs $I_0, \dots, I_n$. The extended mapping class group Mod_S° acts on $A(S)$ in an obvious way.

We can collapse each boundary component of S into a point (different for different boundary components) and then consider these points as punctures. Let R be the resulting surface with punctures. There is a natural map $S \rightarrow R$, establishing a diffeomorphism between the complements of the boundary and of the punctures. Clearly, this map induces an isomorphism between $A(S)$ and the complex $A(R)$ from 5.5. Using this isomorphism and Corollary 5.5.B, we immediately get the following result.

Lemma 8.4.A *Every automorphism of $A(S)$ is determined by its action on any single top dimensional simplex. In particular, if an automorphism of $A(S)$ agrees with the action of some element of Mod_S° on some simplex of codimension 0, then this automorphism agrees with this element of Mod_S° on the whole complex $A(S)$. $\square$*

In view of the last Lemma, the automorphisms of $A(S)$ seem to be much more accessible than the automorphisms of $C(S)$. The main idea of the remaining part of the proof of Theorem 8.1.A is to reduce the study of automorphisms of $C(S)$ to a study of automorphisms of $A(S)$. The main tool is a coding of the vertices of $A(S)$ in terms of the vertices of $C(S)$ and some additional data.

Let us describe this coding. The vertices of $A(S)$ are naturally divided into three types, and a vertex is coded by its type and one or two vertices of $C(S)$, depending on its type. The types of vertices are the following: (a) the isotopy classes of arcs I connecting two different components D_1, D_2 of ∂S ; (b) the

isotopy classes of arcs I connecting a component D of ∂S with itself such that the image of this arc in R (this image is a circle passing through a puncture) does not bound a disc with one puncture; (c) the isotopy classes of arcs I connecting a component D of ∂S with itself such that the image of this arc in R does bound a disc with one puncture. In addition to the type, a vertex of type (a) is coded by $\langle C \rangle$, where C is the boundary of some regular neighborhood of $D_1 \cup I \cup D_2$ in S , a vertex of type (b) is coded by the pair $\langle C_1 \rangle, \langle C_2 \rangle$, where C_1, C_2 are two components of a regular neighborhood of $D \cup I$, and a vertex of type (c) is coded by $\langle C \rangle$, where C is the non-trivial component of a regular neighborhood of $D \cup I$ (one of the components is trivial in this case).

Now, we need to prove that every automorphism of $C(S)$ maps vertices coding vertices of $A(S)$ of type (a) into similar vertices, behaves similarly with respect to types (b) and (c) and also maps codings of pairs of vertices connected by an edge in $A(S)$ into similar codings. Fig 8.4 may serve as an illustration of codings of a pair of vertices connected by an edge. This is the most technically difficult part of the proof, which splits into many cases according to the types of the vertices involved. After this is done, we can assign an automorphism of $A(S)$ to any automorphism of $C(S)$. Moreover, one can prove that on an explicitly given codimension 0 simplex of $A(S)$ every automorphism of $A(S)$ induced by an automorphism of $C(S)$ agrees with an element of Mod_S° , at least if the number of boundary components is ≥ 2 . In this case Lemma 8.4.A implies that the induced automorphism of $A(S)$ is equal to some element of Mod_S° . And if the induced automorphism of $A(S)$ is equal to some element of Mod_S° , then the original automorphism of $C(S)$ is equal to some element of Mod_S° , as it is easy to see.

This proves Theorem 8.1.A for surfaces with ≥ 2 boundary components. The case of surfaces with ≤ 1 boundary components can be reduced to the case of surfaces with ≥ 2 boundary components as follows. Note first that every automorphism of $C(S)$ takes the isotopy classes of non separating circles into isotopy classes of non separating circles (for example, because C is non separating if and only if there exist a vertex β such that $i(\alpha, \beta) = 1$, where $\alpha = \langle C \rangle$). Since all non separating circles are in the same orbit of the group of diffeomorphisms of S , we can assume that our automorphism of $C(S)$ fixes some vertex $\alpha = \langle C \rangle$, where C is non separating. Such an automorphism induces an automorphism of the link L_α of α and, hence, of the complex $C(S_C)$, where S_C is the result of cutting S along C (compare with the proof of Corollary 8.2.B). Since S_C has ≥ 2 boundary components, this automorphism of $C(S_C)$ is equal to some element of $\text{Mod}_{S_C}^\circ$. Considering different non separating circles C (in fact, all of them), one can deduce that the original automorphism of $C(S)$ is equal to some element of Mod_S° . (If the genus of S is 2, an additional difficulty is caused by the fact that the genus of S_C will be < 2 . But our automorphisms of $C(S_C)$ are induced by automorphisms of $C(S)$ and, hence, retain all the nice properties of the latter, such as the preservation of the intersection number 1.)

This completes our outline of the proof of Theorem 8.1.A.

8.5 An application to subgroups of finite index

As the first application of Theorem 8.1.A, we give a complete description of isomorphisms between subgroups of finite index in $\text{Mod}_S^\diamond$. This result will find its own applications in 9.2; cf. Corollary 9.2.B.

Theorem 8.5.A *Suppose that S is not a sphere with ≤ 4 holes and not a torus with ≤ 2 holes. Let Γ_1 and Γ_2 be two subgroups of finite index of $\text{Mod}_S^\diamond$. If S , in addition, is not a closed surface of genus 2, then all isomorphisms $\Gamma_1 \rightarrow \Gamma_2$ have the form $x \mapsto gxg^{-1}$ for some $g \in \text{Mod}_S^\diamond$. If S is a closed surface of genus 2, then all isomorphisms $\Gamma_1 \rightarrow \Gamma_2$ have the form $x \mapsto gxg^{-1}\chi(x)$ for some $g \in \text{Mod}_S^\diamond$ and some homomorphism χ from Γ_1 to the center of $\text{Mod}_S^\diamond$, i.e. the subgroup of $\text{Mod}_S^\diamond$ of order 2 generated by the hyperelliptic involution (see Theorem 7.5.D); χ can be non-trivial only if Γ_2 contains the hyperelliptic involution.*

Proof. First, note that for some natural number $N \neq 0$ the N -th powers t_α^N of all Dehn twists t_α are contained in Γ_1 , because Γ_1 is of finite index. Next, Theorem 7.5.B easily implies that any isomorphism $\varphi : \Gamma_1 \rightarrow \Gamma_2$ maps sufficiently high powers of Dehn twists into powers of Dehn twists. Taking into account the fact that two non-trivial powers of Dehn twists commute if and only if the corresponding isotopy classes of circles have geometric intersection number 0 (by Theorem 7.5.C), we see that every such isomorphism φ induces an automorphism $C(S) \rightarrow C(S)$. By Theorem 8.1.A, this automorphism of $C(S)$ is given by some element $g \in \text{Mod}_S^\diamond$. This means that for some sufficiently large N , we have

$$\varphi(t_\alpha^N) = t_{g(\alpha)}^{M_\alpha}$$

for some $M_\alpha \neq 0$, for all vertices α of $C(S)$. The potential dependence of M_α on α is irrelevant in what follows, and we write simply M for M_α .

Now, let $f \in \Gamma_1$. Then, for any vertex α ,

$$\varphi(ft_\alpha^N f^{-1}) = \varphi(t_{f(\alpha)}^N) = t_{g(f(\alpha))}^M.$$

On the other hand,

$$\varphi(ft_\alpha^N f^{-1}) = \varphi(f)\varphi(t_\alpha^N)\varphi(f)^{-1} = \varphi(f)t_{g(\alpha)}^M\varphi(f)^{-1} = t_{\varphi(f)(g(\alpha))}^M.$$

Comparing the results of these two computations, we conclude (note that two nontrivial powers t_α^n, t_β^m of right Dehn twists are equal if and only if $\alpha = \beta$ and $m = n$) that $\varphi(f)(g(\alpha)) = g(f(\alpha))$ for all α and (after putting $\alpha = g^{-1}(\beta)$) that $\varphi(f)(\beta) = gfg^{-1}(\beta)$ for all vertices β of $C(S)$. If S is not a closed surface of genus 2, this implies that $\varphi(f) = gfg^{-1}$, i.e. φ has the required form; see Corollary 7.5.E. If S is a closed surface of genus 2, then $\varphi(f)$ can differ from gfg^{-1} by the hyperelliptic involution. It is easy to see that this difference can be described by some homomorphism χ as stated. $\square$

Corollary 8.5.B *Suppose that S is not a sphere with ≤ 4 holes and not a torus with ≤ 2 holes. Let Γ be a subgroup of finite index of $\text{Mod}_S^\diamond$. Then the outer automorphism group $\text{Out}(\Gamma)$ is finite. $\square$*

Theorem 8.5.A was proved first by the author [I2] for the case $\Gamma_1 = \Gamma_2 = \text{Mod}_S^\diamond$ or Mod_S and for closed surfaces only. Another, but closely related, proof in this case was provided by J. McCarthy [Mc3], who also noticed additional automorphisms resulting from the hyperelliptic involution in the genus 2 case (which were overlooked in [I2]). An alternative approach was suggested by R. T. Tchangang [T]. Shortly after [I2] the author extended these results to surfaces with boundary and to some natural subgroups of $\text{Mod}_S^\diamond$ like $\Gamma_1 = \Gamma_2 = \text{PMod}_S$; see [I8]. A crucial element of these early approaches, namely the fact that any isomorphism φ takes powers of Dehn twists to powers of Dehn twists, also plays a crucial role in the above proof of Theorem 8.5.A.

Recently, J. McCarthy and the author extended the results of [I8], [Mc3] to some injective homomorphisms between mapping class groups. Let us measure the *size* of a surface S by the number $3g - 3 + b$, where g is the genus and b is the number of boundary components of S (this number is equal to the maximal number of pairwise nonisotopic disjoint circles on S). The main result of [IMc1], [IMc2] asserts that if the sizes of two surfaces S, S' differ by at most 1 and they have the genus ≥ 1 , then injective homomorphisms $\text{Mod}_S \rightarrow \text{Mod}_{S'}$ are, in fact isomorphisms and S is diffeomorphic to S' with, possibly, a few exceptions.

8.6 An application to Teichmüller spaces

In this section we consider the Teichmüller spaces of *punctured* surfaces (cf. 5.1).

The Teichmüller space T_R of a punctured surface R carries a natural structure of a metric space, given by its *Teichmüller metric* (we note in passing that Teichmüller spaces have a couple of other natural metrics). This structure of a metric space is not derived from a riemannian metric (although it can be derived from a finslerian metric), but nevertheless has some nice geometric properties. In particular, T_R is a complete metric space, every two points can be connected by a geodesic (i.e., a locally shortest path), and, moreover, geodesics have a nice description in terms of the (conformal) geometry of surfaces. For an introduction to this theory, see W. Abikoff's book [A]. The fact that the metric is naturally defined implies that $\text{Mod}_R^\diamond$ acts on T_R by isometries. A remarkable result proved by H. Royden [R] for closed surfaces (and isometries preserving the natural complex structure of T_S), and then extended by C. Earle and I. Kra [EK] to the surfaces with boundary (and general isometries), asserts that there are no other isometries. More precisely, we have the following theorem.

Theorem 8.6.A *Suppose that R is not a sphere with ≤ 4 punctures and not a torus with ≤ 1 punctures. Then any isometry of T_R is induced by some element of $\text{Mod}_R^\diamond$.*

Note that the conclusion of this theorem is not true for a sphere with 4 punctures and a torus with 0 or 1 punctures, because for these surfaces the Teichmüller space is isometric to the hyperbolic plane, which has a continuous group of isometries. For a sphere with ≤ 3 punctures the conclusion is vacuous, because the Teichmüller space consists of just one point. For a torus with 2 punctures the conclusion of the theorem is not true, because if $S_{1,2}$ is a torus with 2 punctures and $S_{0,5}$ is a sphere with 5 punctures, then $T_{S_{1,2}}$ is isometric to $T_{S_{0,5}}$, but $\text{Mod}_{S_{1,2}}$ is not isomorphic to $\text{Mod}_{S_{0,5}}$ (cf. 8.1). See [EK].

H. Royden's proof (and its extension by C. Earl and I. Kra) is analytic and is based on a detailed investigation of the non-smoothness properties of the finislerian metric underlying the Teichmüller metric. It is essentially local in nature. A detailed exposition of this proof is presented in F. Gardiner's book [Ga]; cf. [Ga], Chapter 9. Our Theorem 8.1.A can be used to prove this theorem in a completely different manner (in all cases except that of a torus with two punctures). This new proof, outlined in [I10], reveals a deep analogy between Royden's theorem and G. D. Mostow's rigidity theorems [Most1], [Most2]. In fact, Theorem 8.1.A plays a role in this proof similar to the role played by the theorem of Tits mentioned in 8.1 in Mostow's proof [Most2]. In addition to Theorem 8.1.A this proof of Theorem 8.6.A, naturally, relies heavily on the theory of the Teichmüller spaces and, in particular, on the results of S. Kerckhoff [Ke1] and H. Masur [Mas1]–[Mas3]. Unfortunately, this theory falls outside the scope of the present paper.

9 Mapping class groups and arithmetic groups

9.1 Arithmetic groups

The analogy between arithmetic groups and mapping class groups was probably suggested first by Harvey [Harv1], and had guided much of research about the mapping class groups since then. Let us give a down-to-earth version of the definition of arithmetic groups.

The basic example of an arithmetic group is $SL_n(\mathbf{Z})$, the group of all integral $n \times n$ matrices with the determinant 1 (note that the inverse of an integral matrix with the determinant 1 is automatically an integral matrix). All other arithmetic groups can be, in some sense, constructed from this one. Suppose that G is a *semisimple $\mathbf{Q}$ -algebraic* subgroup of $SL_n(\mathbf{R})$. This means that G is simultaneously a semisimple Lie subgroup of $SL_n(\mathbf{R})$ and a subset of the space of all real $n \times n$ matrices which can be described by polynomial equations for the matrix entries with *rational* coefficients. For any such G the intersection $G_{\mathbf{Z}}$ of G with the set of integral matrices is, by definition, an arithmetic group. Next, let G_0 be the connected component of the identity of G and $\varphi : G_0 \rightarrow H$ be a homomorphism from G_0 onto a connected semisimple Lie group H with compact kernel (to be honest, we should also assume that

H has trivial center and no compact factors, but the reader may ignore this remark). Then $\Gamma = \varphi(G_{\mathbf{Z}} \cap G_0)$ is, by definition, an arithmetic group, as is also any subgroup Γ' of H *commensurable* with it, i.e. any subgroup Γ' such that the intersection $\Gamma \cap \Gamma'$ has finite index both in Γ and Γ' . More precisely, we say that any such Γ' is an *arithmetic subgroup of H* . We also call *arithmetic* any group isomorphic to any such Γ' .

So, any arithmetic group can be obtained from $SL_n(\mathbf{Z})$ in three steps: taking an intersection with a semisimple $\mathbf{Q}$ -algebraic real Lie group, then taking the image under a surjective homomorphism with compact kernel (this step is the most difficult to control), and, finally, taking any subgroup commensurable with this image. In particular, any subgroup of finite index of an arithmetic group is also an arithmetic group.

In addition to $SL_n(\mathbf{Z})$, a typical example of an arithmetic group is the symplectic group $Sp_{2g}(\mathbf{Z})$, which shows up in the topology of surfaces as the group of all automorphisms of the first homology group $H_1(S_g)$ of a closed orientable surface S_g of genus g preserving the intersection pairing. In particular, we have a natural homomorphism $\text{Mod}_{S_g} \rightarrow Sp_{2g}(\mathbf{Z})$, which is well known to be surjective.

In [Harv1], W. Harvey asked if the mapping class groups are arithmetic. They turned out to not be, as was proved by the author [I2], [I7]. On the other hand, the years after [Harv1] profoundly demonstrated that there is indeed a deep analogy between mapping class groups and arithmetic groups, especially in their cohomology properties (the main results of Section 6 are, in fact, motivated by analogous results about arithmetic groups). Although the first proof of the non-arithmeticity of the mapping class groups was quite short (cf. 9.3), it did not exhaust the problem, and the question of *why* the mapping class groups are not arithmetic led to further results. In the next section (cf. 9.2) we will describe what seems to be the best answer to this question.

9.2 Abstract commensurators

The notion of the abstract commensurator of a group has its roots in the theory of arithmetic groups. The author learned it from [NeR]; see also [BasK], Appendix B, where it is attributed to J.-P. Serre and W. D. Neumann. It allows us to express one of the most striking differences between arithmetic and non-arithmetic groups; cf. Theorem 9.2.A below and the remarks after it.

Let Γ be a group. Let us consider all possible isomorphisms $\varphi : \Gamma_1 \rightarrow \Gamma_2$ between subgroups Γ_1, Γ_2 of finite index of Γ . Let us identify two such isomorphisms φ, φ' defined on Γ_1, Γ'_1 respectively, if they agree on a subgroup of finite index in the intersection $\Gamma_1 \cap \Gamma'_1$. We can compose them in an obvious manner; the composition $\varphi \circ \varphi'$ of $\varphi : \Gamma_1 \rightarrow \Gamma_2$ with $\varphi' : \Gamma'_1 \rightarrow \Gamma'_2$ is defined on $\varphi^{-1}(\Gamma_2 \cap \Gamma'_1)$. Under this composition operation, the classes of such isomorphisms form a group, which is called the *abstract commensurator* of Γ and is denoted by $\text{Comm}(\Gamma)$. There is a natural map $i : \Gamma \rightarrow \text{Comm}(\Gamma)$, sending an

element γ of Γ to the (class of the) inner automorphism $g \mapsto \gamma g \gamma^{-1}$. This map is injective if the centralizers of subgroups of finite index in Γ are trivial, as is the case for arithmetic groups and for mapping class groups (see Theorem 7.5.D for the latter).

As a nice example, let us mention that $\text{Comm}(\mathbf{Z}^n) = SL_n(\mathbf{Q})$.

Theorem 9.2.A *If Γ is an arithmetic group, then $i(\Gamma)$ is of infinite index in $\text{Comm}(\Gamma)$.*

Proof. We only outline the main idea. Let us consider $SL_n(\mathbf{Z})$ and show that any element of $SL_n(\mathbf{Q})$ naturally defines an element of $\text{Comm}(SL_n(\mathbf{Z}))$.

Let $g \in SL_n(\mathbf{Q})$, and let m be the least common multiple of all denominators of the entries of the matrices g, g^{-1} . Let $\Gamma_1 = \{\gamma \in SL_n(\mathbf{Z}) \mid \gamma \equiv I \pmod{m^2}\}$, where I is the identity matrix. Clearly, Γ_1 is of finite index in $SL_n(\mathbf{Z})$. If $\gamma \in \Gamma_1$, then $\gamma = I + m^2 h$, where h is an integral matrix, and hence $g \gamma g^{-1} = I + m^2 g h g^{-1}$ is also an integral matrix. Thus $g \Gamma_1 g^{-1} \subset SL_n(\mathbf{Z})$. Let us show that $\Gamma_2 = g \Gamma_1 g^{-1}$ is a subgroup of finite index in $SL_n(\mathbf{Z})$. Consider the subgroup $\Gamma' = \{\gamma \in SL_n(\mathbf{Z}) \mid \gamma \equiv I \pmod{m^4}\}$. If $\gamma \in \Gamma'$, then $\gamma = I + m^4 h$, where h is an integral matrix, and hence $g^{-1} \gamma g = I + m^2 (m^2 g^{-1} h g) \in \Gamma_1$, because $m^2 g^{-1} h g$ is an integral matrix. It follows that $g^{-1} \Gamma' g \subset \Gamma_1$, and hence $\Gamma' \subset g \Gamma_1 g^{-1}$. Since Γ' obviously has finite index in $SL_n(\mathbf{Z})$, we conclude that $\Gamma_2 = g \Gamma_1 g^{-1} \supset \Gamma'$ is of finite index.

Now, it is clear that the inner automorphism $x \mapsto g x g^{-1}$ induces an isomorphism $\Gamma_1 \rightarrow \Gamma_2$ between two subgroups of finite index of $SL_n(\mathbf{Z})$, and hence an element of $\text{Comm}(SL_n(\mathbf{Z}))$. This leads to a homomorphism $SL_n(\mathbf{Q}) \rightarrow \text{Comm}(SL_n(\mathbf{Z}))$. Clearly, its image contains $i(SL_n(\mathbf{Z}))$ as a subgroup of infinite index. This proves the theorem for $SL_n(\mathbf{Z})$.

The same argument works for the groups $G_{\mathbf{Z}}$, where G is a semisimple $\mathbf{Q}$ -algebraic group, simply because $G_{\mathbf{Z}}$, like $SL_n(\mathbf{Z})$, has a corresponding group of rational matrices associated to it. After this, it is not hard to extend the result to the groups of the form $\varphi(G_{\mathbf{Z}} \cap G_0)$ (cf. 9.1) and then to any group commensurable with such a group. We refer to R. Zimmer's book [Zim] for the details; cf. [Zim], Section 6.2. $\square$

A converse to this theorem is also true: if Γ is a *lattice* in a semisimple Lie group G (i.e. if G/Γ has finite invariant volume) and $i(\Gamma)$ is of infinite index in $\text{Comm}(\Gamma)$, then Γ is arithmetic. This result is an immediate corollary of an arithmeticity theorem of G. A. Margulis (cf. [Zim], Chapter 6) and G. D. Mostow's rigidity theorem [Most1], [Most2]; cf. [NeR], Section 1. This converse is much more deep and difficult than Theorem 9.2.A, and we do not need it for our applications.

Corollary 9.2.B *Suppose that S is not a sphere with ≤ 4 holes and not a torus with ≤ 2 holes. Then the mapping class groups Mod_S , $\text{Mod}_S^\diamond$, and all subgroups of finite index in them, are not arithmetic.*

Proof. It follows from Theorem 8.5.A that $\text{Comm}(\text{Mod}_S) = \text{Comm}(\text{Mod}_S^\diamond) = i(\text{Mod}_S^\diamond)$, and hence $\text{Mod}_S, \text{Mod}_S^\diamond$ cannot be arithmetic by Theorem 9.2.A.

If Γ is a subgroup of finite index in $\text{Mod}_S^\diamond$, then again $\text{Comm}(\Gamma) = i(\text{Mod}_S^\diamond)$ by Theorem 8.5.A. (Note that the additional automorphisms for the closed surfaces of genus 2 disappear when we pass to a subgroup of finite index not containing the hyperelliptic involution.) $\square$

If S is a sphere with ≤ 4 holes or a torus with ≤ 1 holes, then Mod_S is indeed arithmetic: if S is a torus with ≤ 1 holes, then Mod_S is isomorphic to $SL_2(\mathbf{Z})$; if S is a sphere with 4 holes, then Mod_S is commensurable with $PSL_2(\mathbf{Z}) = SL_2(\mathbf{Z})/\{\pm I\}$ (cf., [I8], Section 7 for a detailed discussion of this case); if S is a sphere with ≤ 3 holes, then Mod_S is finite. If S is a torus with two holes, then Mod_S is not arithmetic. The above argument does not (currently) apply to this case, because Theorem 8.1.A does not include it. But older approaches, discussed in the next two sections (cf. 9.3 and 9.4) do apply to this case also.

It is worth stressing that this proof of the non-arithmeticity of the mapping class groups uses only very basic properties of arithmetic groups, going not much further than the definition, and a deep result (Theorem 8.1.A) from the topology of surfaces. This contrasts strongly with the original approach to the non-arithmeticity, which relied on the deepest properties of arithmetic groups; cf. 9.3.

9.3 The original approach to the non-arithmeticity

Arithmetic groups, like representations, can be divided into reducible and irreducible ones. An arithmetic group Γ is called *reducible* if it is “close” to a product of two infinite groups; in particular, if Γ is reducible, then it contains two subgroups Γ_1, Γ_2 of infinite index commuting with each other and generating together a subgroup of finite index in Γ . Using the last remark and Theorem 7.5.A, it is easy to see that no subgroup of finite index in $\text{Mod}_S^\diamond$ can be isomorphic to a reducible arithmetic group. If Γ is not reducible, it is called *irreducible*.

By the definition, any arithmetic group Γ arises as a subgroup of a real semisimple Lie group H (without compact factors). The rank of the latter (i.e., the dimension of a Cartan subalgebra of its Lie algebra) is usually called the *rank* of Γ . In this section we will not address the question of whether this rank depends only on Γ or also on the realization of Γ as an arithmetic subgroup of some semisimple Lie group (cf. 9.4), and will consider our use of this notion as a slight abuse of language. The main point is that the properties of Γ depend strongly on whether the rank of Γ is 1 or ≥ 2 .

In particular, if Γ is an irreducible arithmetic group of rank ≥ 2 , then any normal subgroup of Γ is either a central finite subgroup or a subgroup of finite index. This is the famous Margulis finiteness theorem; see [Zim], Chapter 8 for

the details. Now, for a closed surface S_g of genus g the kernel of the natural homomorphism $\text{Mod}_{S_g} \rightarrow Sp_{2g}(\mathbf{Z})$ from 9.1 is an infinite (and non-central) subgroup of infinite index. Indeed, this kernel, usually called the *Torelli subgroup* of Mod_{S_g} , contains, for example, all Dehn twists about separating circles and the above homomorphism is surjective, as we noted in 9.1. It follows that Mod_{S_g} cannot be an irreducible arithmetic group of rank ≥ 2 . Similar arguments apply to surfaces with non empty boundary; for example, one can use the kernels of homomorphisms from 2.8 instead of the Torelli subgroup or a version of the latter.

It remains to show that Mod_S cannot be an irreducible arithmetic group of rank 1. One way to show this is to use the well known fact that any such arithmetic group contains a subgroup of finite index (actually, any torsion-free subgroup of finite index will work) which is isomorphic to the fundamental group of a complete riemannian manifold of finite volume with curvature pinched between two negative constants, together with the following theorem.

Theorem 9.3.A *If S is not a sphere with ≤ 4 holes and not a torus with ≤ 1 holes, then no subgroup of finite index in Mod_S° is isomorphic to the fundamental group of a complete riemannian manifold of finite volume with curvature pinched between two negative constants.*

Proof. We only indicate the main idea. It is sufficient to prove that any subgroup of finite index in Mod_S contains a subgroup isomorphic to $(\mathbf{Z} * \mathbf{Z}) \times \mathbf{Z}$, and that none of the considered fundamental groups does. Subgroups of Mod_S isomorphic to $(\mathbf{Z} * \mathbf{Z}) \times \mathbf{Z}$ can be constructed in abundance with powers of Dehn twists as generators. The absence of such subgroups in the considered fundamental groups follows easily from the theory of P. Eberlein and B. O’Neill [EbO]; see [I7], Proof of Theorem 2 for the details. $\square$

Irreducible arithmetic groups of rank 1 can also be excluded in a couple of other ways. See [Har5], Section 4.2 for a cohomological approach due to Harer, and still another approach due to W. Goldman. The latter is similar in spirit to the use of the Eberlein–O’Neill theory above.

The ideas outlined in this section, if combined with Mostow’s rigidity theorem [Most1], [Most2], lead also to some non-arithmeticity results for normal subgroups of Mod_S (even for so-called subnormal subgroups). For example, the Torelli subgroup, like the mapping class group itself, is not arithmetic. See [I7].

Finishing the discussion of this original approach to the non-arithmeticity, we note that it uses almost no non-trivial information about mapping class groups, but relies heavily on the Margulis finiteness theorem, one of the deepest results about arithmetic groups at all. Cf. the remarks at the end of 9.2.

9.4 Rank of the mapping class groups

As we mentioned in 9.3, properties of an arithmetic group depend to a considerable extent on whether its rank is 1 or ≥ 2 . Given the analogy between mapping class groups and arithmetic groups, it is important to ascertain what arithmetic groups—of rank 1 or of rank ≥ 2 —would be better analogues of mapping class groups. The best answer to this question would be a notion of rank applicable to both arithmetic groups and mapping class groups, together with a computation of this rank for the mapping class groups.

A suitable notion of rank was proposed by W. Ballmann and P. Eberlein [BalE], who developed the ideas of G. Prasad and M. Raghunathan [PrR]. Let Γ be an abstract group. Let $(\Gamma)_i$ denote the set of elements of Γ whose centralizer contains a free abelian subgroup of rank $\leq i$ as a subgroup of finite index (so that $(\Gamma)_0 \subset (\Gamma)_1 \subset \dots \subset (\Gamma)_i \subset \dots$). Let $r(\Gamma)$ be the least natural number i such that Γ can be presented as a finite union

$$\Gamma = \gamma_1(\Gamma)_i \cup \dots \cup \gamma_m(\Gamma)_i$$

of left translates of the set $(\Gamma)_i$ (the elements $\gamma_1, \dots, \gamma_m \in \Gamma$ are arbitrary). Let us define the *rank* of Γ as the maximum of the numbers $r(\Gamma')$, where Γ' runs over all subgroups of finite index in Γ . We denote it by $\text{rank}(\Gamma)$.

G. Prasad and M. Raghunathan [PrR] showed that if Γ is an arithmetic group, then $r(\Gamma)$ is equal to its rank in the sense of 9.3. In particular, this implies that the rank of an arithmetic group Γ in the sense of 9.3 depends only on the structure of Γ as an abstract group and not on its particular realization as an arithmetic subgroup of a semisimple Lie group. The replacement of $r(\cdot)$ by $\text{rank}(\cdot)$ was suggested by W. Ballmann and P. Eberlein [BalE] in order to get a notion of rank invariant under passing to subgroups of finite index. They proved that if Γ is a discrete group of motions of a complete simply connected riemannian manifold X whose curvature is nonpositive and bounded from below, such that X/Γ has finite volume, then $\text{rank}(\Gamma)$ is equal to a geometrically defined rank of X . This result, in fact, includes the case of arithmetic groups, and means that this rank of an abstract group gives the desired result even for groups more general than arithmetic ones.

Theorem 9.4.A *If S is not a sphere with ≤ 3 holes, then $\text{rank}(\text{Mod}_S) = 1$.*

Proof. We will give only an outline of the proof. Let Γ be a torsion free subgroup of finite index in Mod_S (such subgroups exist in view of Corollary 7.1.B). Since $\text{rank}(\cdot)$ is invariant under passing to subgroups of finite index, it is sufficient to show that $r(\Gamma) = 1$ for any such Γ . It is easy to see that $r(\Gamma) \neq 0$. Let Γ_{pA} denote the set of pseudo-Anosov elements of Γ . Theorem 7.5.A easily implies that the centralizers of pseudo-Anosov elements contain an infinite cyclic subgroup as a subgroup of finite index. Thus $\Gamma_{\text{pA}} \subset (\Gamma)_1$. Hence

it suffices to prove that for some finite number of elements $g_1, \dots, g_m \in \Gamma$ we have

$$\Gamma = g_1 \Gamma_{\text{pA}} \cup \dots \cup g_m \Gamma_{\text{pA}}.$$

In other words, it is sufficient to find a finite number of elements $g_1, \dots, g_m$ such that for any $g \in \Gamma$ one of the elements $g_1^{-1}g, \dots, g_m^{-1}g$ is pseudo-Anosov.

Using the compactness of the Thurston boundary of the Teichmüller space (cf. 7.2), it is possible to show that there exists a finite collection of isotopy classes $\alpha_1, \dots, \alpha_n$ such that for any isotopy class β one of the pairs α_i, β fills S in the sense of 7.1. Fix such a collection $\alpha_1, \dots, \alpha_n$. Let t_i , $1 \leq i \leq n$ be a non-trivial power of the Dehn twist t_{α_i} contained in Γ (remember that Γ is of finite index in Mod_S). Using Fathi's Theorem 7.1.D, one can show that for any $g \in \Gamma$ one of the elements $t_i^d g$, $1 \leq i \leq n$, $1 \leq d \leq 8$ is pseudo-Anosov. It follows that $r(\Gamma) = 1$. We refer to [I9] for the details. $\square$

Corollary 9.4.B *For any compact orientable surface S , the group Mod_S is not isomorphic to any arithmetic group of rank ≥ 2 .*

One can replace all references to the Margulis finiteness theorem in 9.3 by this Corollary. This leads to still another proof of the non-arithmeticity of the mapping class groups. On the arithmetic groups side, it uses the results of G. Prasad and M. Raghunathan [PrR], which are not as difficult as the Margulis finiteness theorem. It also uses some quite non-trivial results on the mapping class groups side (Fathi's Theorem 7.1.D); cf. the remarks at the end of 9.2, 9.3.

The next theorem, as we will explain in a moment, also supports the idea that the mapping class groups are similar to the arithmetic groups of rank 1, as opposed to rank ≥ 2 .

Theorem 9.4.C *Suppose that S is not a sphere with ≤ 3 holes. Let us fix a finite set of generators of Mod_S and let us denote by $|\cdot|_W$ the minimal word length with respect to these generators. If f is an element of infinite order in Mod_S , then there is a constant $c > 0$ such that*

$$|f^n|_W \geq c |n|$$

for all n .

It is easy to see that one can always replace f by a nontrivial power of f in the proof. In particular, we may assume that f is a pure element of Mod_S (see 7.1). Then there are two different cases to consider. If the diffeomorphism induced on the result of cutting S along some realization of $\sigma(f)$ is isotopic to a pseudo-Anosov diffeomorphism on at least one component, then the theorem follows from the results of L. Mosher [Mos3]. Otherwise, f is a composition of Dehn twists about several disjoint circles. This case was recently dealt with by Y. Minsky [Min2], in response to a question posed by the author.

Recall that an arithmetic group Γ arises as a subgroup of a real semisimple Lie group H . The group Γ is called *cocompact* if H/Γ is compact. The similarity of Mod_S with cocompact arithmetic groups can be easily excluded by cohomological reasons (as, for example, in [Har5], Section 4.2; cf. 9.3). But, for noncompact arithmetic groups Γ of rank ≥ 2 , there are always elements $f \in \Gamma$ of infinite order such that the word length $|f^n|_W$ grows slower than linearly (in fact, logarithmically), according to a recent result of A. Lubotzky, S. Mozes and M. S. Raghunathan [LubMR]. Therefore, Theorem 9.4.C supports the analogy between Mod_S and rank 1 arithmetic groups.

9.5 The Mostow-Margulis superrigidity

The Margulis superrigidity theorem [Ma1], (cf. also [Ma2] or [Zim], Chapter 5) extends the Mostow rigidity theorem [Most1], [Most2]. It asserts, in particular, that any homomorphism $\Gamma_1 \rightarrow \Gamma_2$ between arithmetic subgroups Γ_1, Γ_2 of semisimple Lie groups H_1, H_2 respectively (without compact factors) extends to a homomorphism $H_1 \rightarrow H_2$ after, perhaps, passing to a subgroup of finite index in Γ_1 , if $\text{rank } \Gamma_1 \geq 2$. Note that the restriction $\text{rank } \Gamma_1 \geq 2$ in this theorem is necessary. Here we meet one of the situations where arithmetic groups of rank 1 and rank ≥ 2 behave differently.

The analogy between mapping class groups and arithmetic groups suggests that one may extend this theorem to include mapping class groups on the equal footing with arithmetic groups. At first sight, this is impossible, because there are no Lie groups naturally (or, to the best of our knowledge, otherwise) containing mapping class groups. But this simply means that there should be no homomorphisms, or better, taking into account the possible passage to a finite index subgroup in the Margulis theorem, that any homomorphism should have finite image. In addition, since mapping class groups have rank 1 and the source group in the Margulis theorem should have rank ≥ 2 , we are forced to consider only homomorphisms from arithmetic groups of rank ≥ 2 to mapping class groups. Then it is natural to expect that any such homomorphism has finite image.

The first step in the direction of this conjecture was made in [I2], and the conjecture itself was stated explicitly in [I4]. Further steps were made in [I7]. Recently, V. Kaimanovich and H. Masur [KaM] almost completely proved it. Following [KaM], let us call a subgroup of Mod_S *non-elementary* if it leaves no finite subset of the Thurston boundary of the corresponding Teichmüller space invariant.

Theorem 9.5.A *Any non-elementary subgroup of a mapping class group is not isomorphic to an arithmetic group of rank ≥ 2 .*

See [KaM], Theorem 2.4.1. The proof of V. Kaimanovich and H. Masur is based on a theory of random walks on Teichmüller spaces and mapping class

groups developed by H. Masur [Mas4] and V. Kaimanovich and H. Masur [KaM] and the results of H. Furstenberg [Fu1], [Fu2] about random walks and harmonic functions on discrete subgroups of Lie groups ([Fu2] provides an excellent introduction to this circle of ideas). More precisely, H. Furstenberg introduced a property of harmonic functions (which are closely related to the random walks) on discrete groups which holds for all arithmetic groups of rank ≥ 2 and does not hold for (at least some) arithmetic groups of rank 1. V. Kaimanovich and H. Masur show that this property does not hold also for all non-elementary subgroups of mapping class groups; thus they cannot be arithmetic groups of rank ≥ 2 . Note that these results of H. Furstenberg (as also the results of G. D. Mostow) preceded and influenced G. A. Margulis' approach to his superrigidity theorem.

If combined with the Margulis finiteness theorem and the results of [I15] (cf. 7.4) about subgroups of mapping class groups, Theorem 9.5.A leads to the following Corollary, which proves the above conjecture.

Corollary 9.5.B *Let $\varphi : \Gamma \rightarrow \text{Mod}_S$ be a homomorphism from an arithmetic group Γ to a mapping class group Mod_S . If $\text{rank } \Gamma \geq 2$, then the image of φ is finite.*

A proof of this Corollary along similar lines was also found by B. Farb and H. Masur [FM]. It seems that it is possible to prove this Corollary more directly, avoiding the use of the Margulis finiteness theorem, of random walks (and, hence, of Furstenberg's results), and even of Teichmüller spaces (but one still needs their Thurston's boundaries). Such an approach would follow the Margulis proof of his superrigidity theorem. I hope to return to these ideas in a future paper.

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